

# Non-path Toeplitz trees

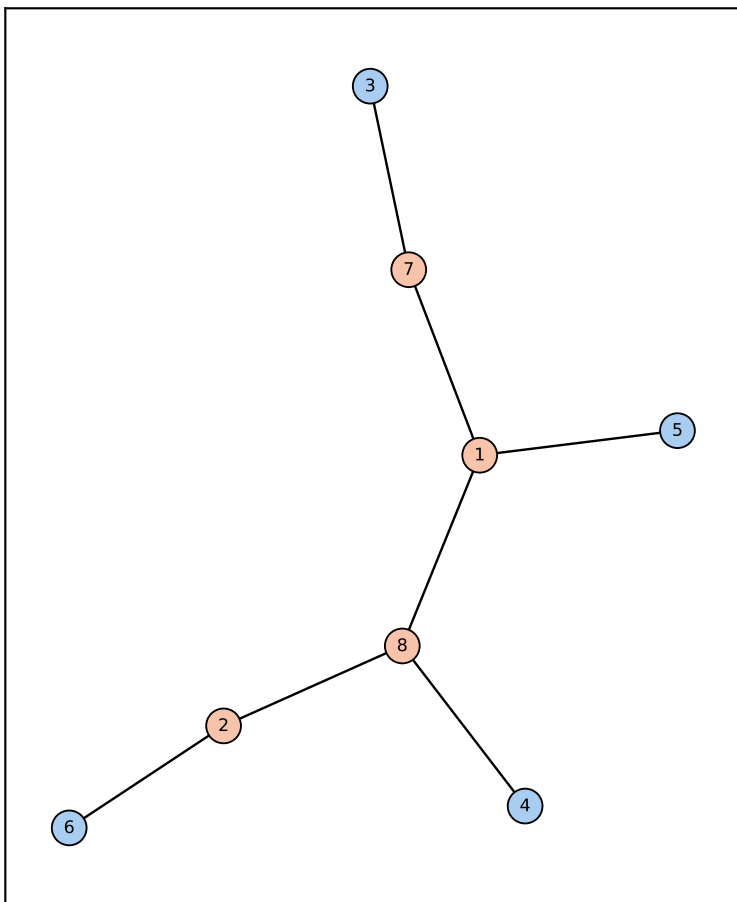
all  $T_n(t_1, \dots, t_k)$  that are trees but not paths,  $n \leq 32$

103 stripe sets, grouped by  $n$ . Equal letters would mark isomorphic trees, none coincide,  
so the 103 trees are pairwise non-isomorphic.

$s_j = n - t_j$  form a partition of  $n - 1$  into distinct parts summing to the edge count

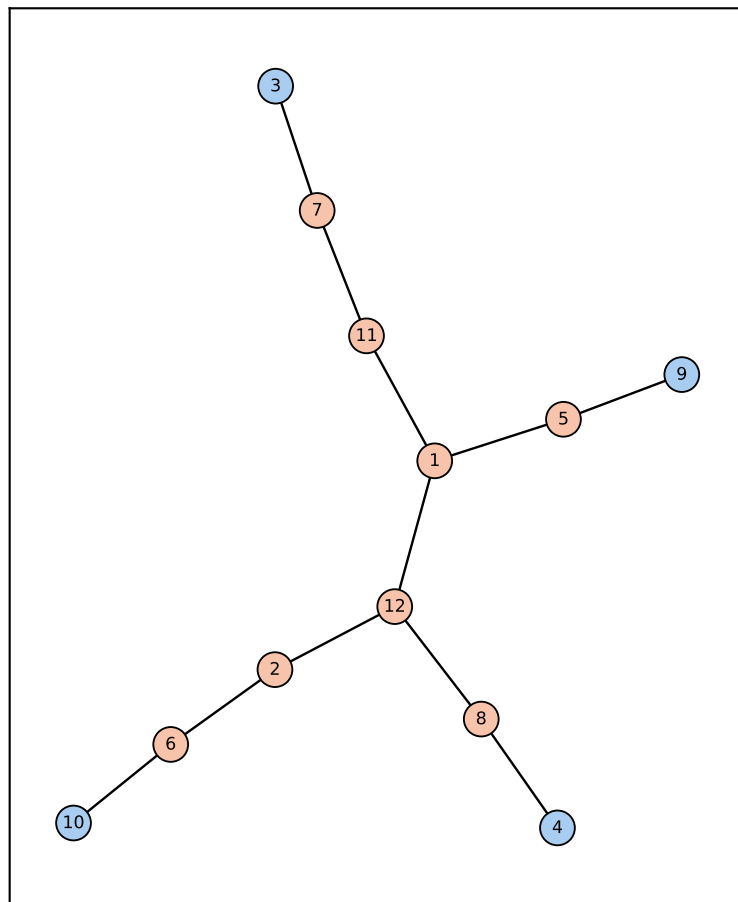
 leaf       internal

$T_8\langle 4, 6, 7 \rangle$  (A)



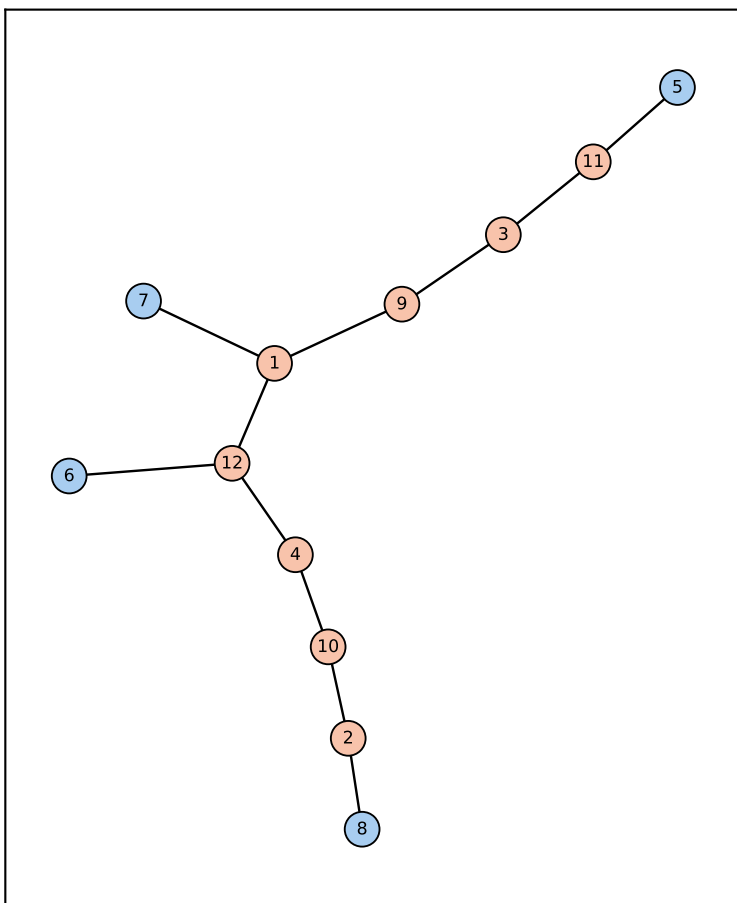
$s = (4, 2, 1)$ ,  $\Delta = 3$ , caterpillar

$T_{12}\langle 4, 10, 11 \rangle$  (A)



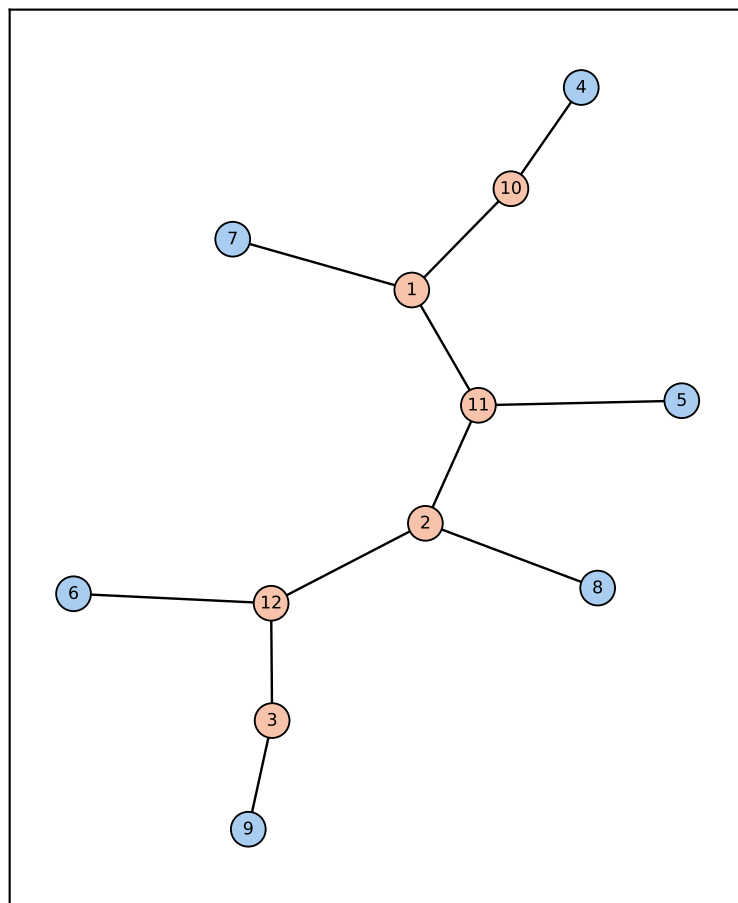
$s = (8, 2, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{12}\langle 6, 8, 11 \rangle$  (B)



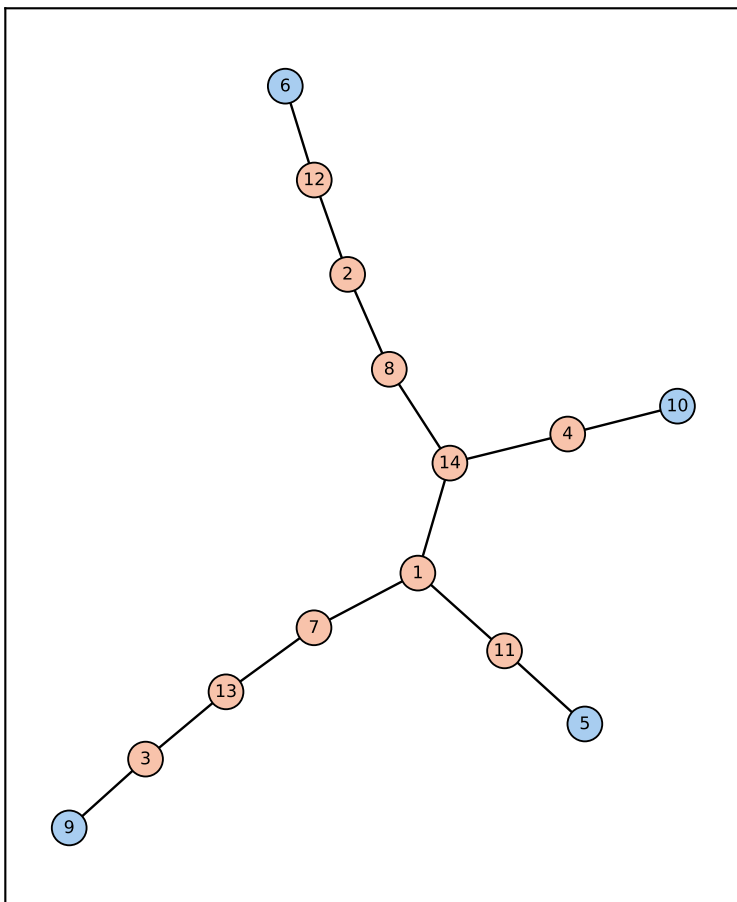
$s = (6, 4, 1)$ ,  $\Delta = 3$ , caterpillar

$T_{12}\langle 6, 9, 10 \rangle$  (C)



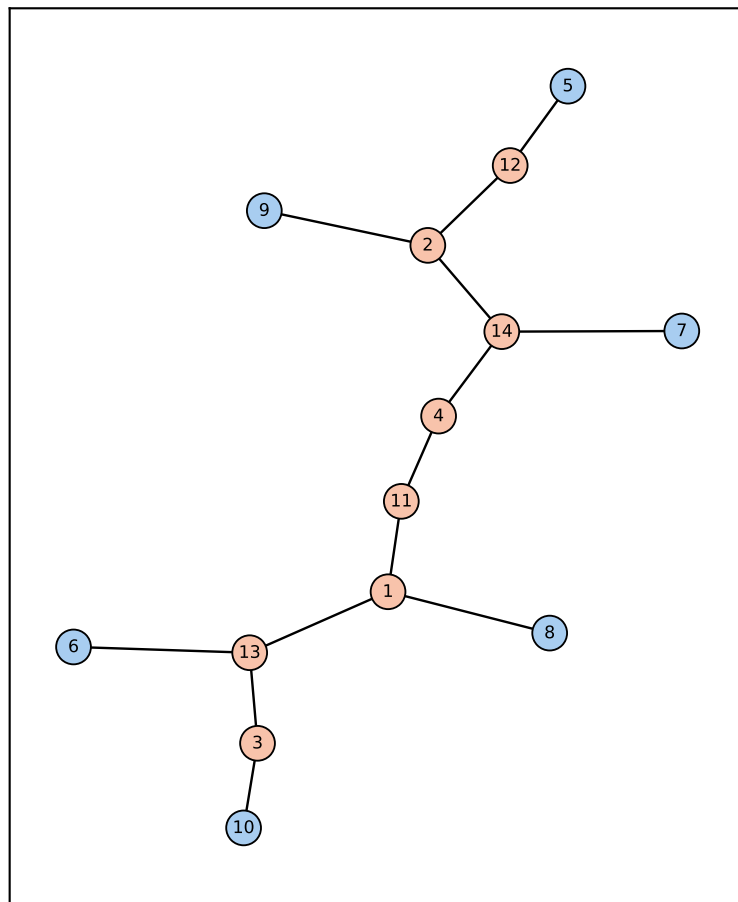
$s = (6, 3, 2)$ ,  $\Delta = 3$ , caterpillar

$T_{14}(6, 10, 13)$  (A)



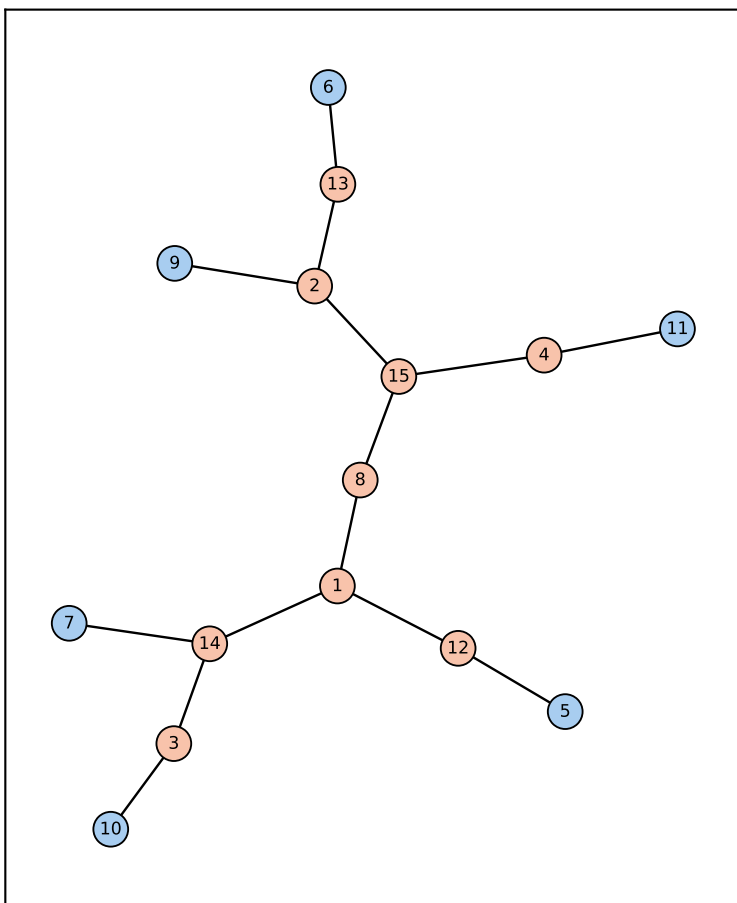
$s = (8, 4, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{14}(7, 10, 12)$  (B)



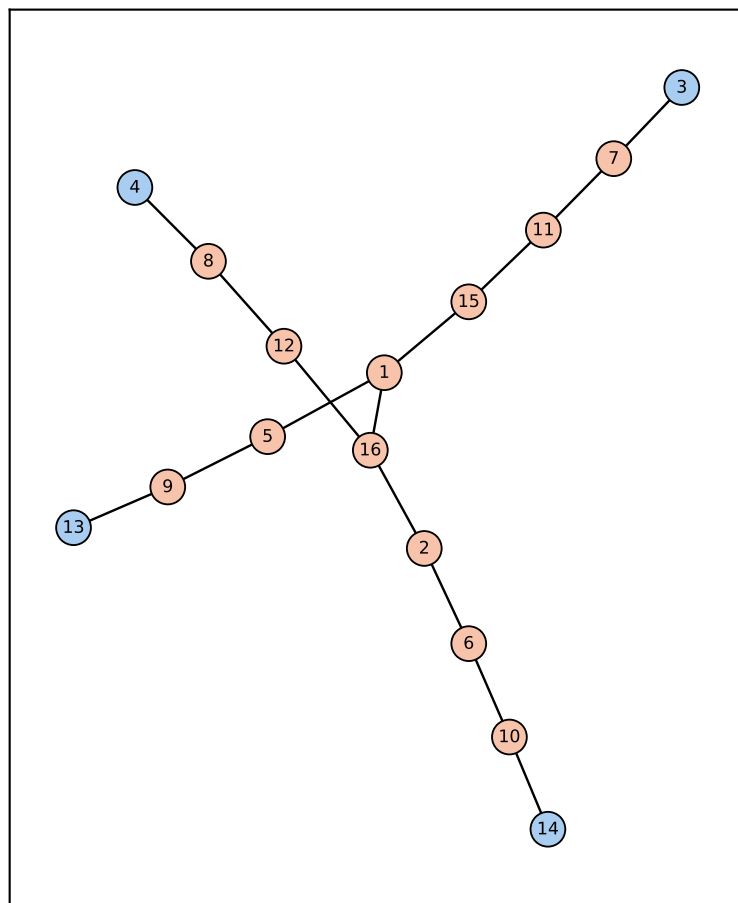
$s = (7, 4, 2)$ ,  $\Delta = 3$ , caterpillar

$T_{15}(7, 11, 13)$  (A)



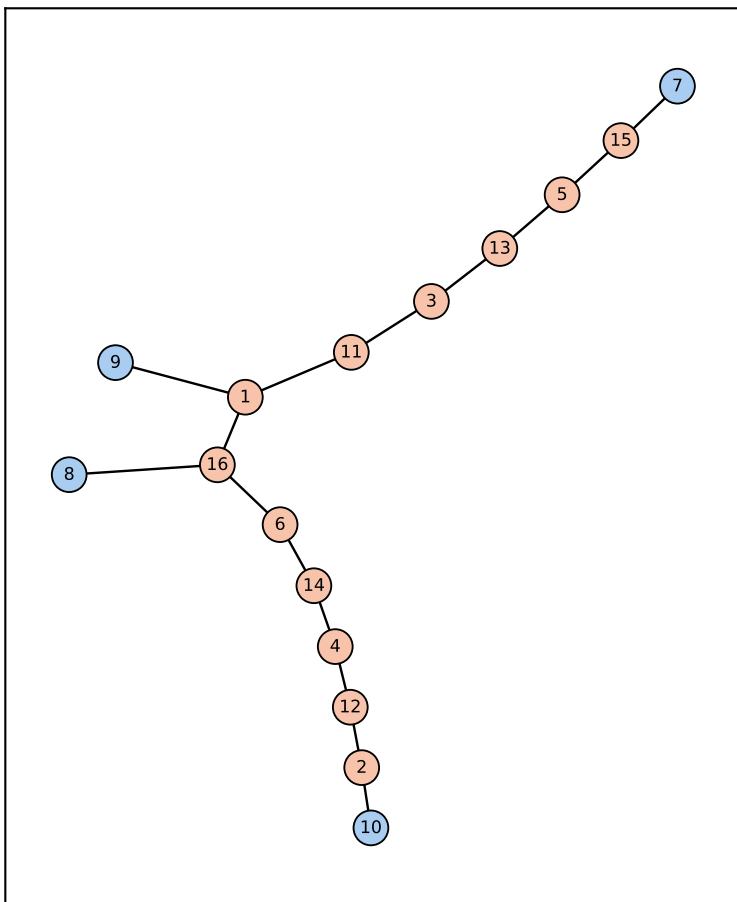
$s = (8, 4, 2)$ ,  $\Delta = 3$ , not caterpillar

$T_{16}(4, 14, 15)$  (A)



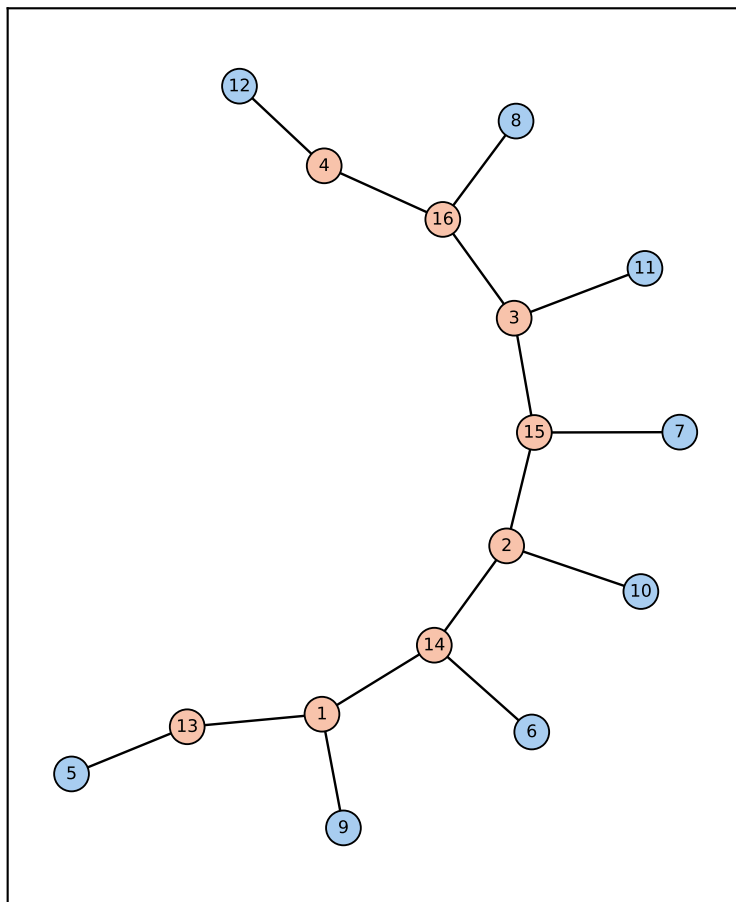
$s = (12, 2, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{16}(8, 10, 15)$  (B)



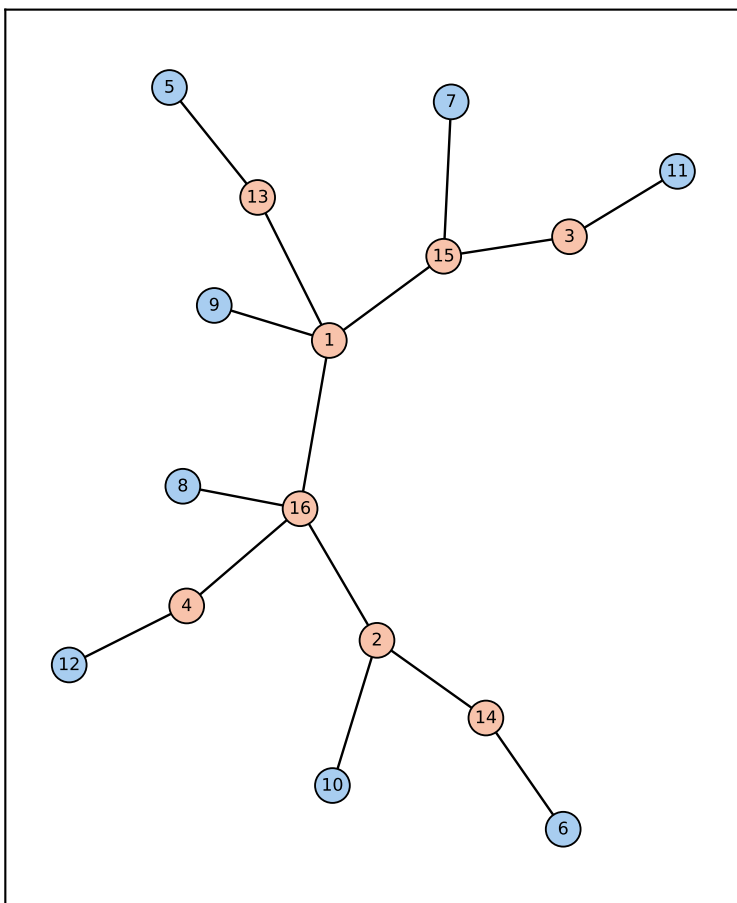
$s = (8, 6, 1)$ ,  $\Delta = 3$ , caterpillar

$T_{16}(8, 12, 13)$  (C)



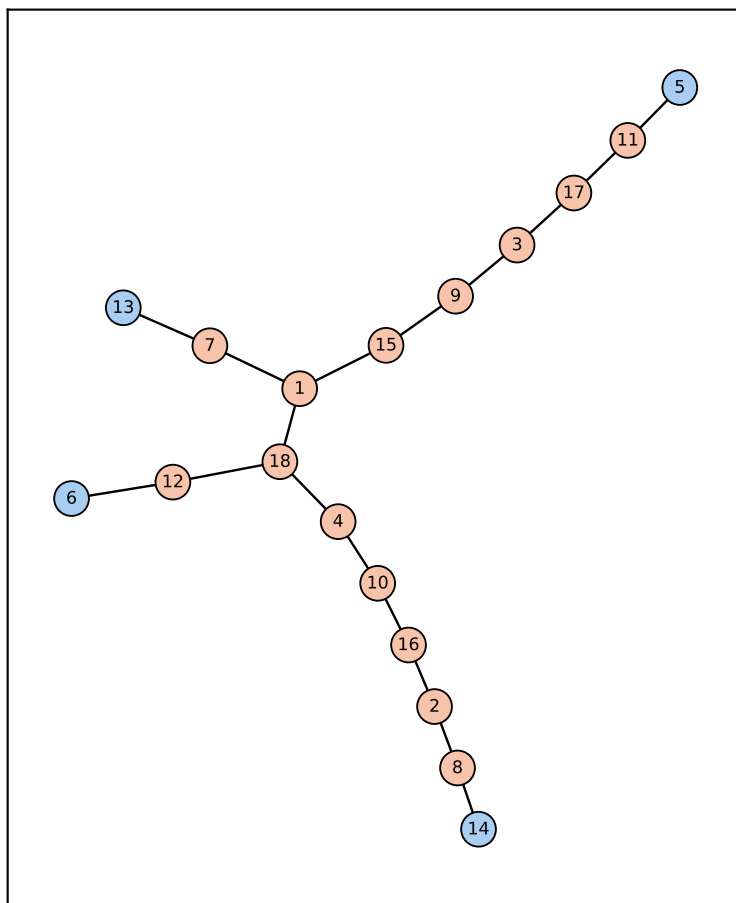
$s = (8, 4, 3)$ ,  $\Delta = 3$ , caterpillar

$T_{16}(8, 12, 14, 15)$  (D)



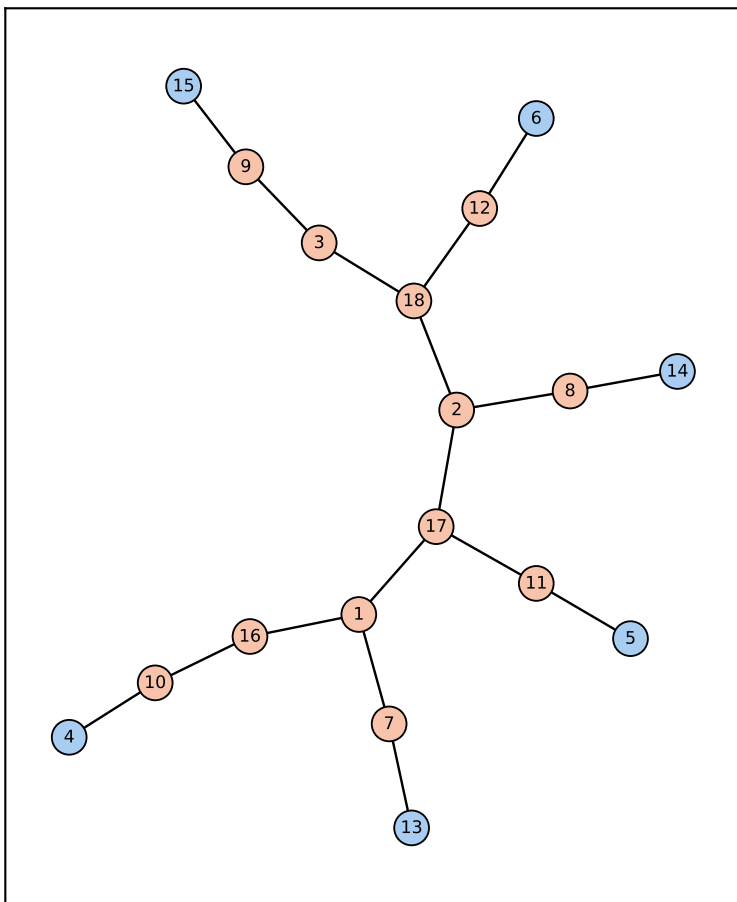
$s = (8, 4, 2, 1)$ ,  $\Delta = 4$ , not caterpillar

$T_{18}(6, 14, 17)$  (A)



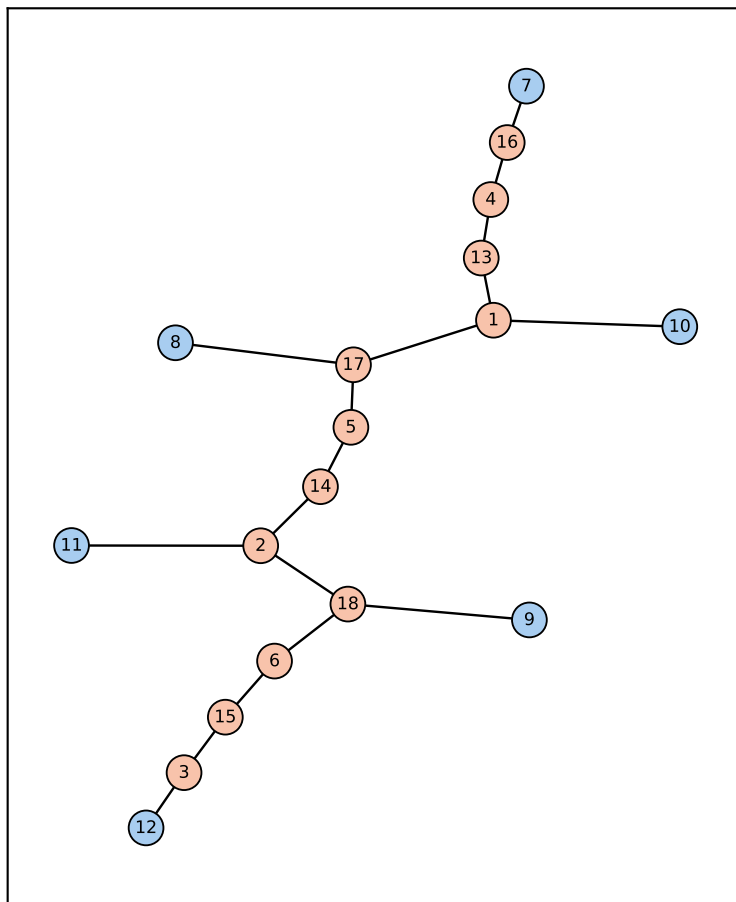
$s = (12, 4, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{18}(6, 15, 16)$  (B)



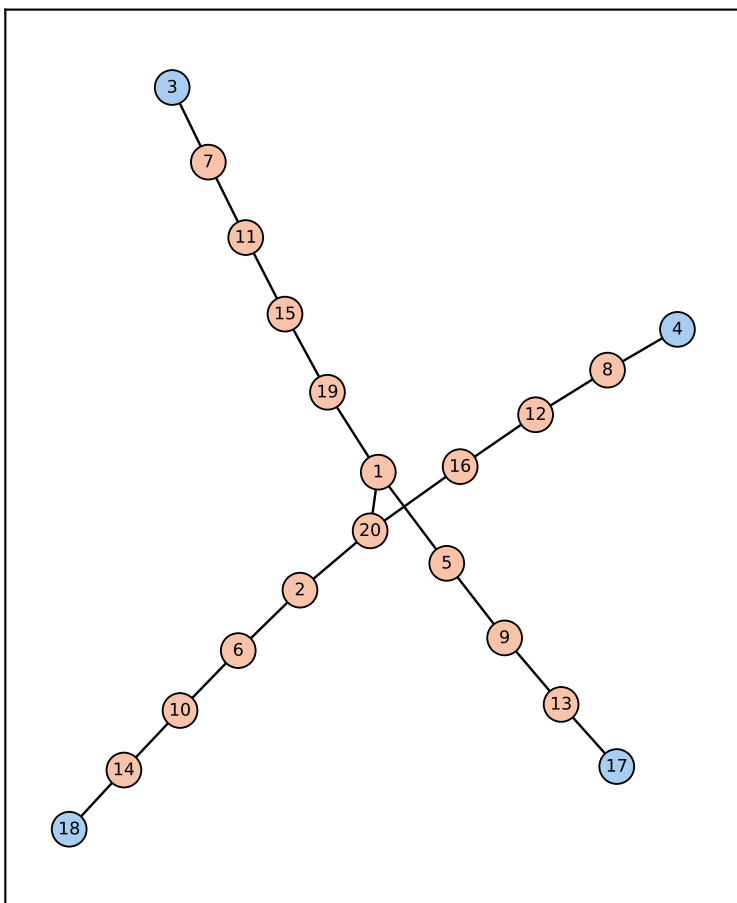
$s = (12, 3, 2)$ ,  $\Delta = 3$ , not caterpillar

$T_{18}(9, 12, 16)$  (C)



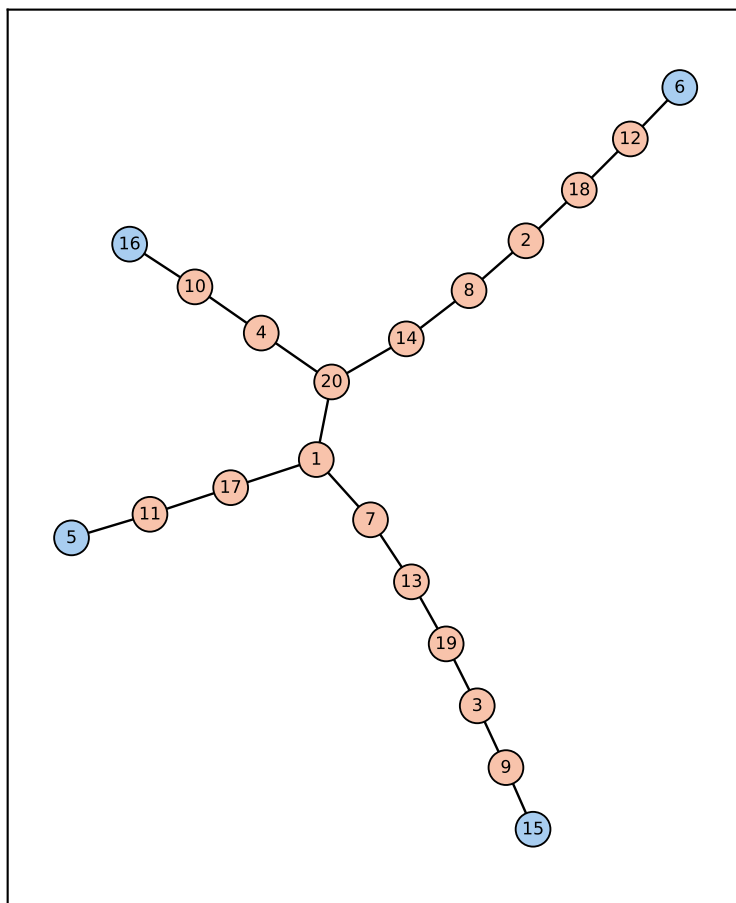
$s = (9, 6, 2)$ ,  $\Delta = 3$ , caterpillar

$T_{20}(4, 18, 19)$  (A)



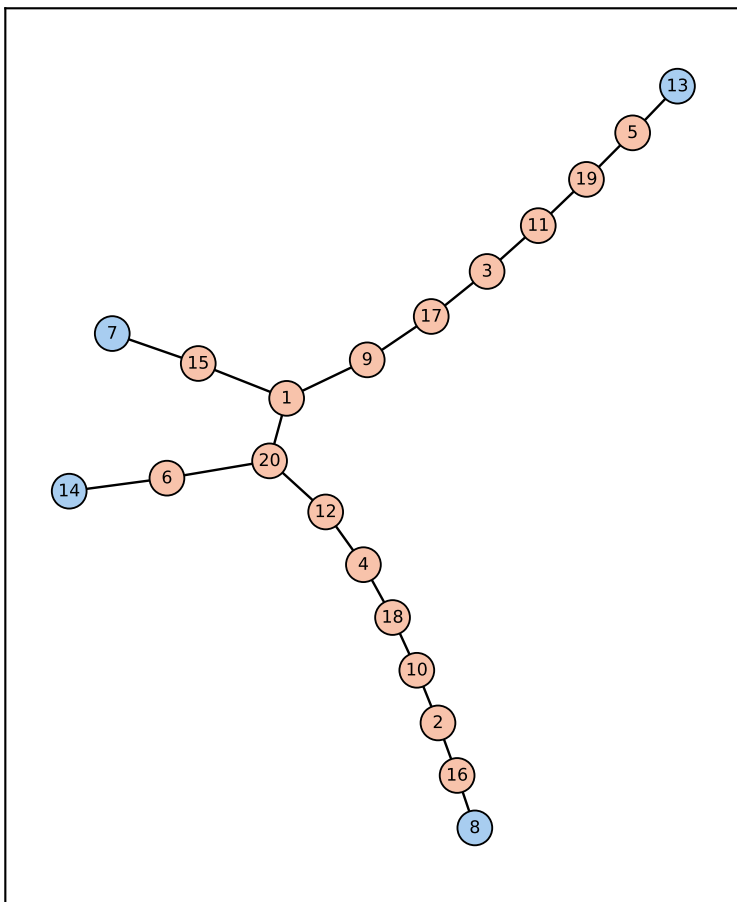
$s = (16, 2, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{20}(6, 16, 19)$  (B)



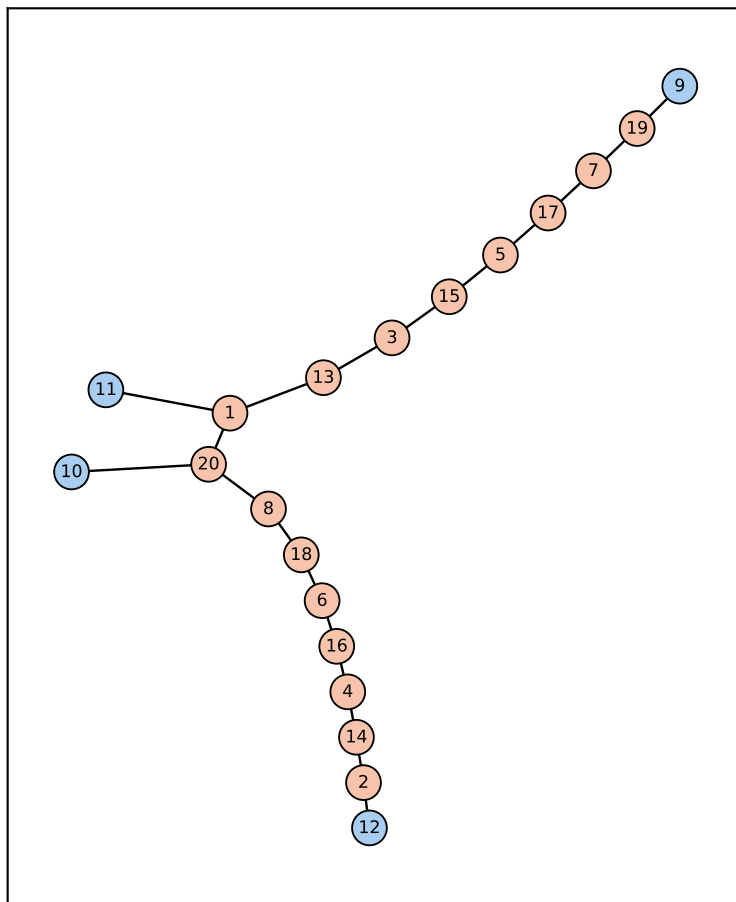
$s = (14, 4, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{20}\langle 8, 14, 19 \rangle$  (C)



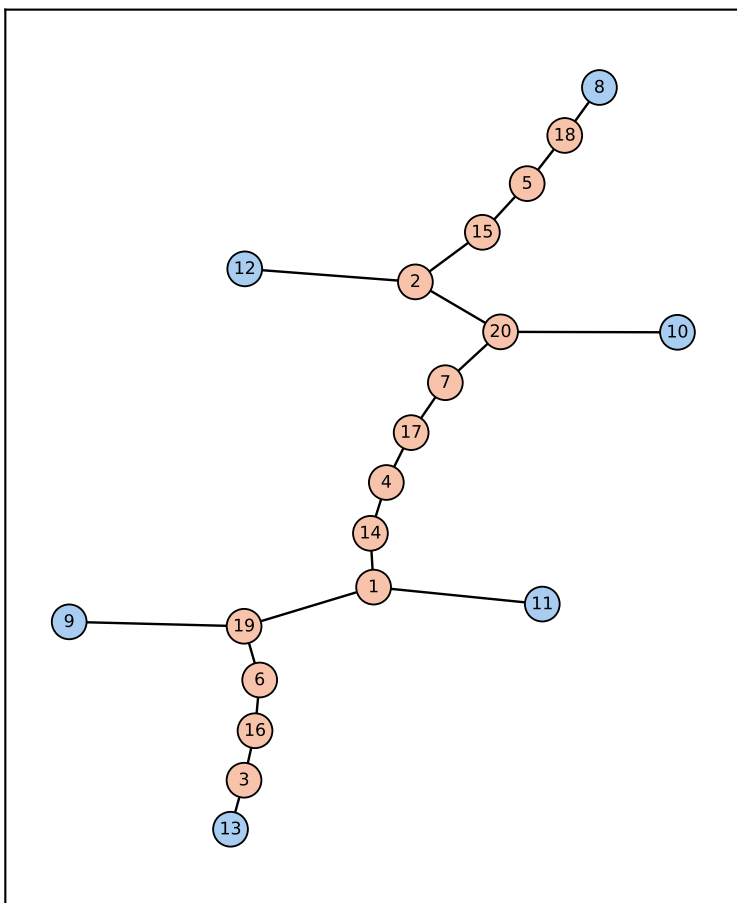
$s = (12, 6, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{20}\langle 10, 12, 19 \rangle$  (D)



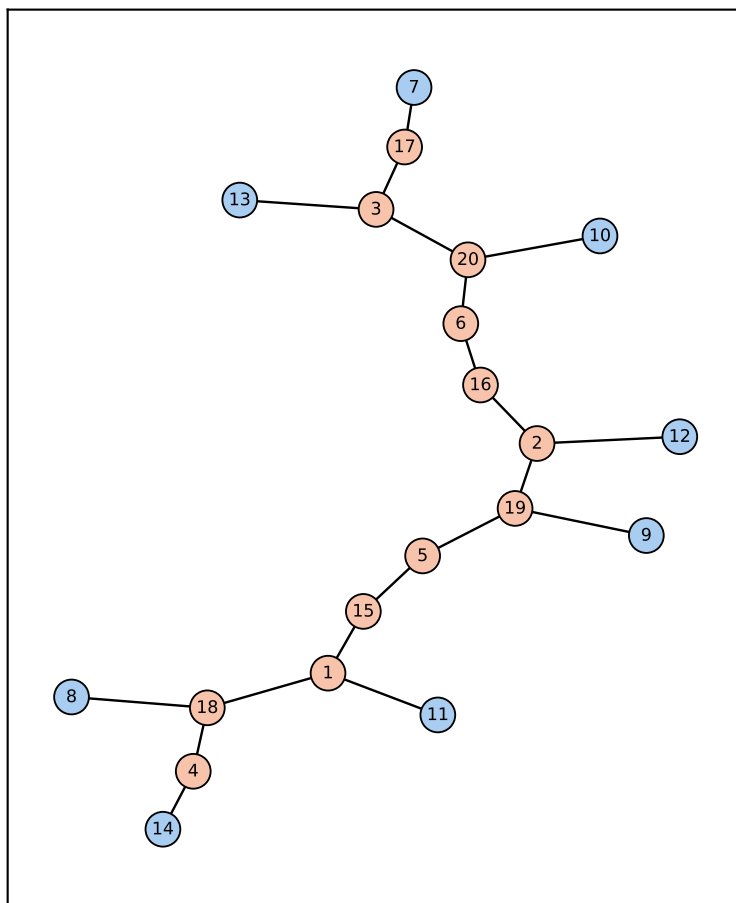
$s = (10, 8, 1)$ ,  $\Delta = 3$ , caterpillar

$T_{20}\langle 10, 13, 18 \rangle$  (E)



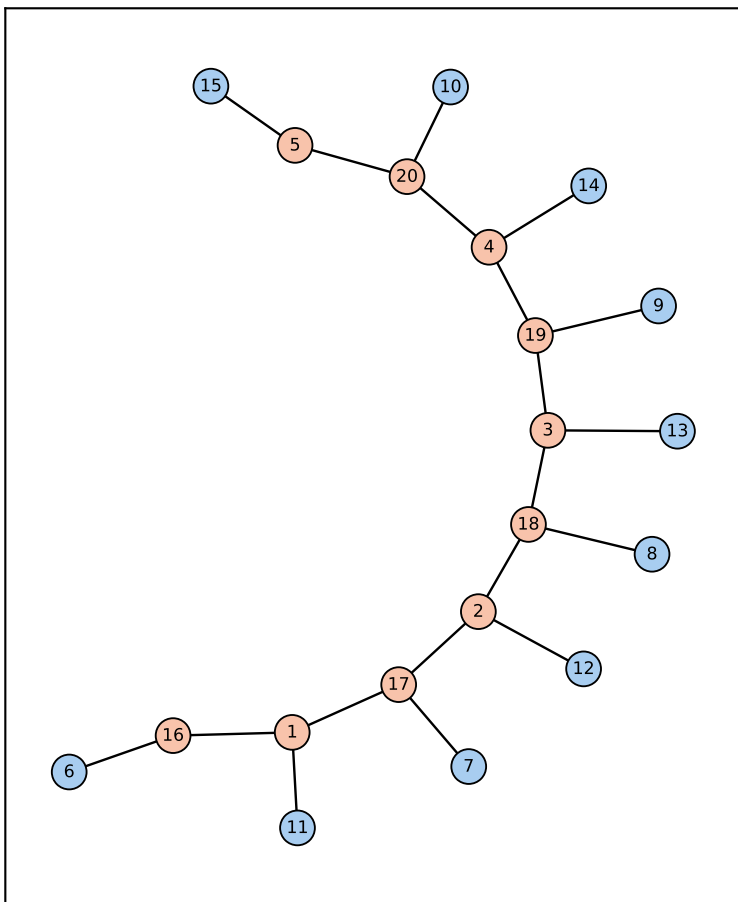
$s = (10, 7, 2)$ ,  $\Delta = 3$ , caterpillar

$T_{20}\langle 10, 14, 17 \rangle$  (F)



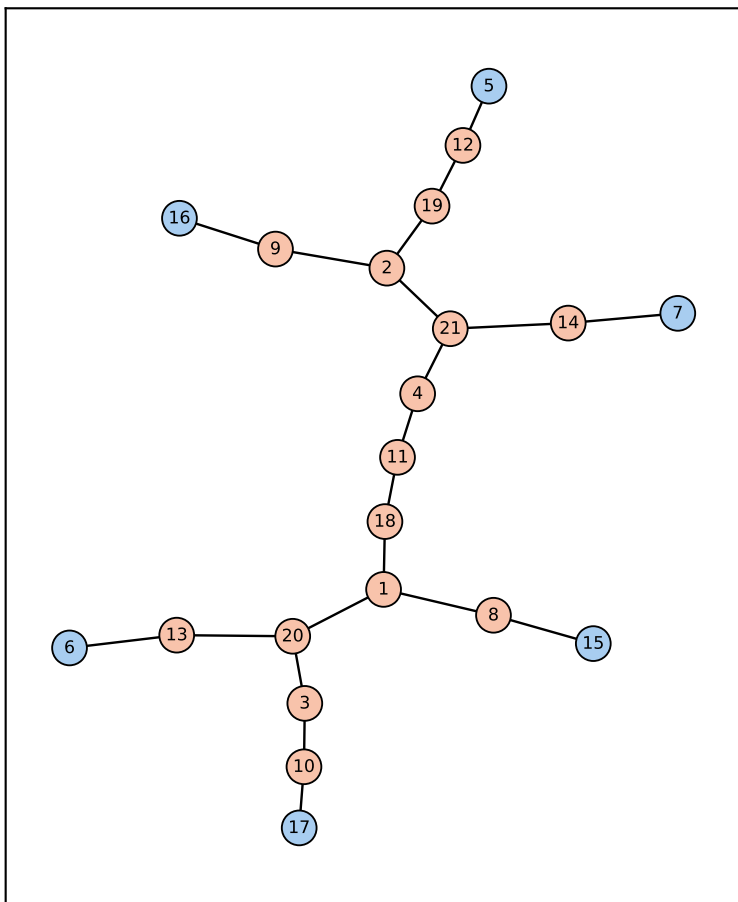
$s = (10, 6, 3)$ ,  $\Delta = 3$ , caterpillar

$T_{20}(10, 15, 16)$  (G)



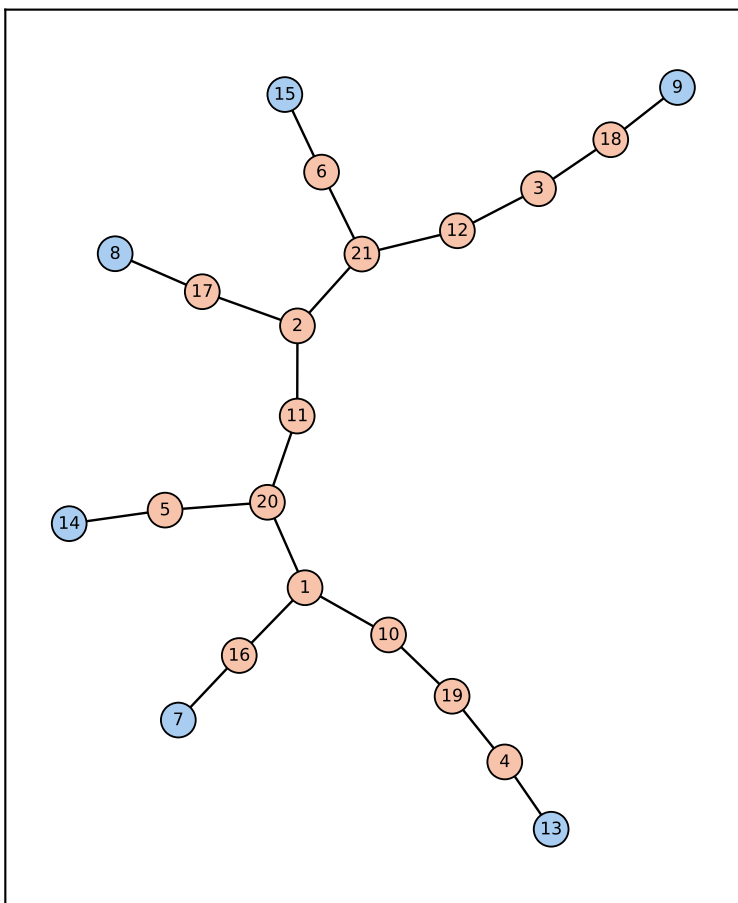
$s = (10, 5, 4)$ ,  $\Delta = 3$ , caterpillar

$T_{21}(7, 17, 19)$  (A)



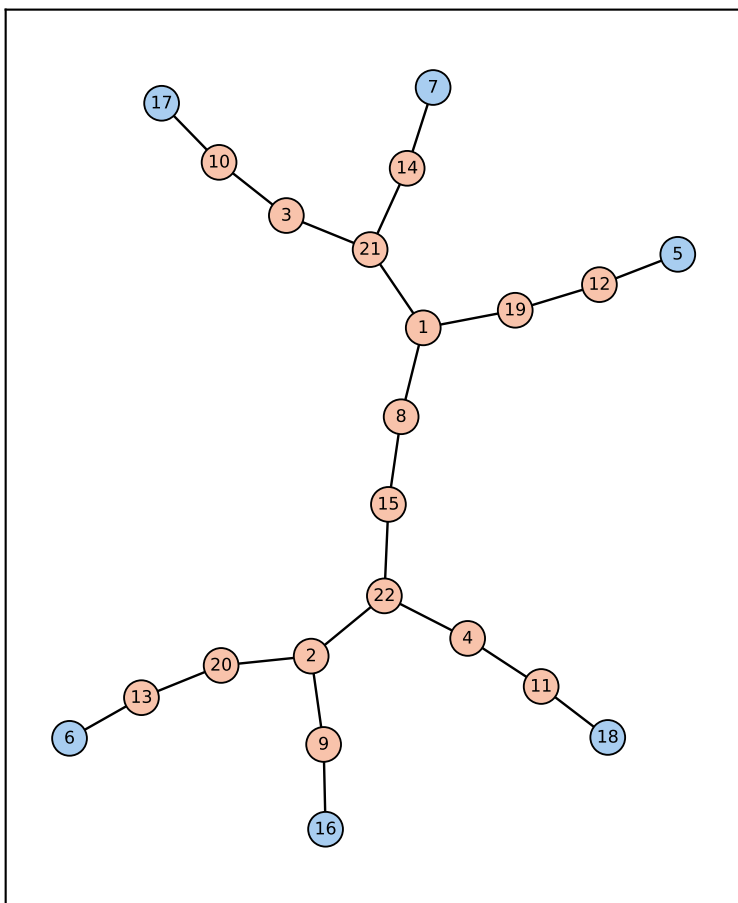
$s = (14, 4, 2)$ ,  $\Delta = 3$ , not caterpillar

$T_{21}(9, 15, 19)$  (B)

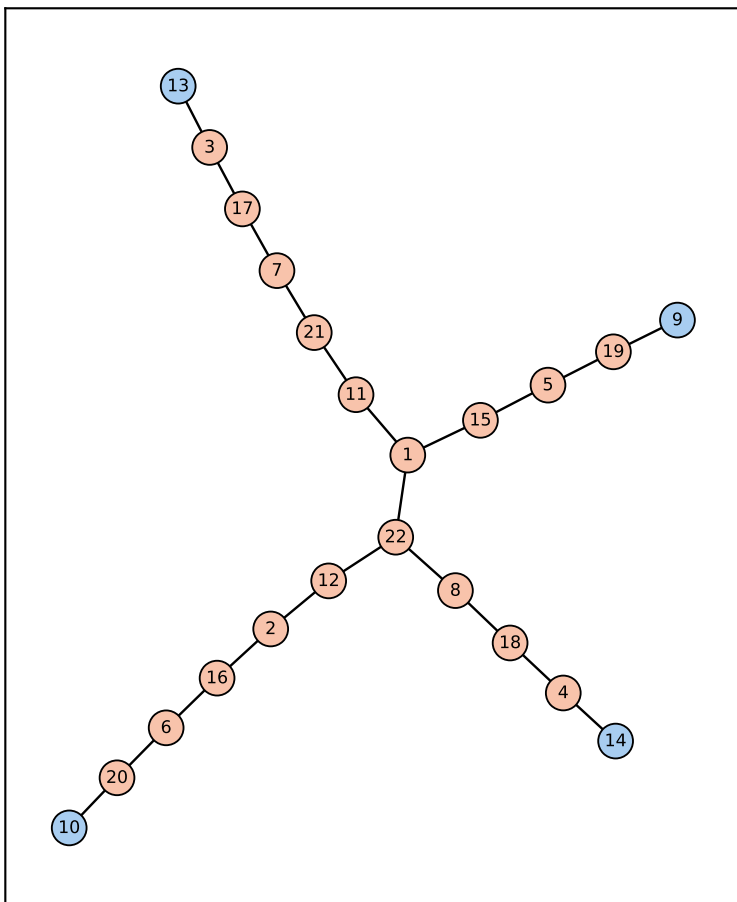


$s = (12, 6, 2)$ ,  $\Delta = 3$ , not caterpillar

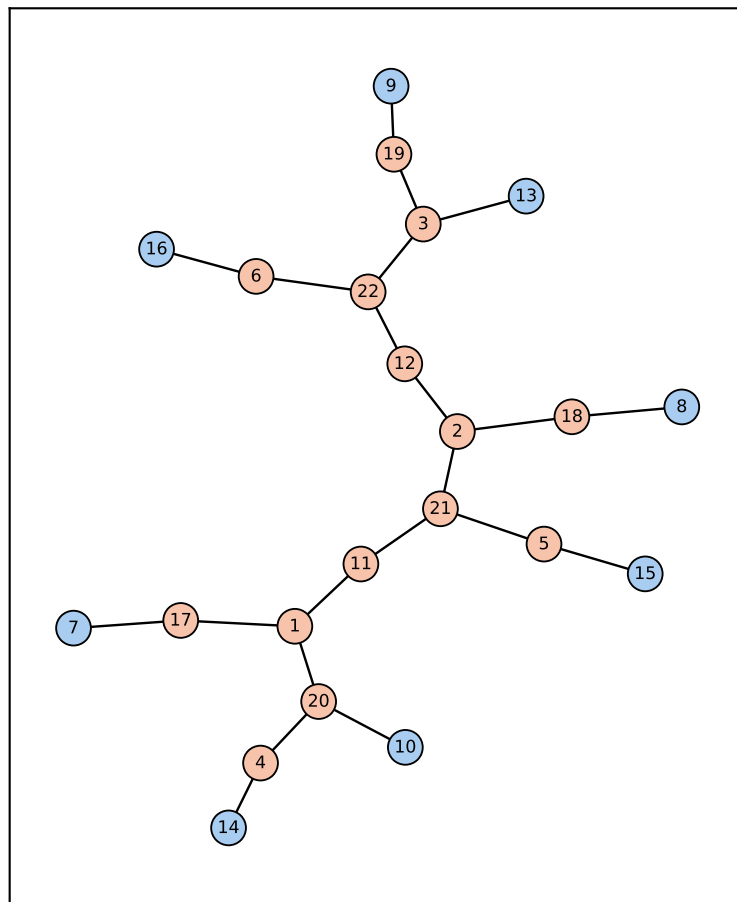
$T_{22}(7, 18, 20)$  (A)



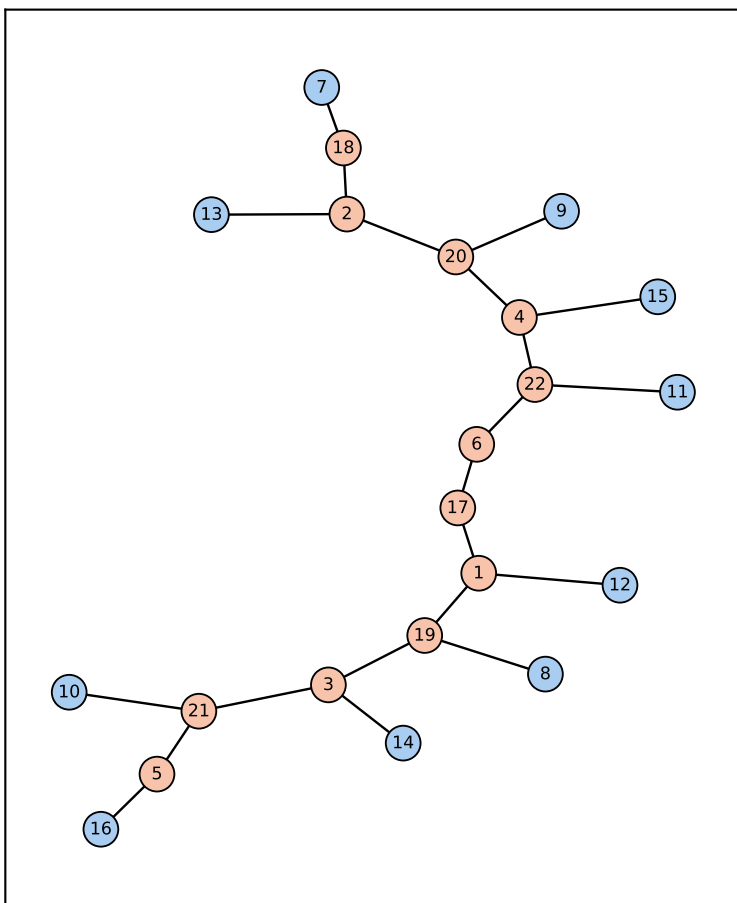
$s = (15, 4, 2)$ ,  $\Delta = 3$ , not caterpillar

$T_{22}\langle 10, 14, 21 \rangle$  (B)


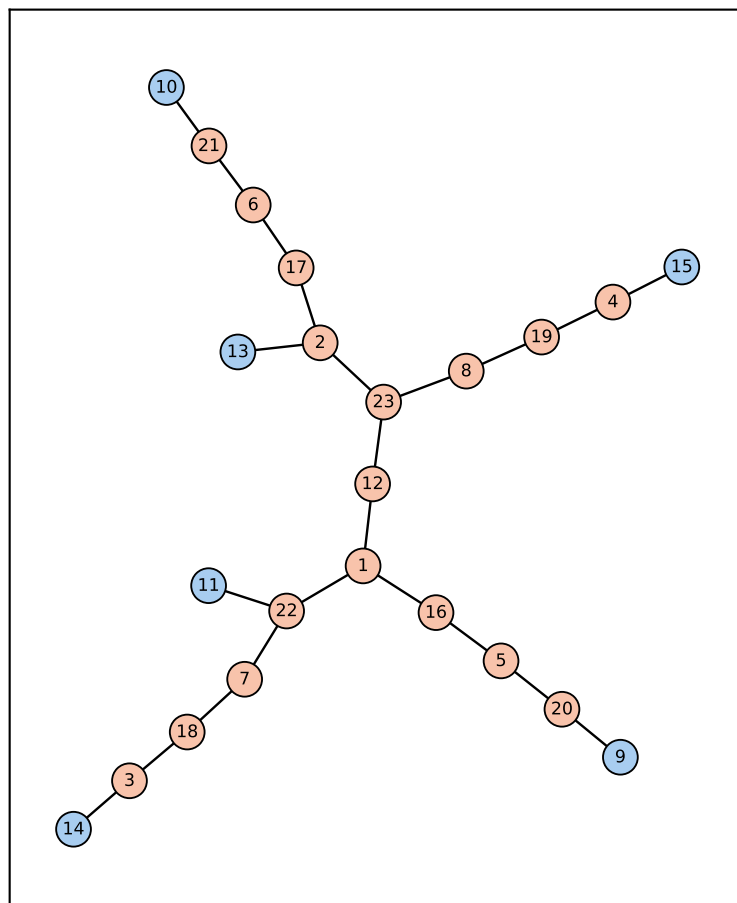
$s = (12, 8, 1)$ ,  $\Delta = 3$ , not caterpillar

 $T_{22}\langle 10, 16, 19 \rangle$  (C)


$s = (12, 6, 3)$ ,  $\Delta = 3$ , not caterpillar

 $T_{22}\langle 11, 16, 18 \rangle$  (D)


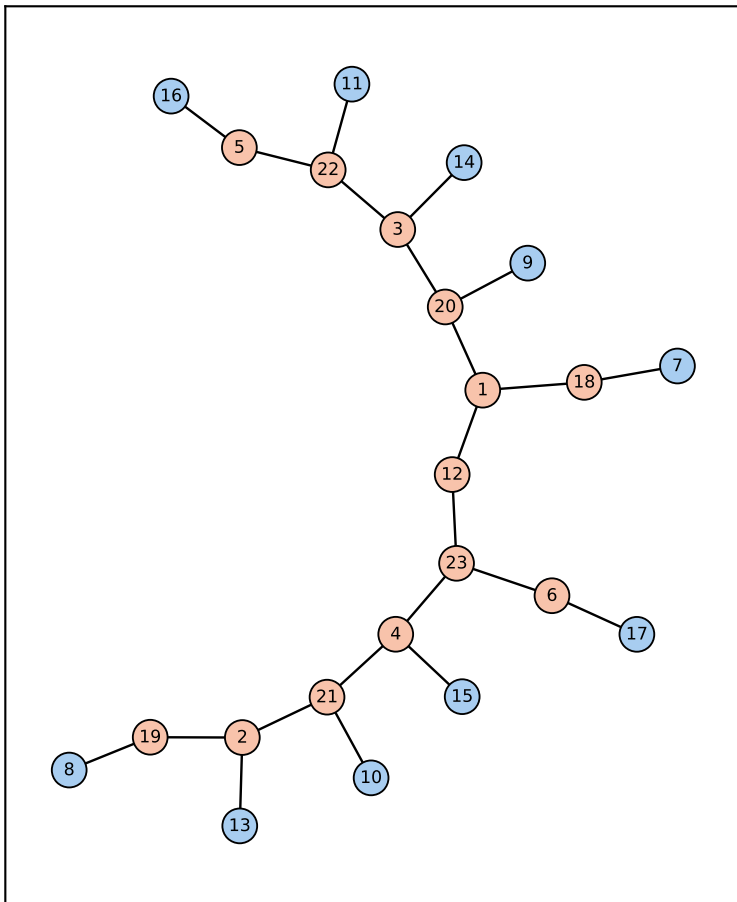
$s = (11, 6, 4)$ ,  $\Delta = 3$ , caterpillar

 $T_{23}\langle 11, 15, 21 \rangle$  (A)


$s = (12, 8, 2)$ ,  $\Delta = 3$ , not caterpillar

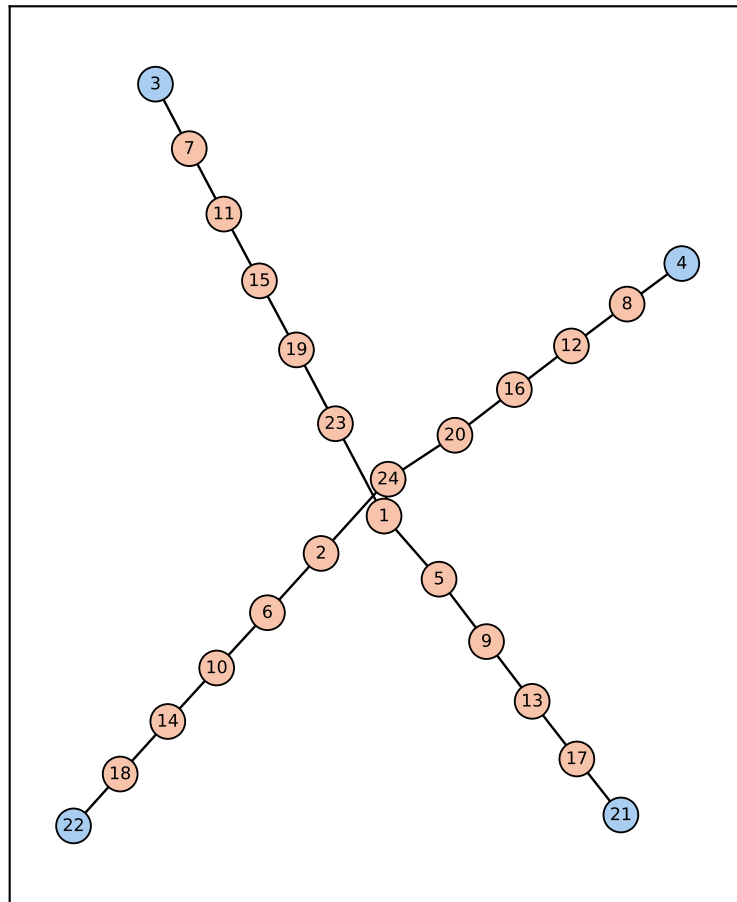


$T_{23}(11, 17, 19)$  (B)



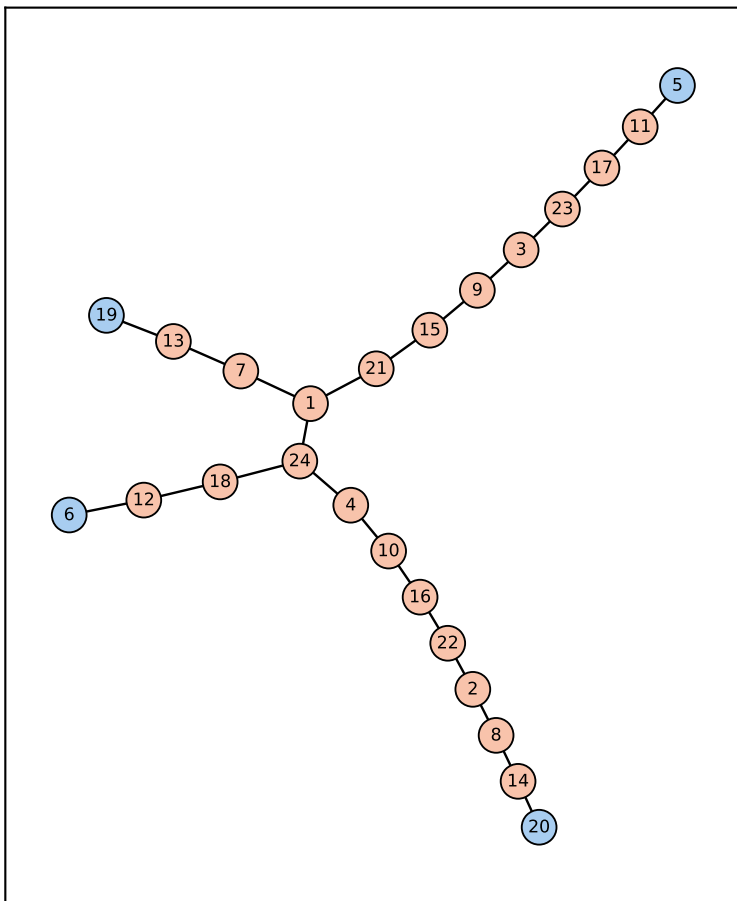
$s = (12, 6, 4)$ ,  $\Delta = 3$ , not caterpillar

$T_{24}(4, 22, 23)$  (A)



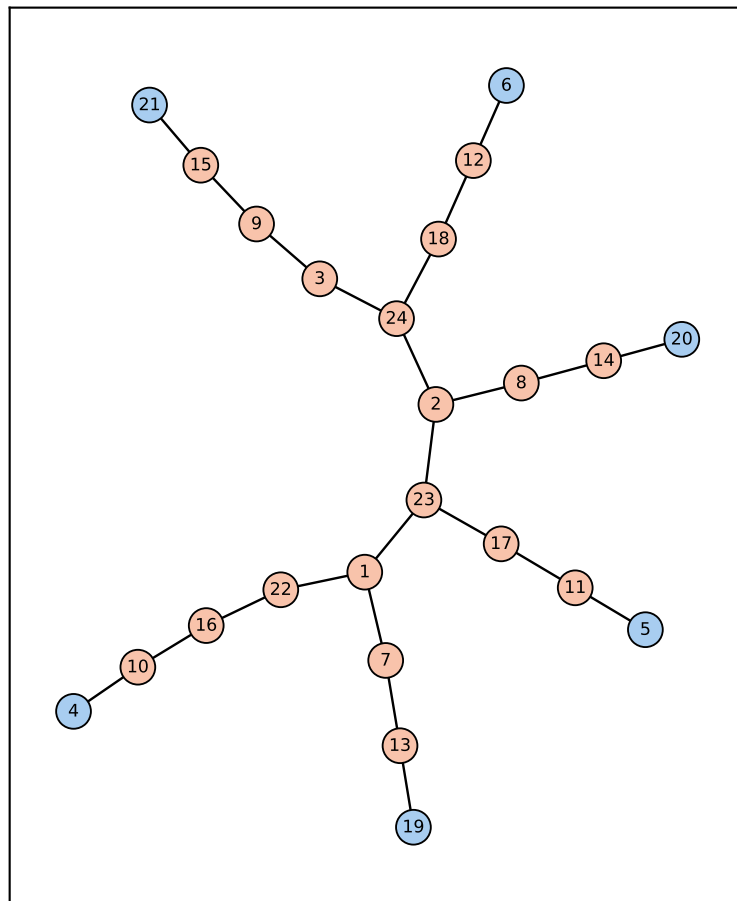
$s = (20, 2, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{24}(6, 20, 23)$  (B)



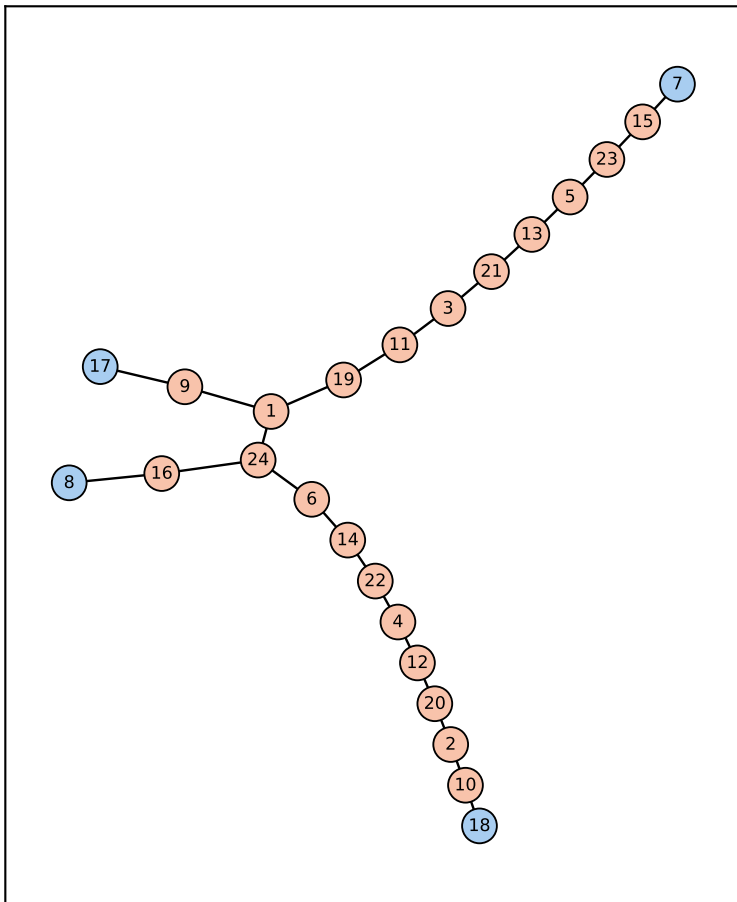
$s = (18, 4, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{24}(6, 21, 22)$  (C)



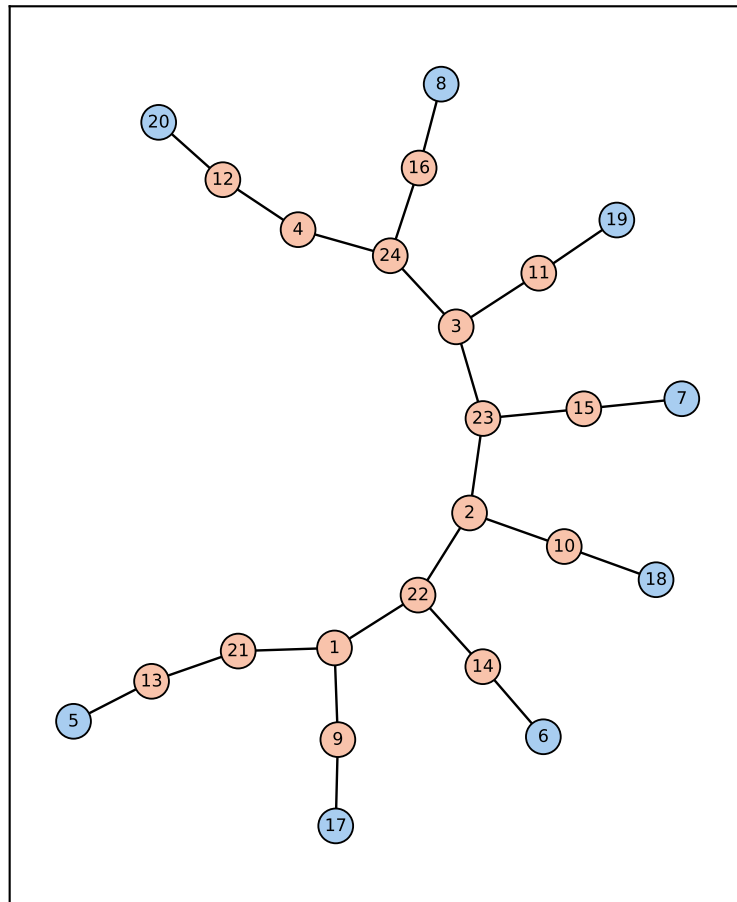
$s = (18, 3, 2)$ ,  $\Delta = 3$ , not caterpillar

$T_{24}\langle 8, 18, 23 \rangle$  (D)



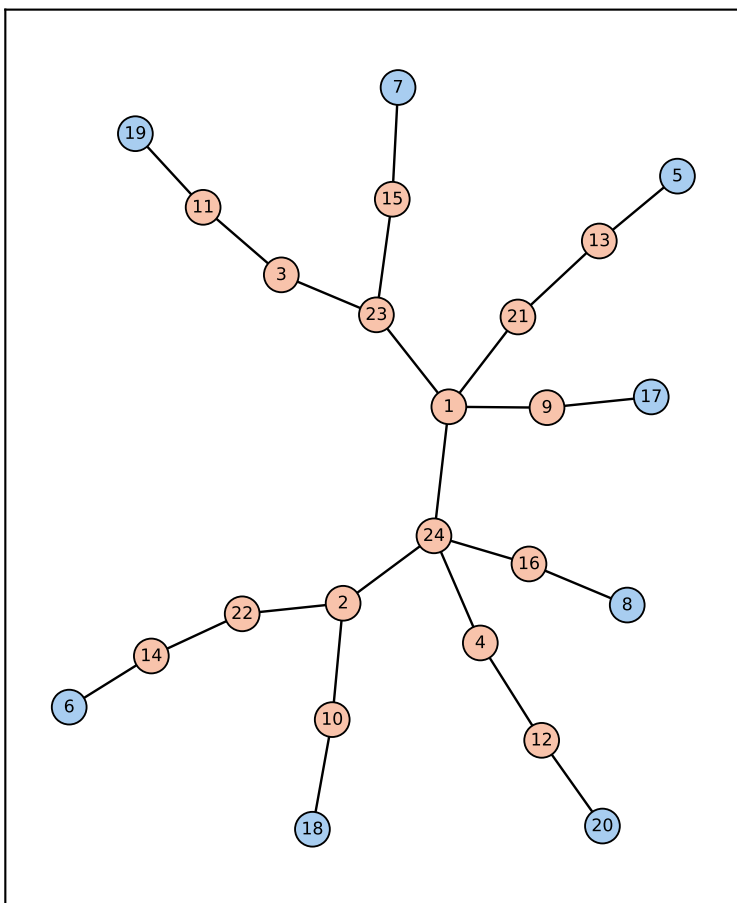
$s = (16, 6, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{24}\langle 8, 20, 21 \rangle$  (E)



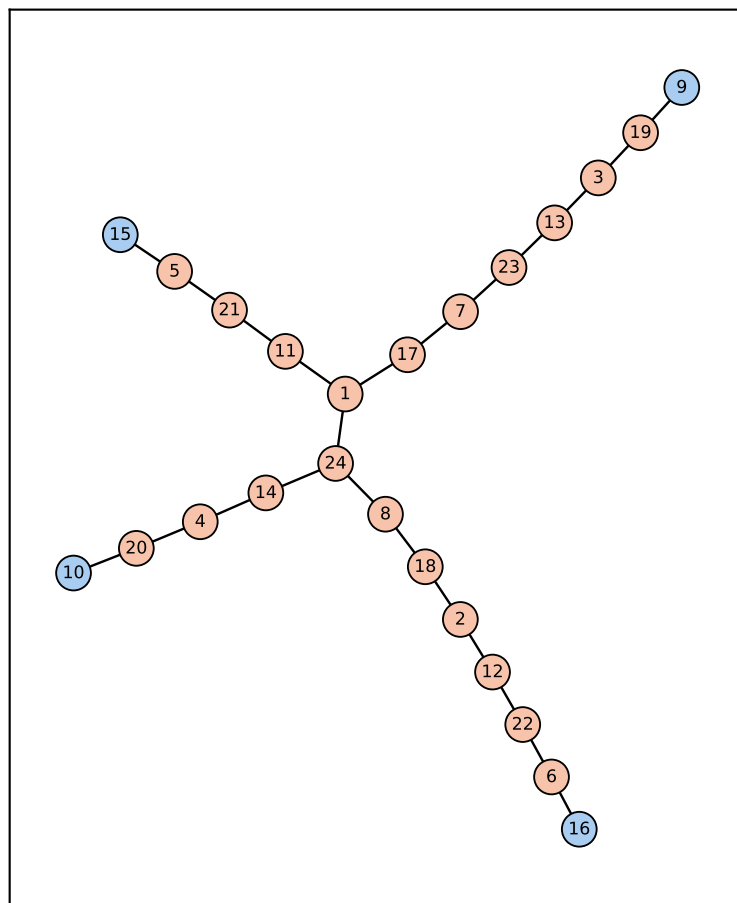
$s = (16, 4, 3)$ ,  $\Delta = 3$ , not caterpillar

$T_{24}\langle 8, 20, 22, 23 \rangle$  (F)

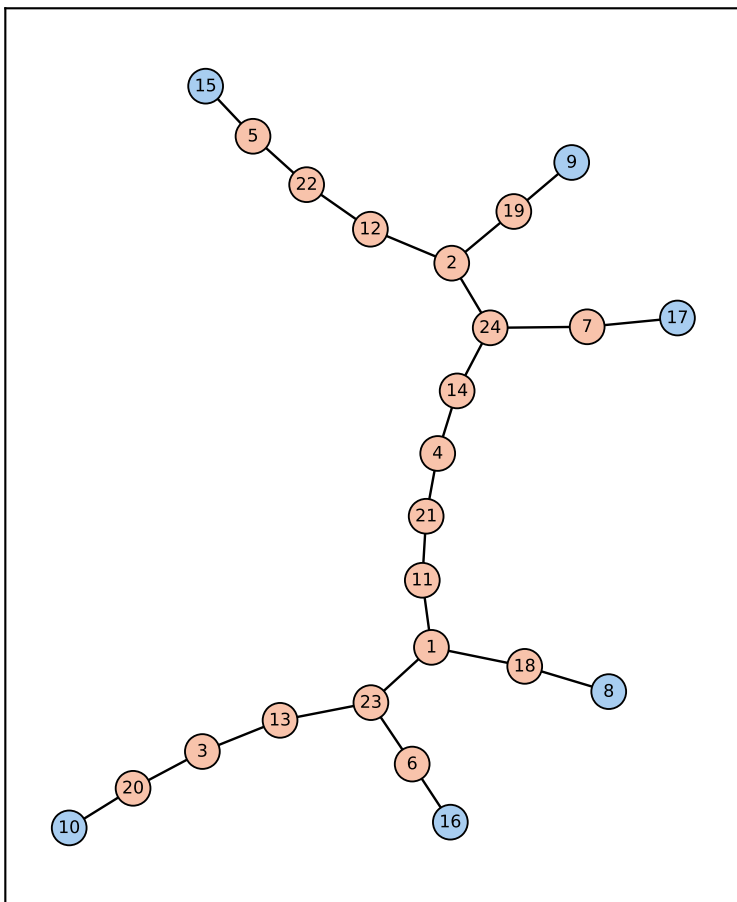


$s = (16, 4, 2, 1)$ ,  $\Delta = 4$ , not caterpillar

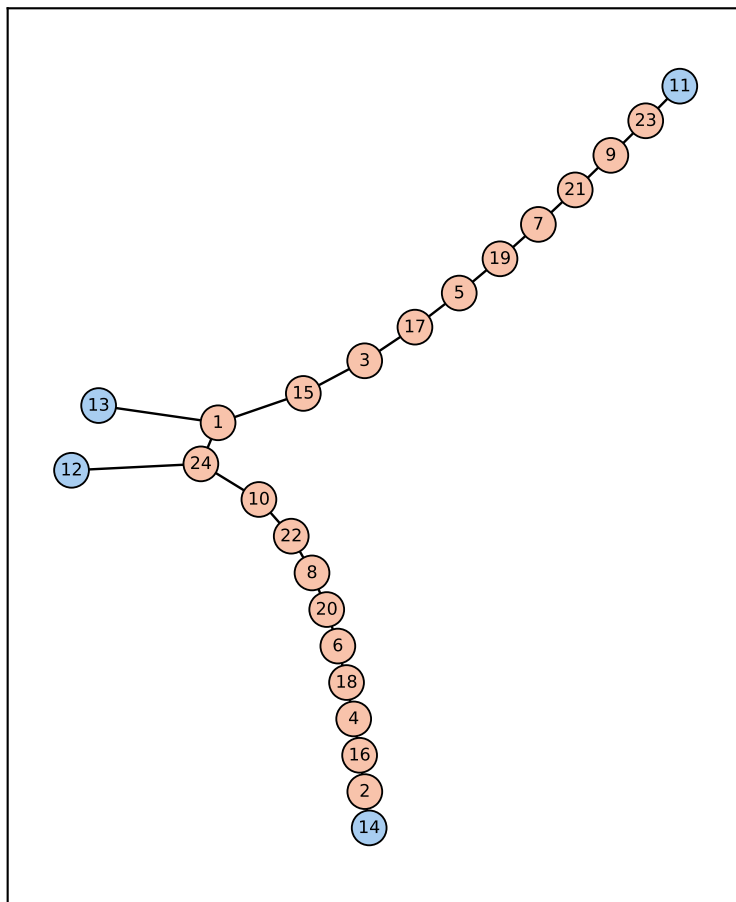
$T_{24}\langle 10, 16, 23 \rangle$  (G)



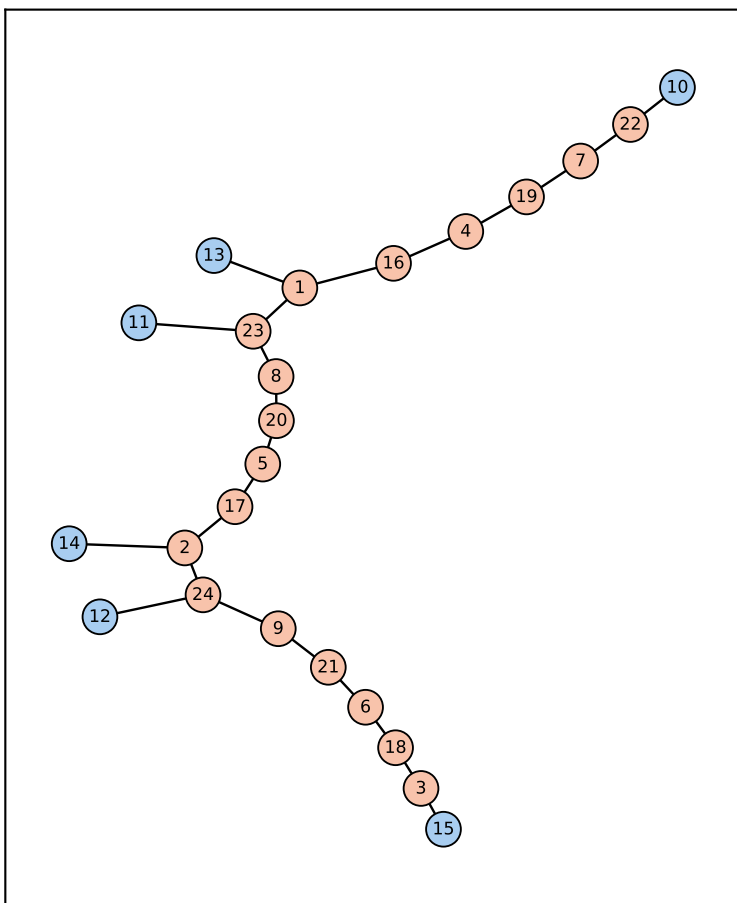
$s = (14, 8, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{24}\langle 10, 17, 22 \rangle \quad (\text{H})$ 


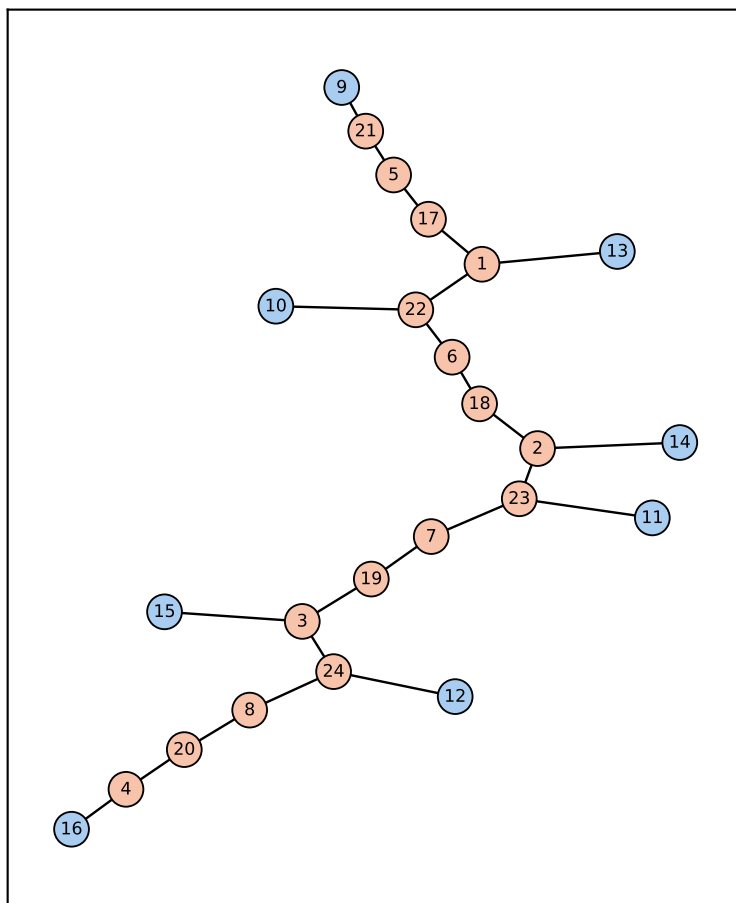
$s = (14, 7, 2)$ ,  $\Delta = 3$ , not caterpillar

 $T_{24}\langle 12, 14, 23 \rangle \quad (\text{I})$ 


$s = (12, 10, 1)$ ,  $\Delta = 3$ , caterpillar

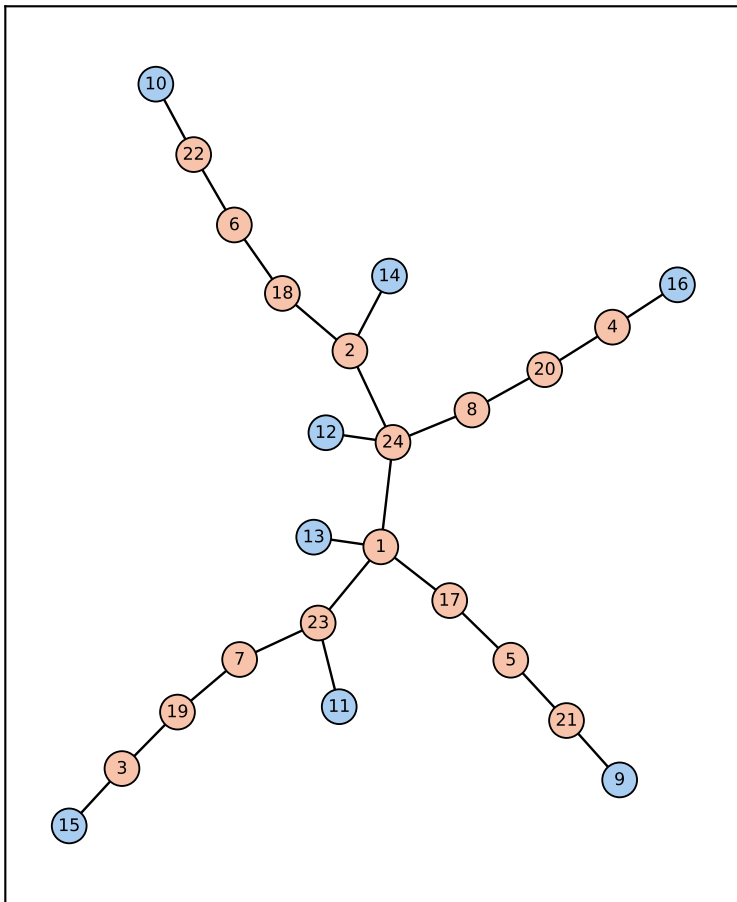
 $T_{24}\langle 12, 15, 22 \rangle \quad (\text{J})$ 


$s = (12, 9, 2)$ ,  $\Delta = 3$ , caterpillar

 $T_{24}\langle 12, 16, 21 \rangle \quad (\text{K})$ 


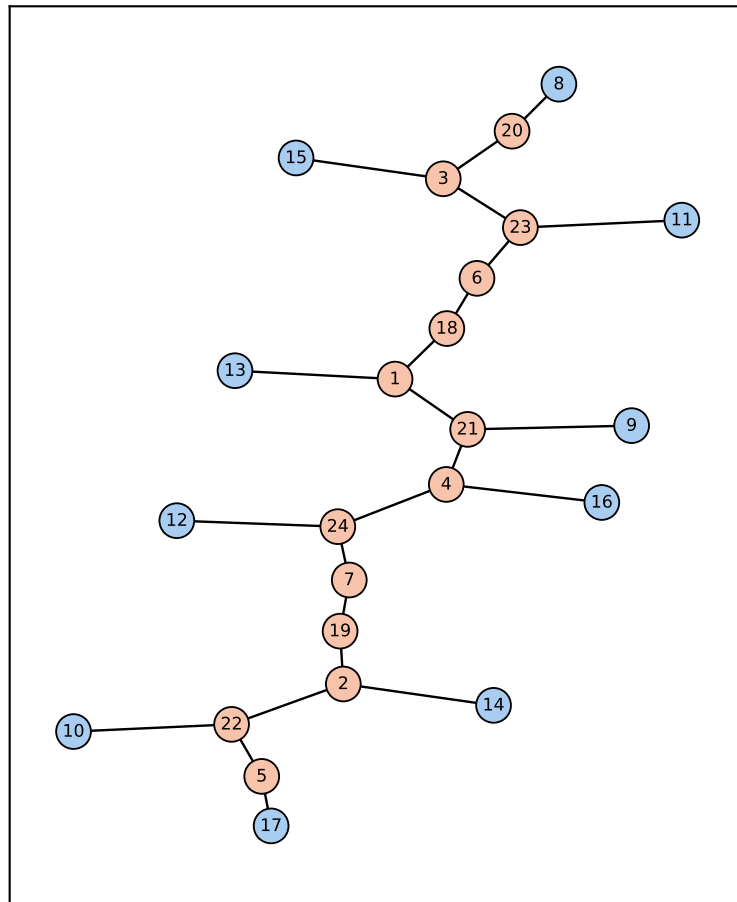
$s = (12, 8, 3)$ ,  $\Delta = 3$ , caterpillar

$T_{24}(12, 16, 22, 23)$  (L)



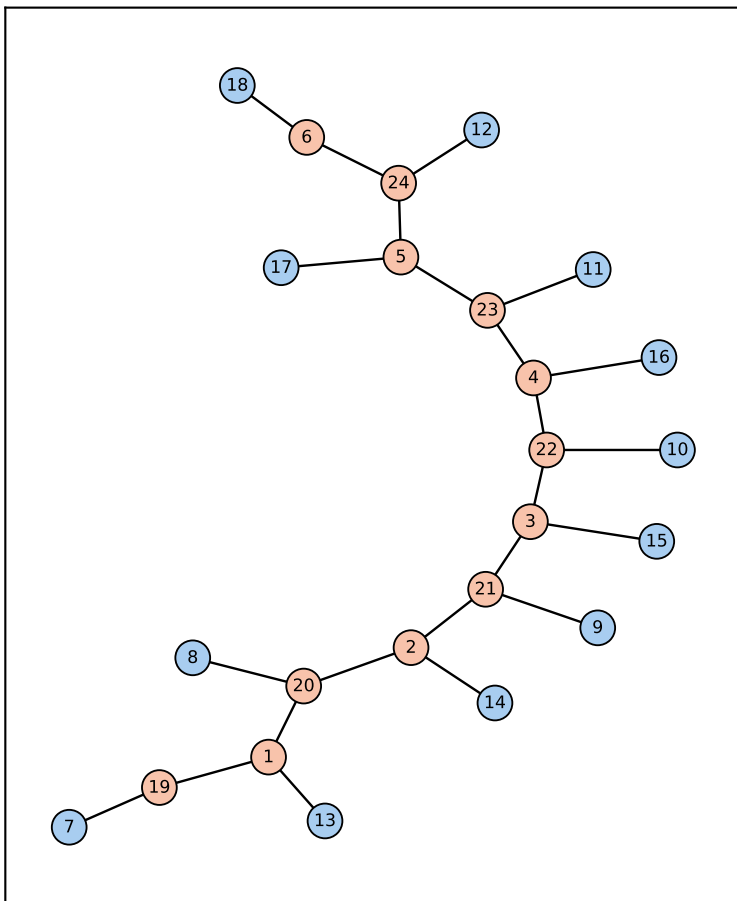
$s = (12, 8, 2, 1)$ ,  $\Delta = 4$ , not caterpillar

$T_{24}(12, 17, 20)$  (M)



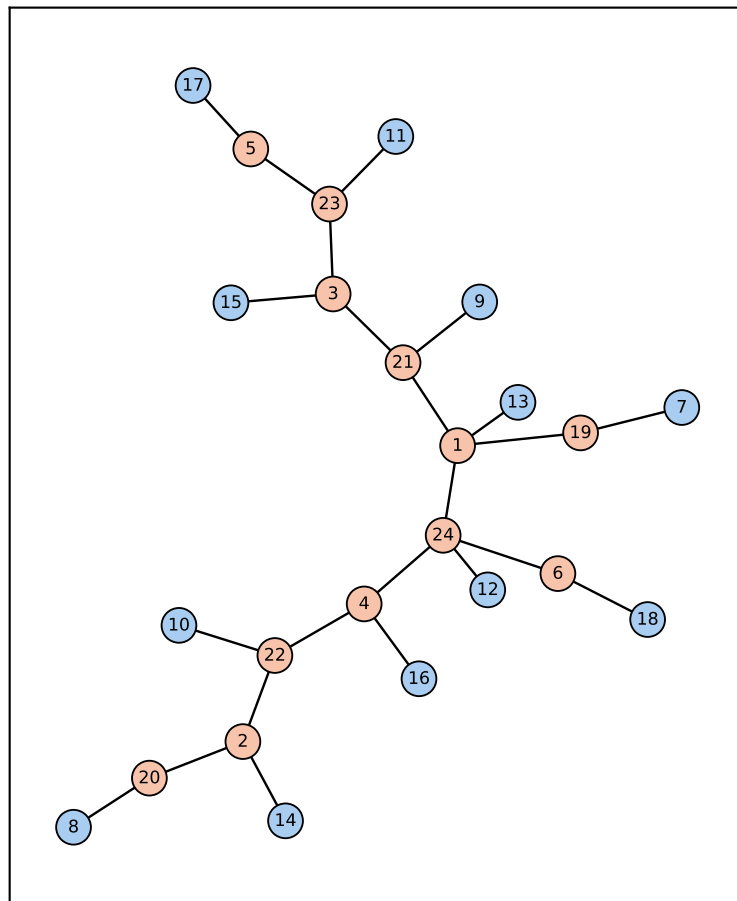
$s = (12, 7, 4)$ ,  $\Delta = 3$ , caterpillar

$T_{24}(12, 18, 19)$  (N)



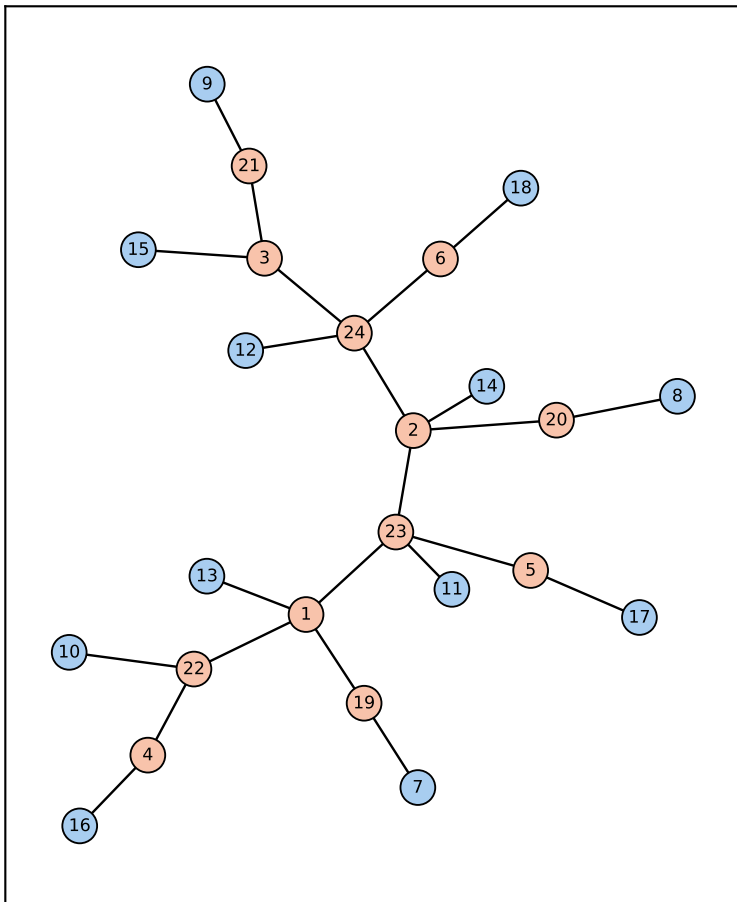
$s = (12, 6, 5)$ ,  $\Delta = 3$ , caterpillar

$T_{24}(12, 18, 20, 23)$  (O)



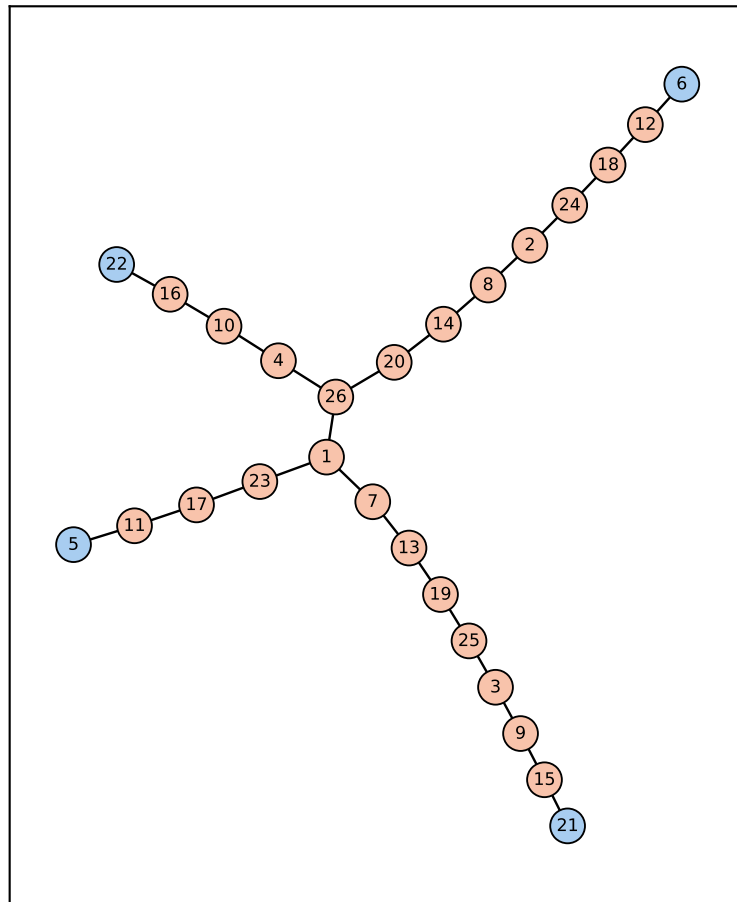
$s = (12, 6, 4, 1)$ ,  $\Delta = 4$ , not caterpillar

$T_{24}(12, 18, 21, 22)$  (P)



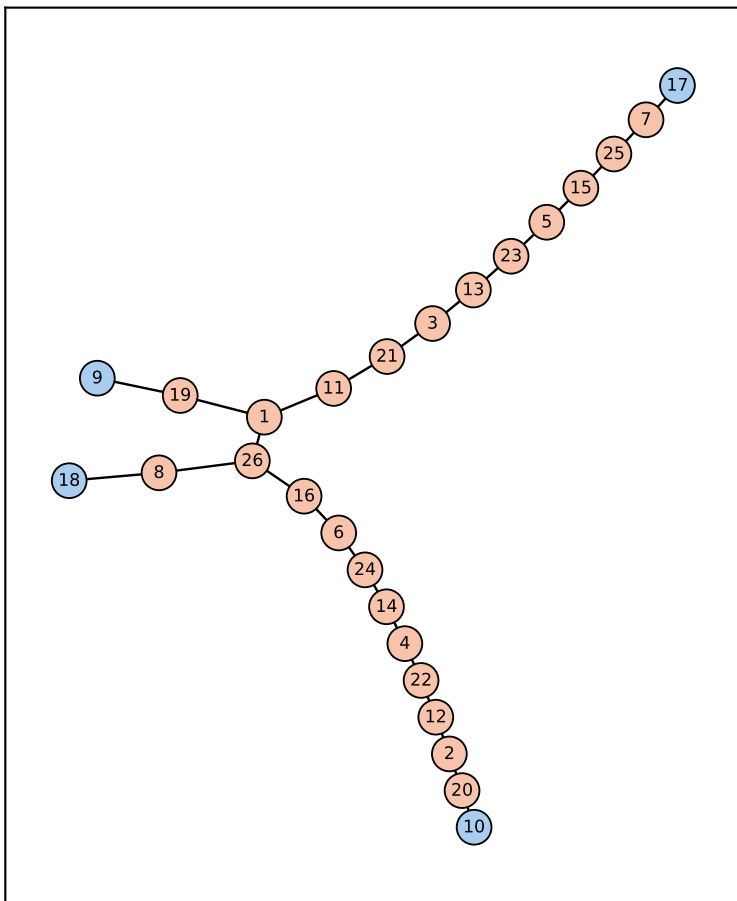
$s = (12, 6, 3, 2)$ ,  $\Delta = 4$ , not caterpillar

$T_{26}(6, 22, 25)$  (A)



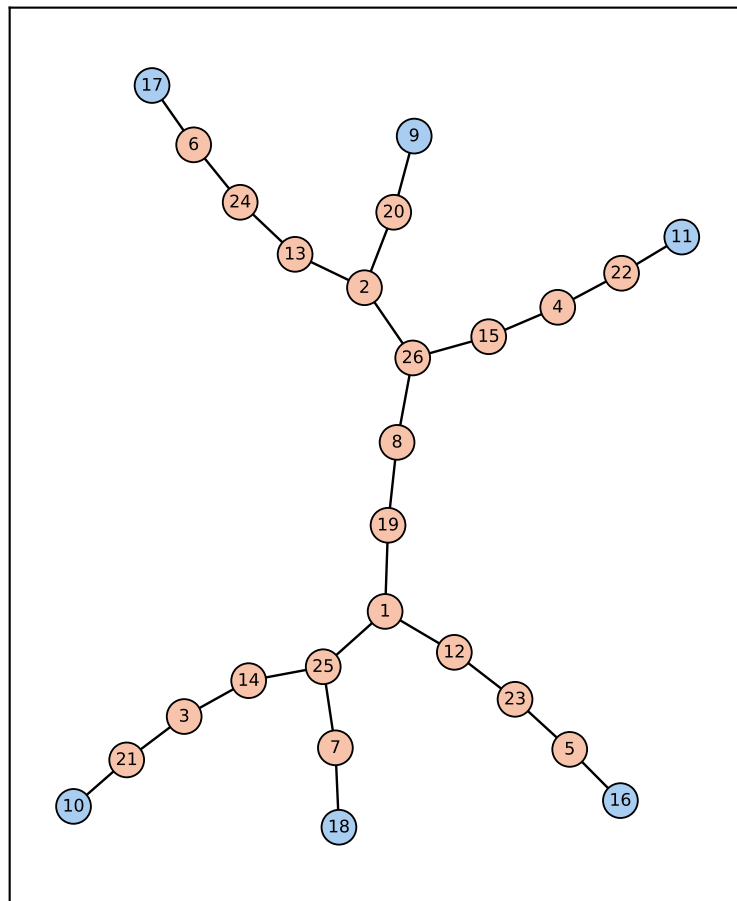
$s = (20, 4, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{26}(10, 18, 25)$  (B)



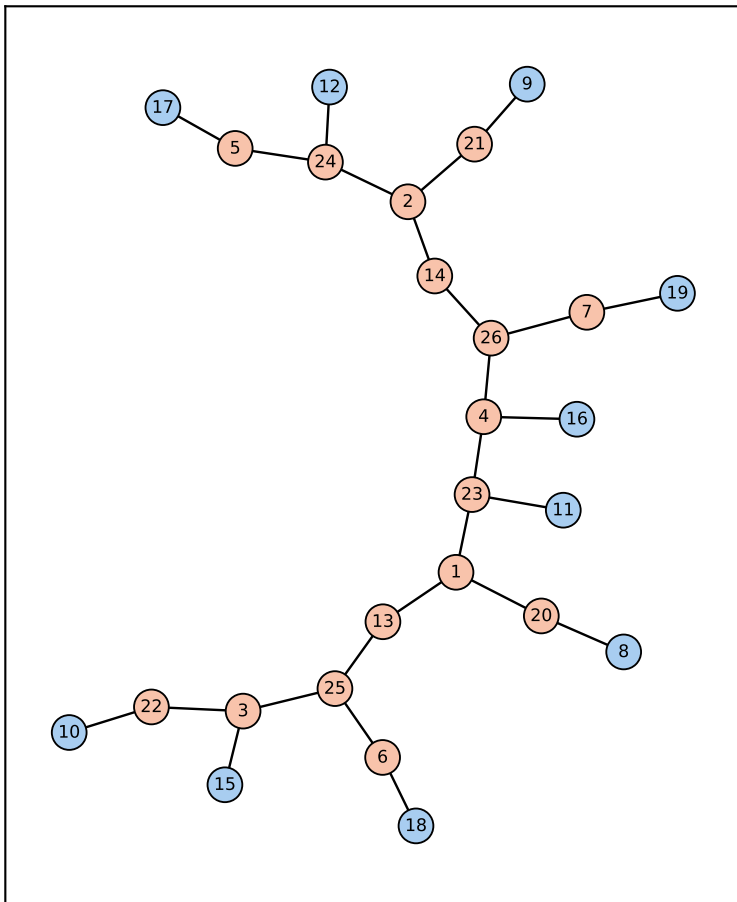
$s = (16, 8, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{26}(11, 18, 24)$  (C)



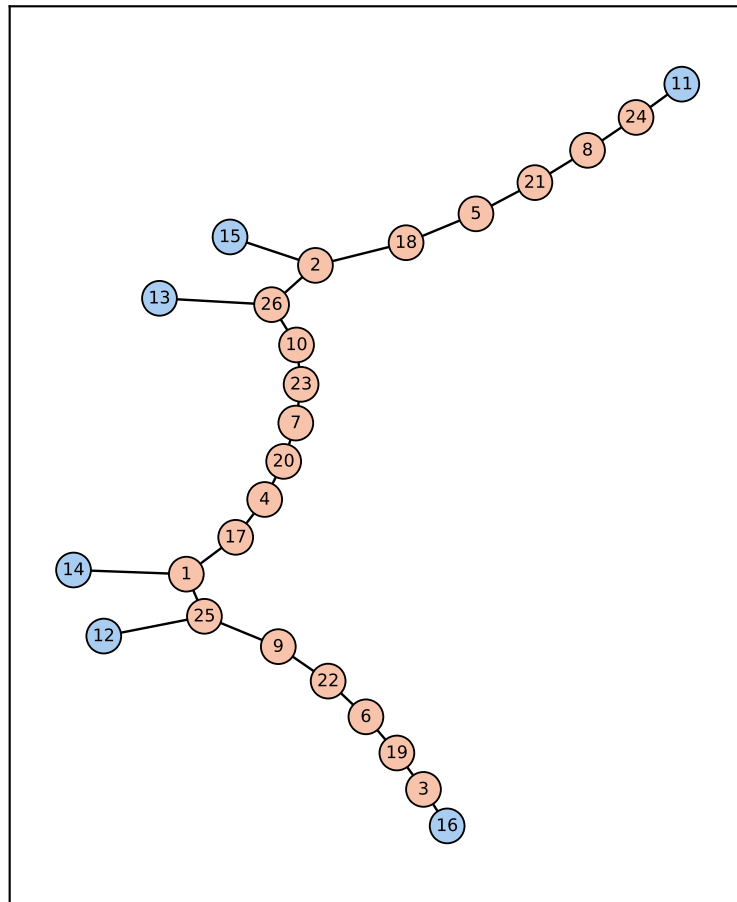
$s = (15, 8, 2)$ ,  $\Delta = 3$ , not caterpillar

$T_{26}(12, 19, 22)$  (D)



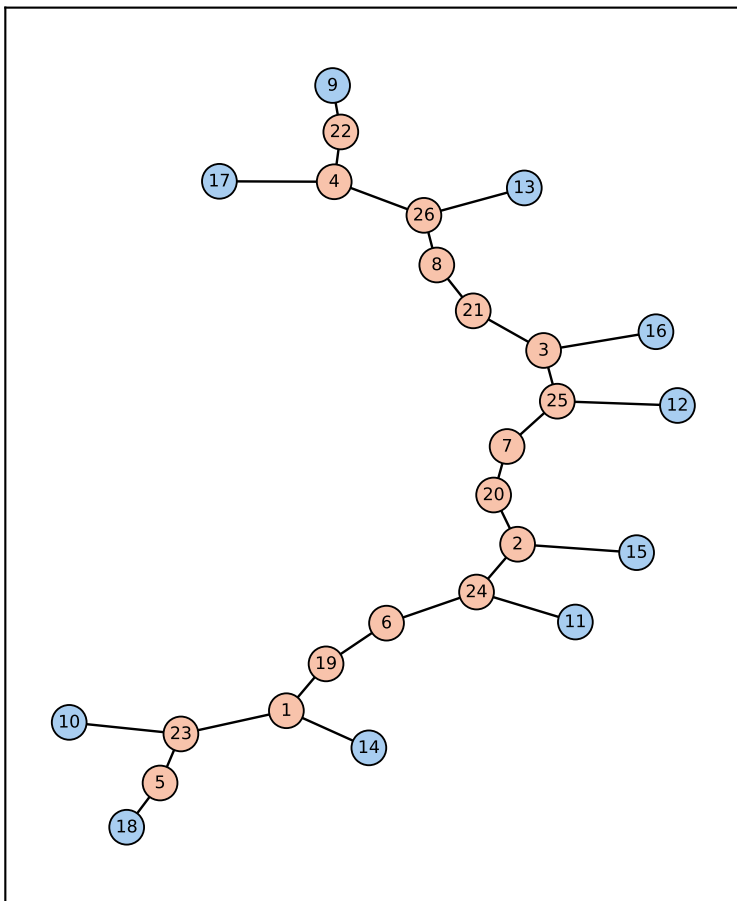
$s = (14, 7, 4)$ ,  $\Delta = 3$ , not caterpillar

$T_{26}(13, 16, 24)$  (E)



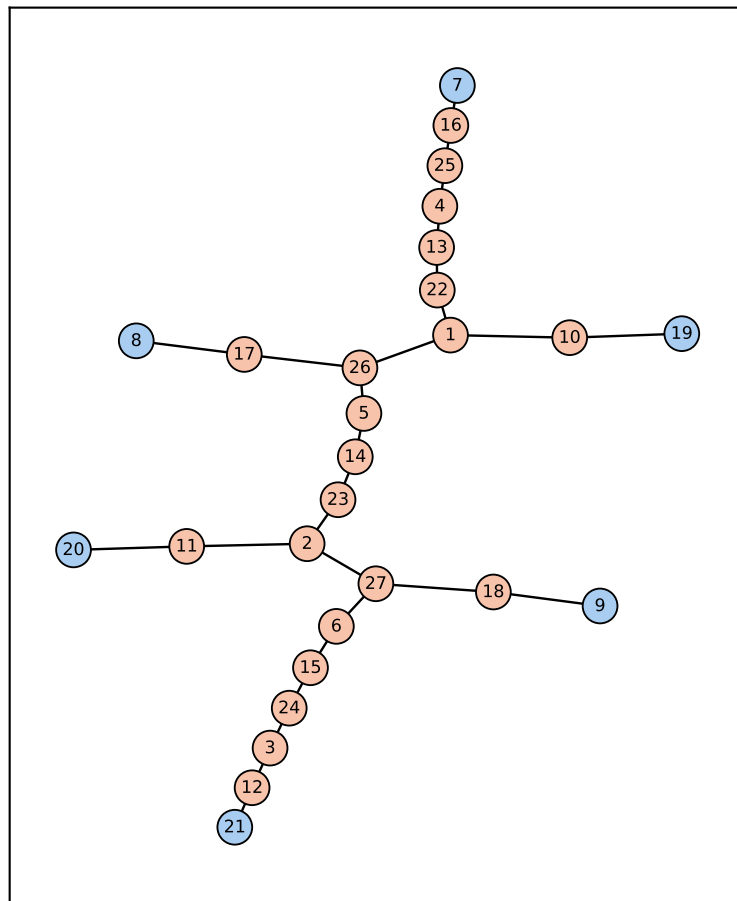
$s = (13, 10, 2)$ ,  $\Delta = 3$ , caterpillar

$T_{26}(13, 18, 22)$  (F)

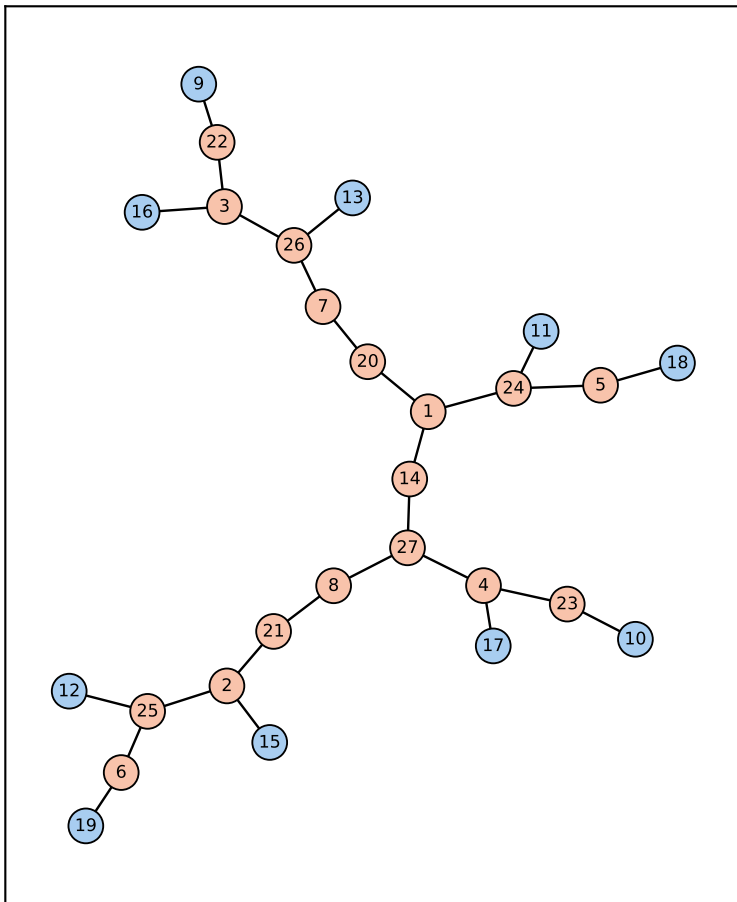


$s = (13, 8, 4)$ ,  $\Delta = 3$ , caterpillar

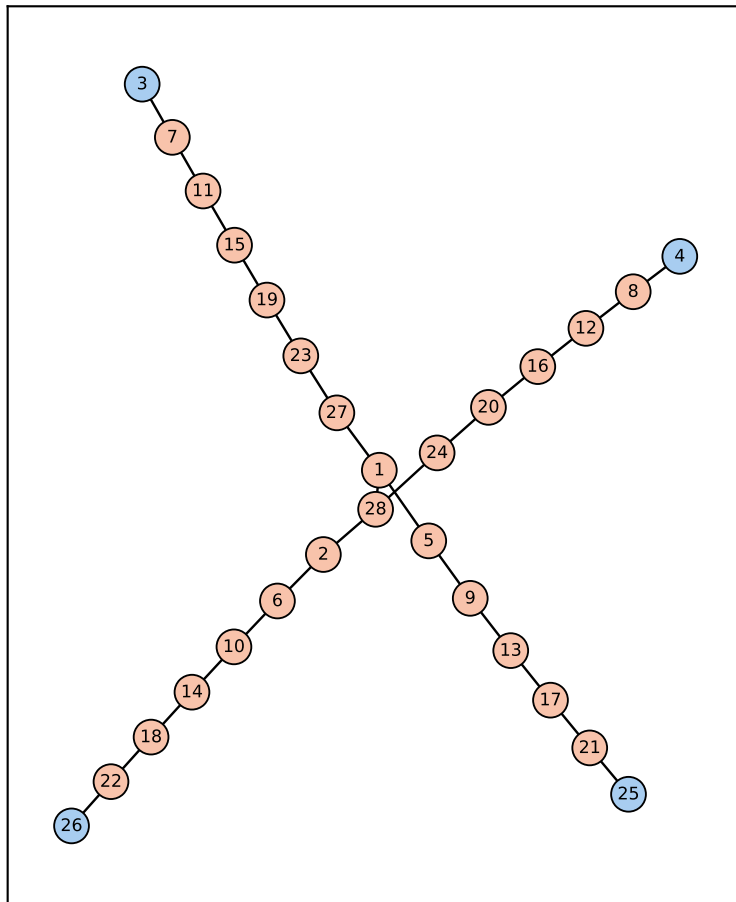
$T_{27}(9, 21, 25)$  (A)



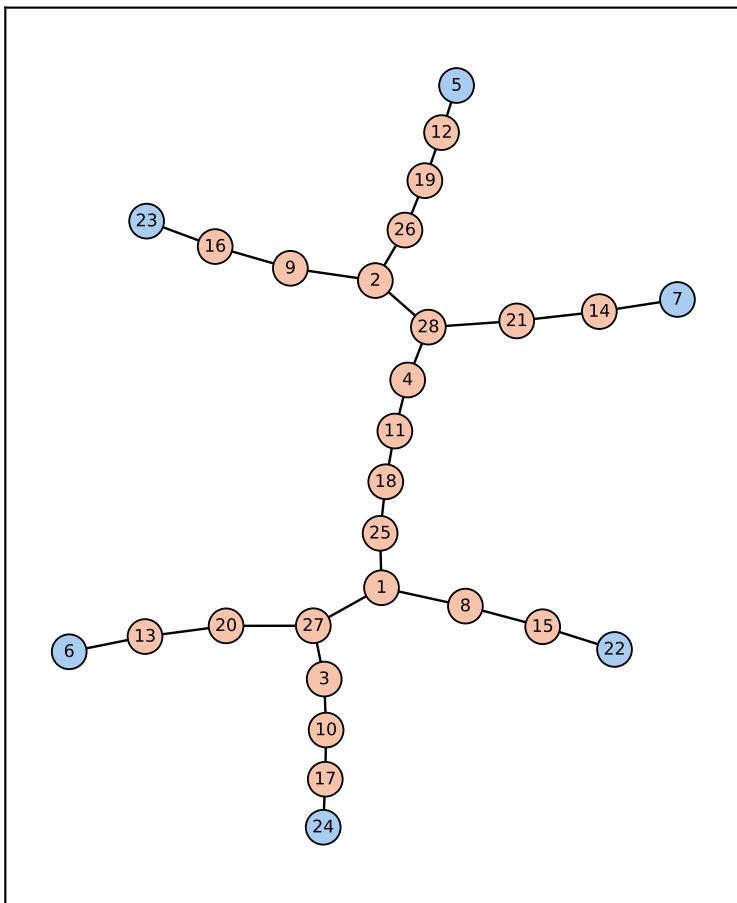
$s = (18, 6, 2)$ ,  $\Delta = 3$ , not caterpillar

$$T_{27}\langle 13, 19, 23 \rangle \quad (\text{B})$$


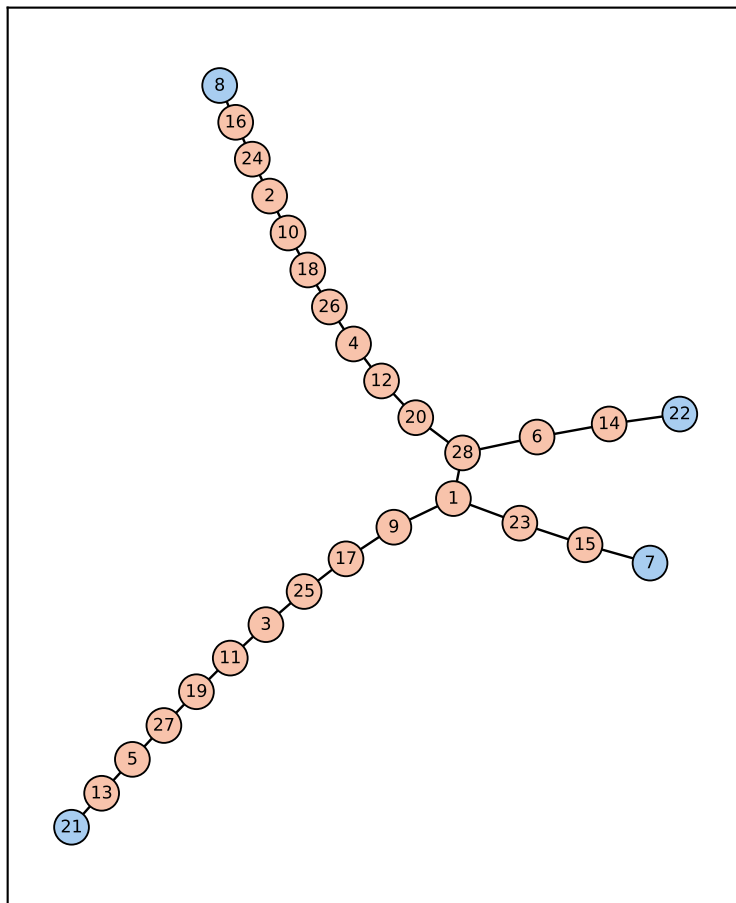
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 $s = (14, 8, 4), \Delta = 3$ , not caterpillar
$$T_{28}\langle 4, 26, 27 \rangle \quad (\text{A})$$


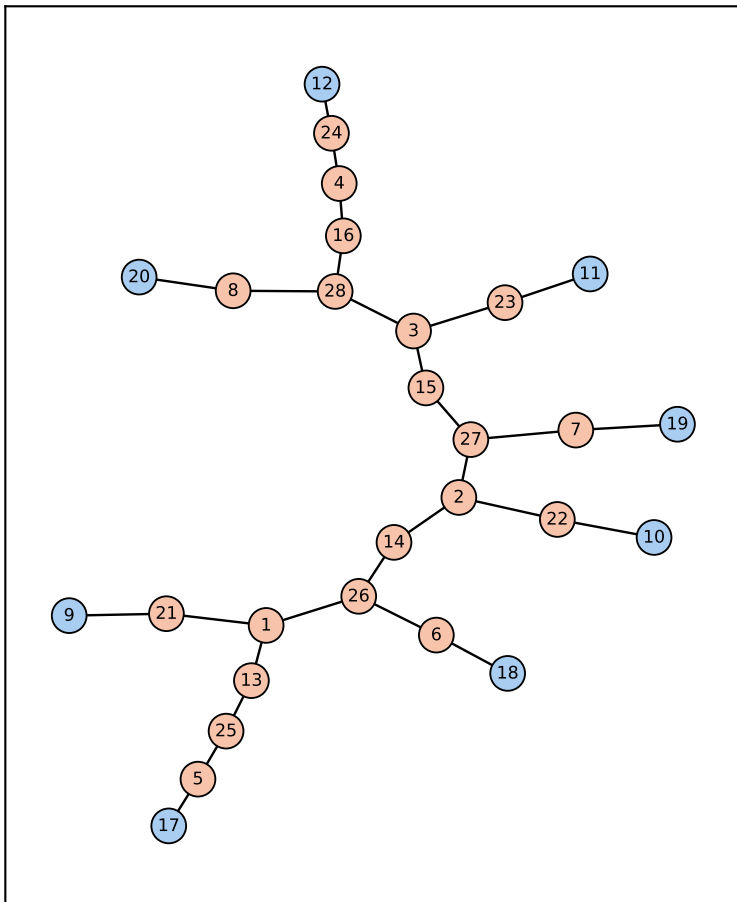
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 $s = (24, 2, 1), \Delta = 3$ , not caterpillar
$$T_{28}\langle 7, 24, 26 \rangle \quad (\text{B})$$


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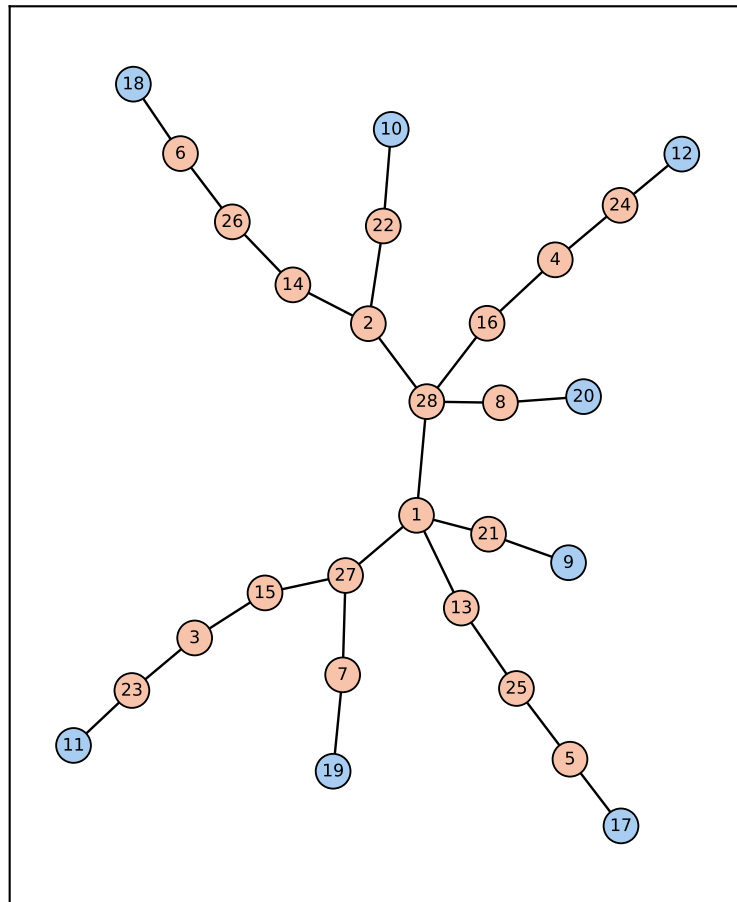
 $s = (21, 4, 2), \Delta = 3$ , not caterpillar
$$T_{28}\langle 8, 22, 27 \rangle \quad (C)$$
 $s = (20, 6, 1), \Delta = 3$ , not caterpillar

$T_{28}(12, 20, 25)$  (D)



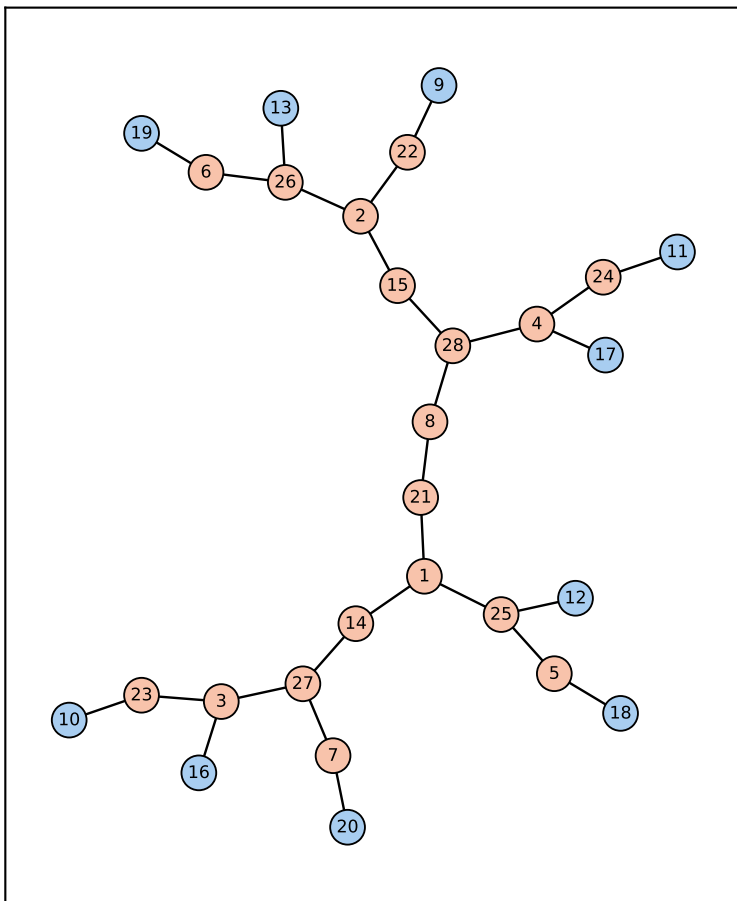
$s = (16, 8, 3)$ ,  $\Delta = 3$ , not caterpillar

$T_{28}(12, 20, 26, 27)$  (E)



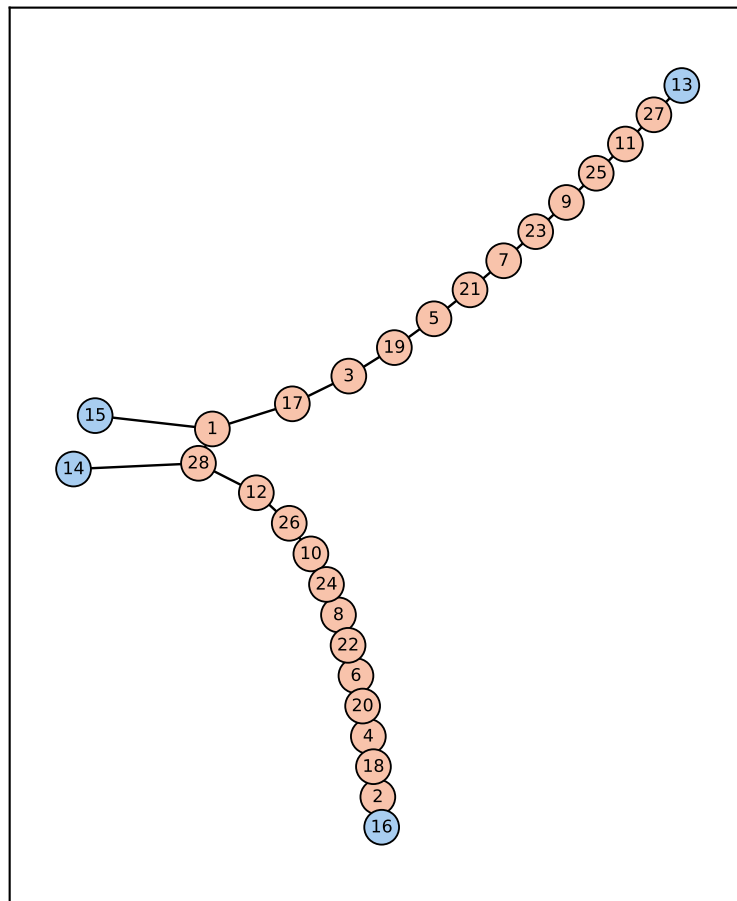
$s = (16, 8, 2, 1)$ ,  $\Delta = 4$ , not caterpillar

$T_{28}(13, 20, 24)$  (F)



$s = (15, 8, 4)$ ,  $\Delta = 3$ , not caterpillar

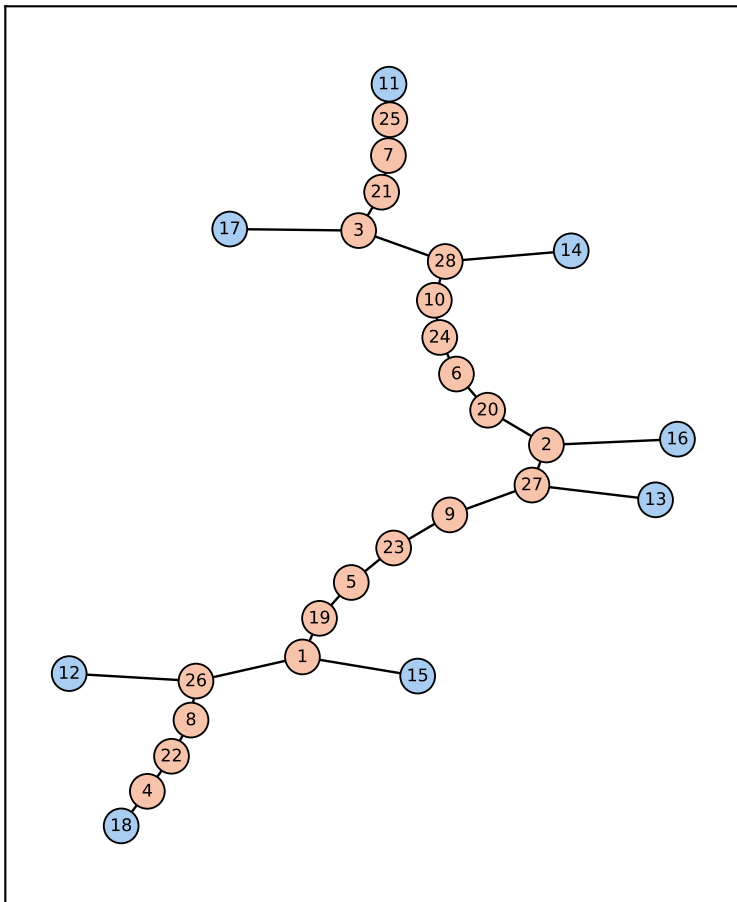
$T_{28}(14, 16, 27)$  (G)



$s = (14, 12, 1)$ ,  $\Delta = 3$ , caterpillar

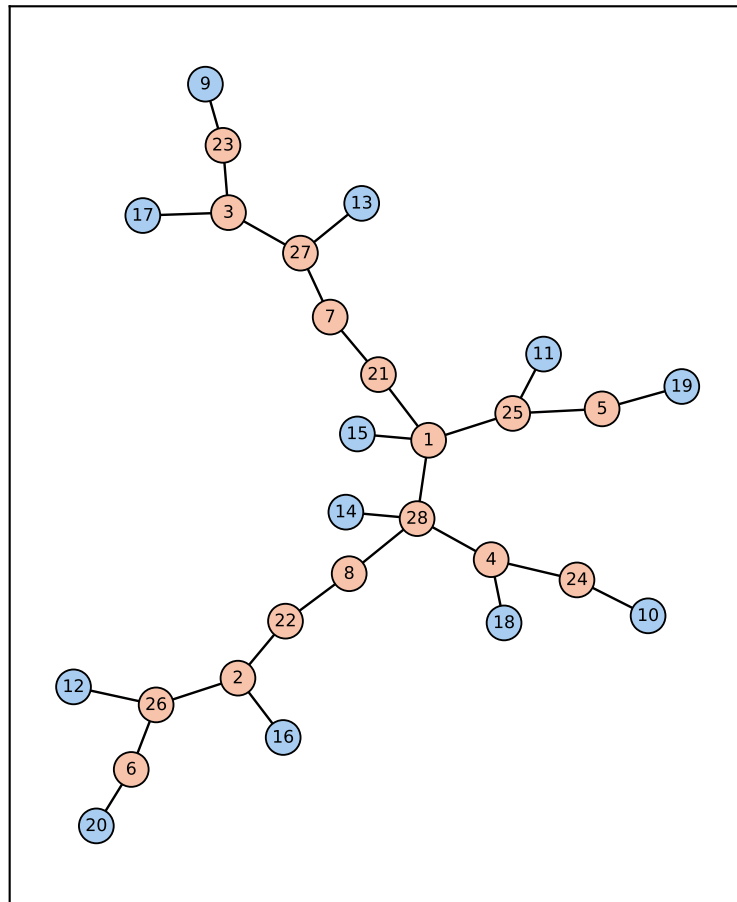


$T_{28}\langle 14, 18, 25 \rangle$  (H)



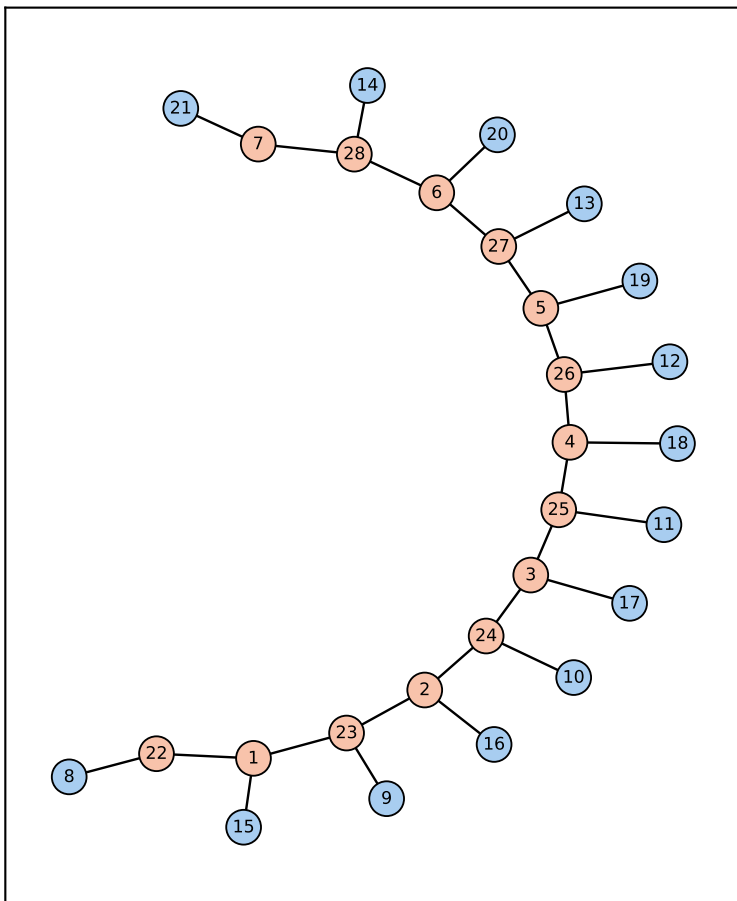
$s = (14, 10, 3)$ ,  $\Delta = 3$ , caterpillar

$T_{28}\langle 14, 20, 24, 27 \rangle$  (I)



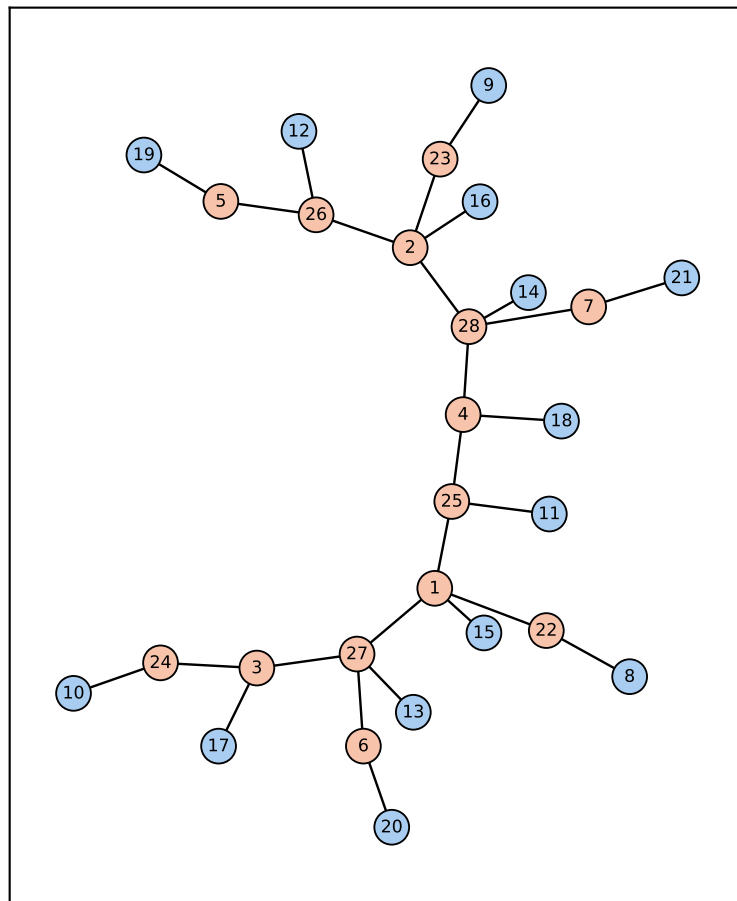
$s = (14, 8, 4, 1)$ ,  $\Delta = 4$ , not caterpillar

$T_{28}\langle 14, 21, 22 \rangle$  (J)



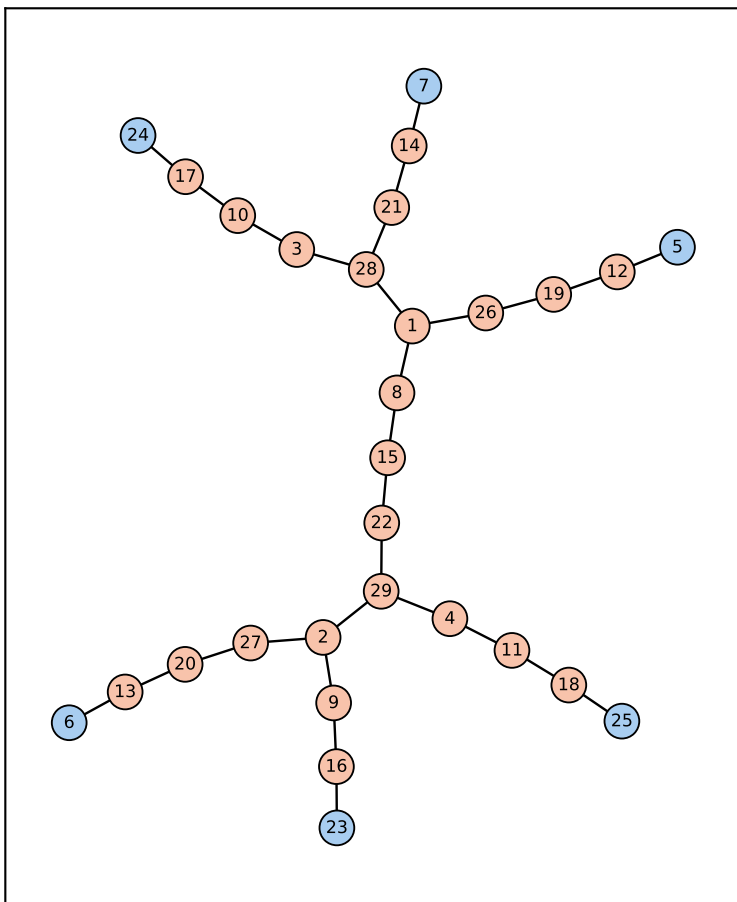
$s = (14, 7, 6)$ ,  $\Delta = 3$ , caterpillar

$T_{28}\langle 14, 21, 24, 26 \rangle$  (K)



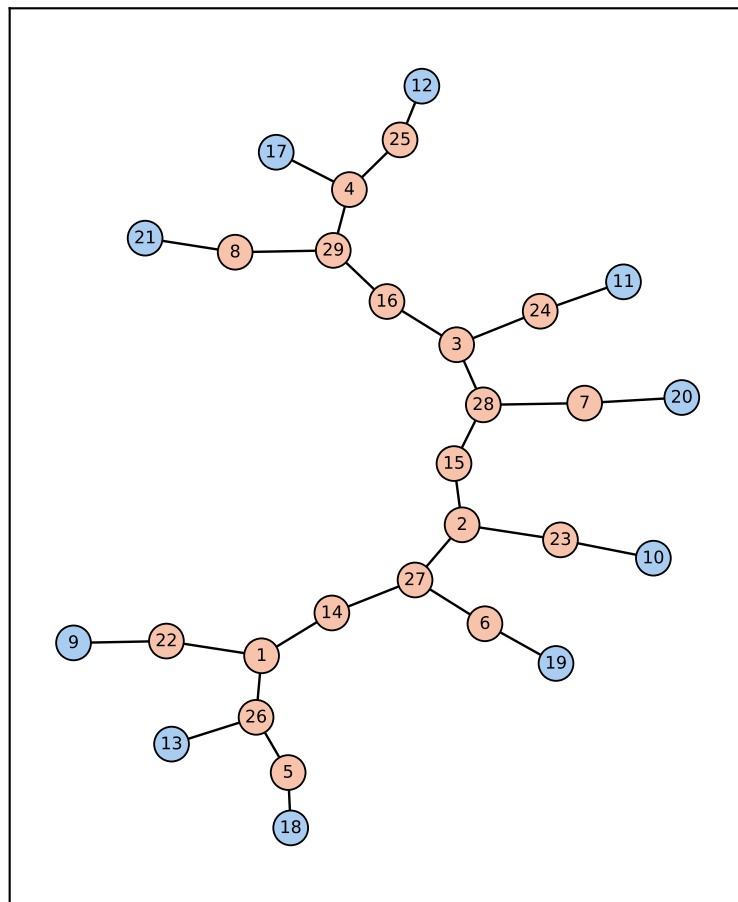
$s = (14, 7, 4, 2)$ ,  $\Delta = 4$ , not caterpillar

$T_{29}\langle 7, 25, 27 \rangle$  (A)



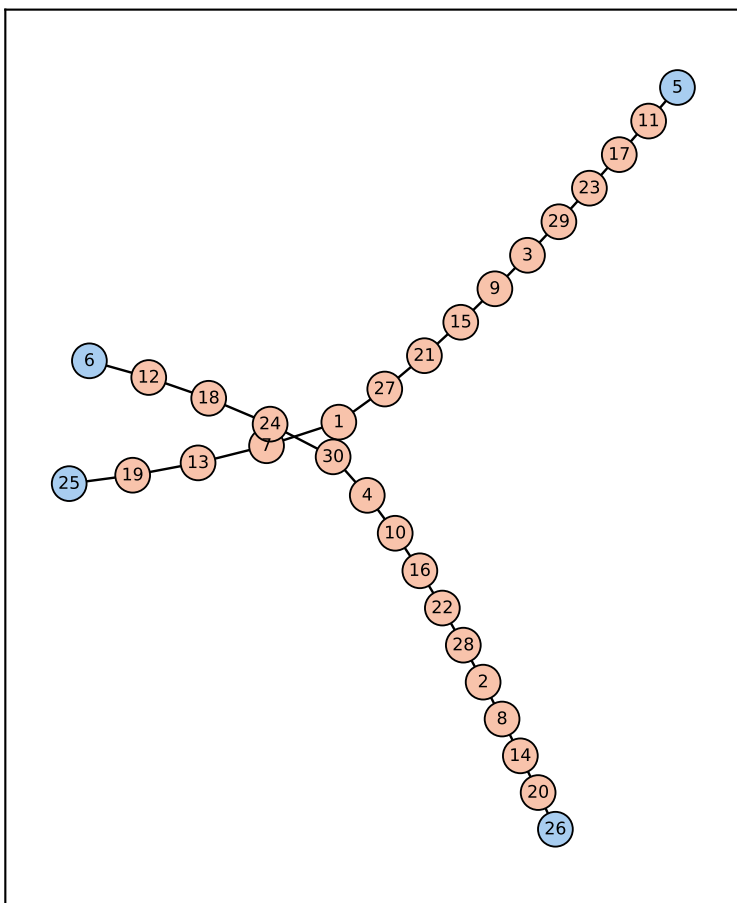
$s = (22, 4, 2)$ ,  $\Delta = 3$ , not caterpillar

$T_{29}\langle 13, 21, 25 \rangle$  (B)



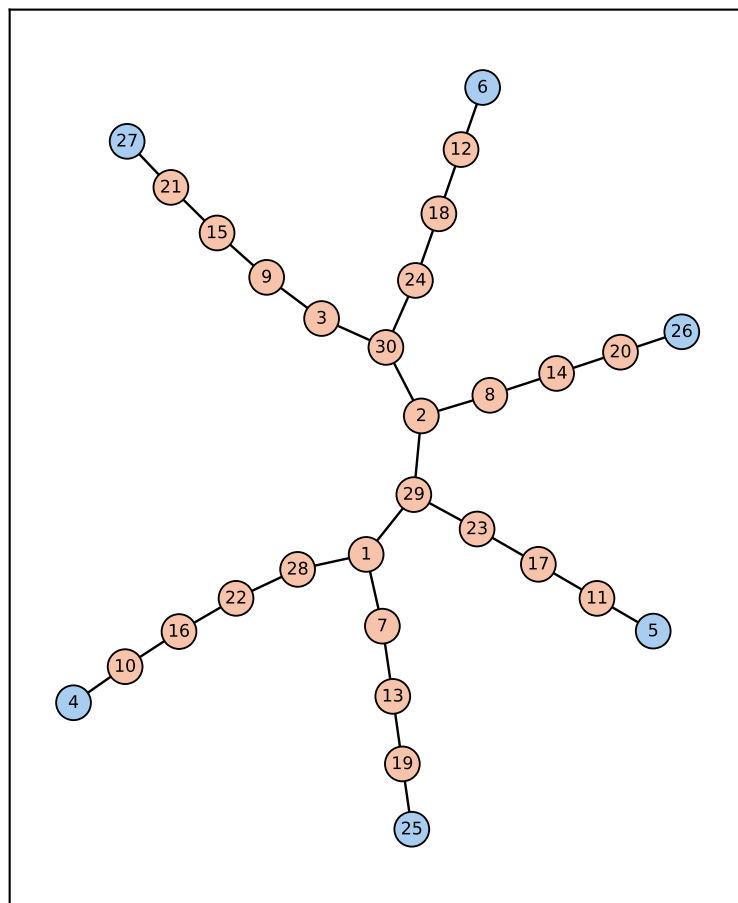
$s = (16, 8, 4)$ ,  $\Delta = 3$ , not caterpillar

$T_{30}\langle 6, 26, 29 \rangle$  (A)



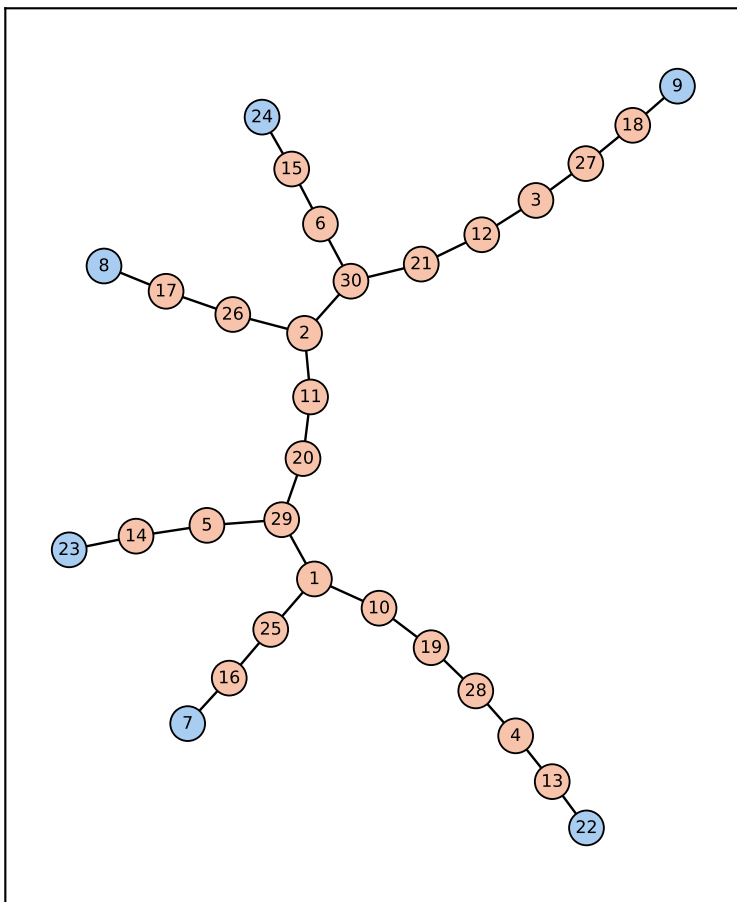
$s = (24, 4, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{30}\langle 6, 27, 28 \rangle$  (B)



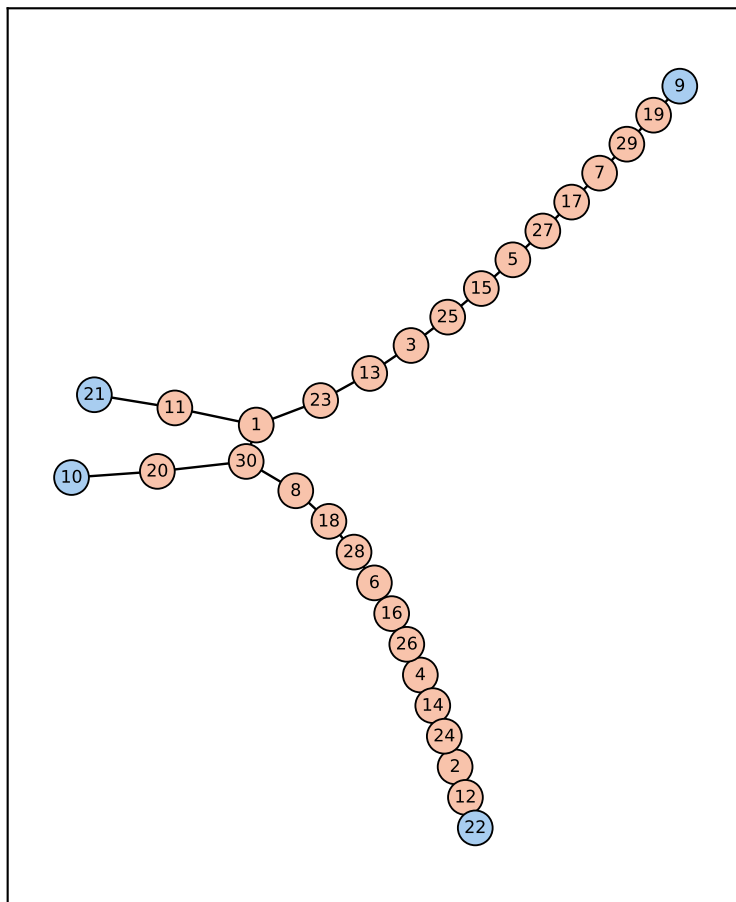
$s = (24, 3, 2)$ ,  $\Delta = 3$ , not caterpillar

$T_{30}\langle 9, 24, 28 \rangle$  (C)



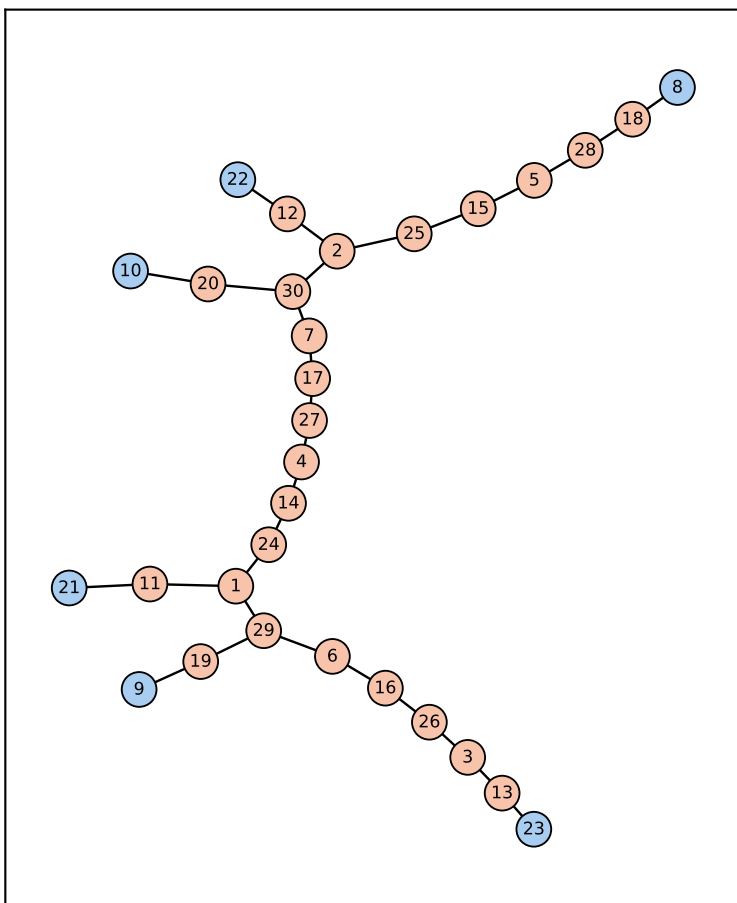
$s = (21, 6, 2)$ ,  $\Delta = 3$ , not caterpillar

$T_{30}\langle 10, 22, 29 \rangle$  (D)



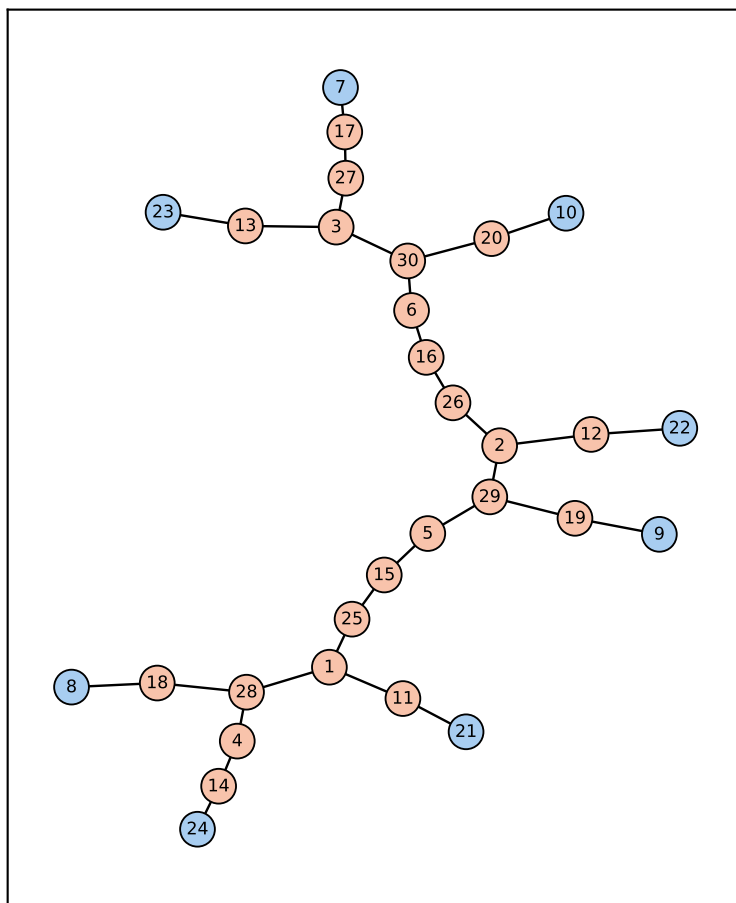
$s = (20, 8, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{30}\langle 10, 23, 28 \rangle$  (E)



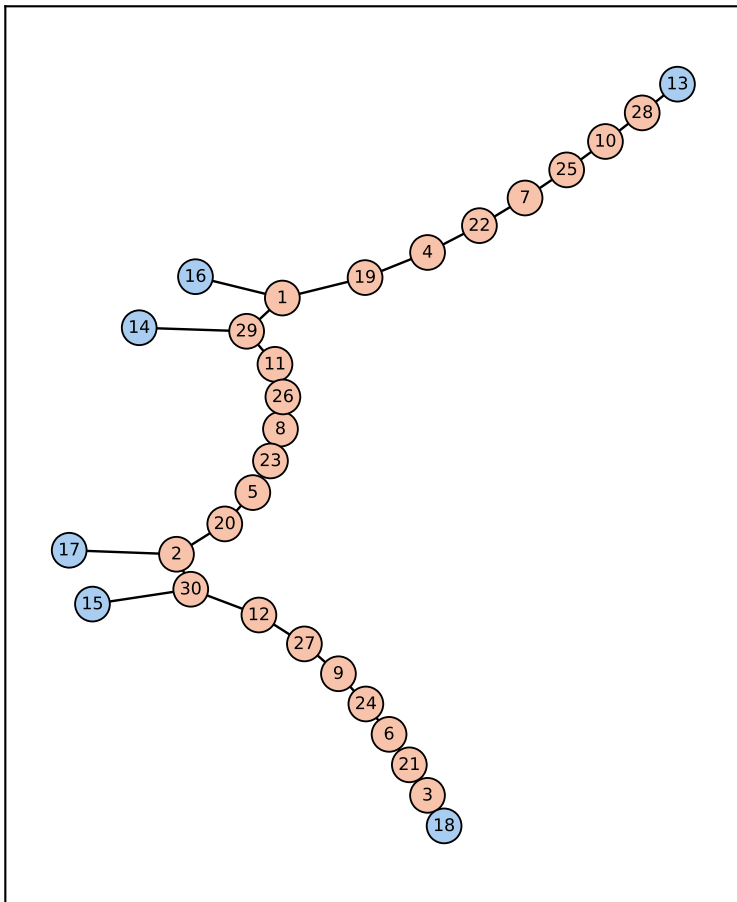
$s = (20, 7, 2)$ ,  $\Delta = 3$ , not caterpillar

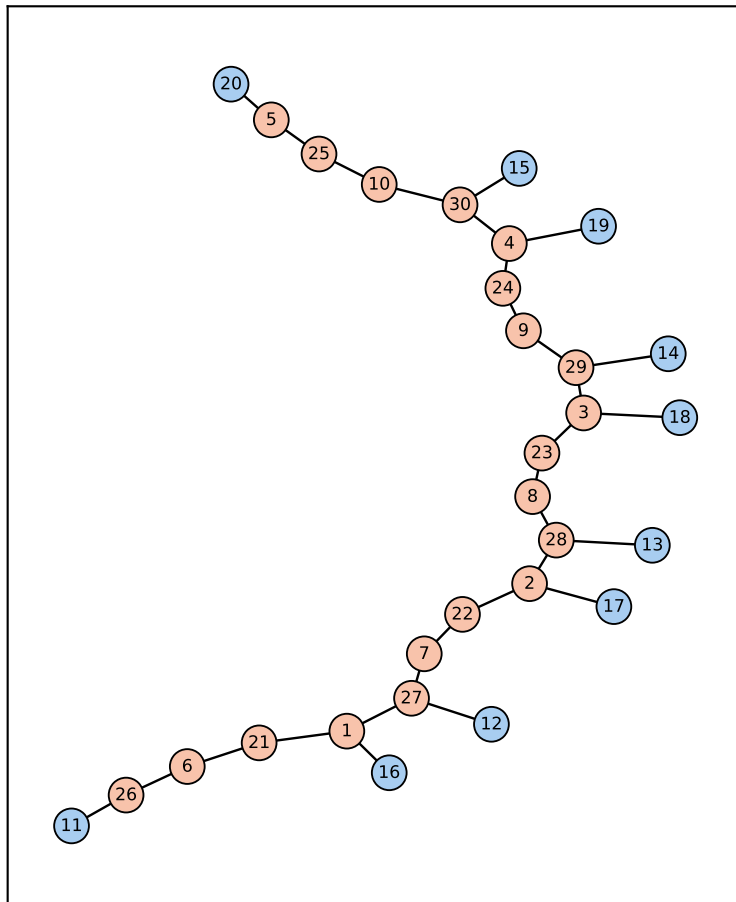
$T_{30}\langle 10, 24, 27 \rangle$  (F)

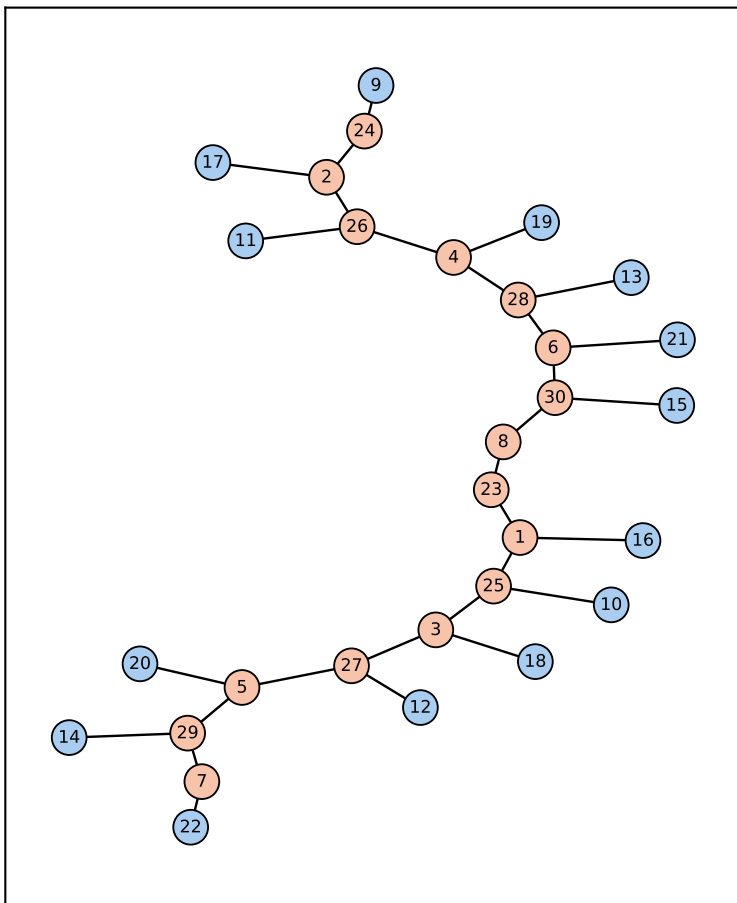


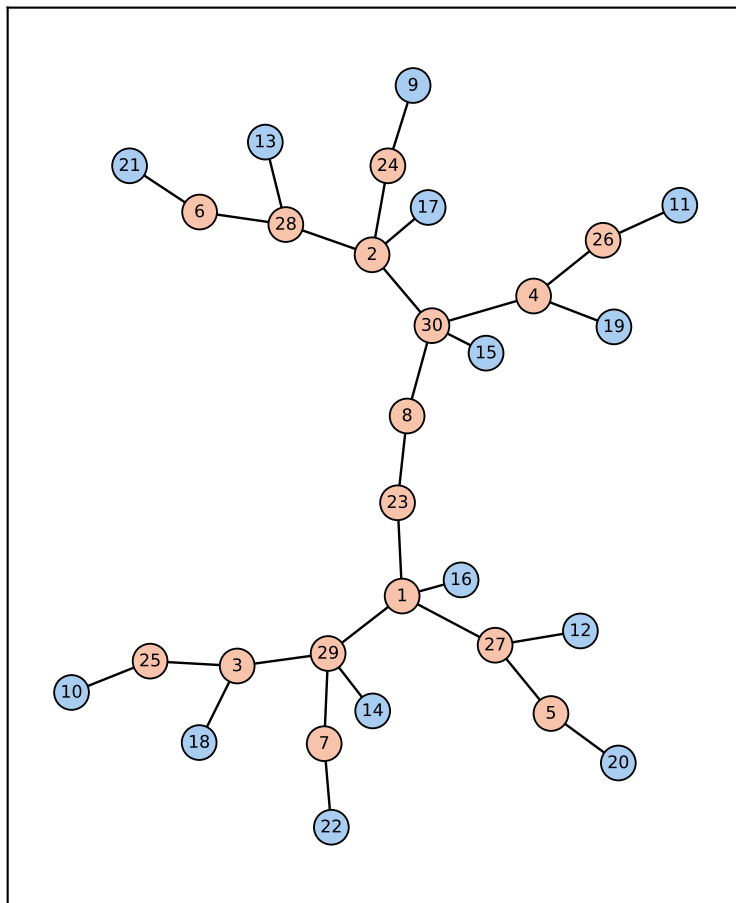
$s = (20, 6, 3)$ ,  $\Delta = 3$ , not caterpillar

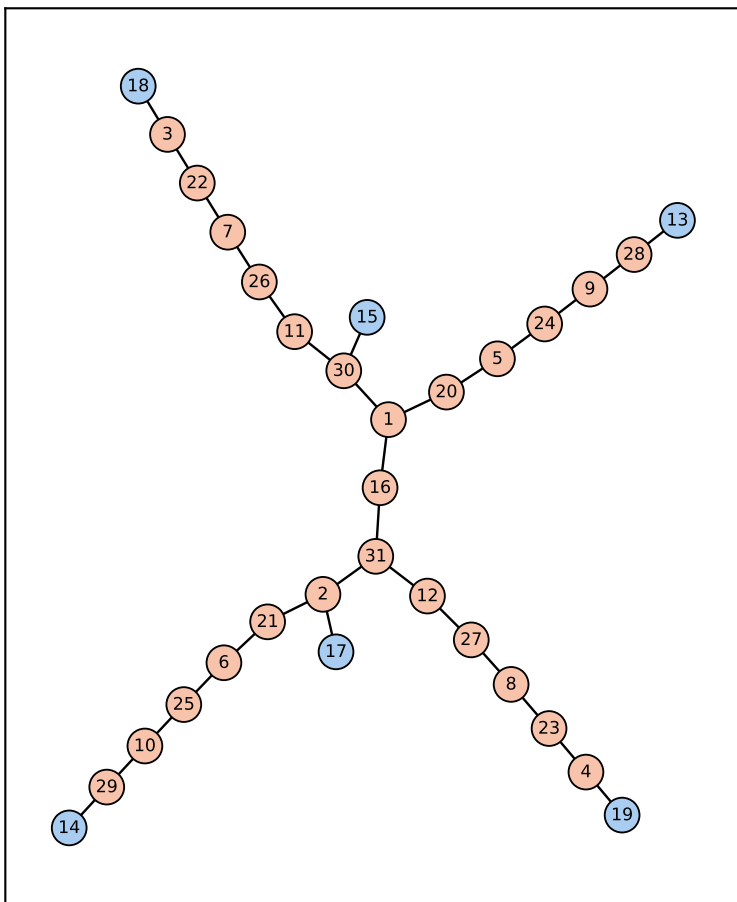


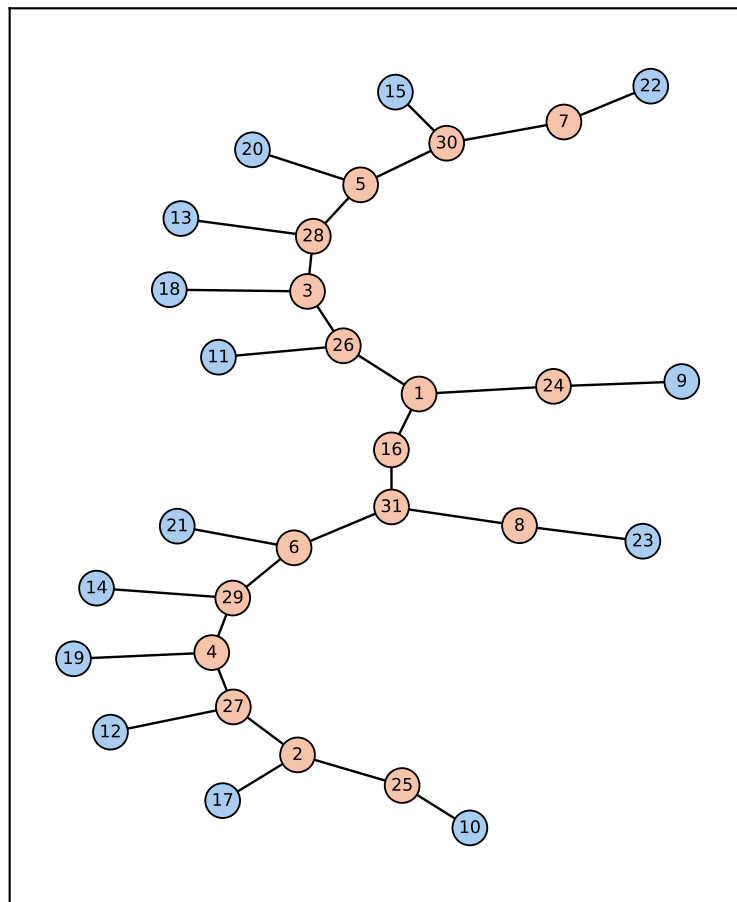
$T_{30}(15, 18, 28)$  (K)

 $s = (15, 12, 2)$ ,  $\Delta = 3$ , caterpillar

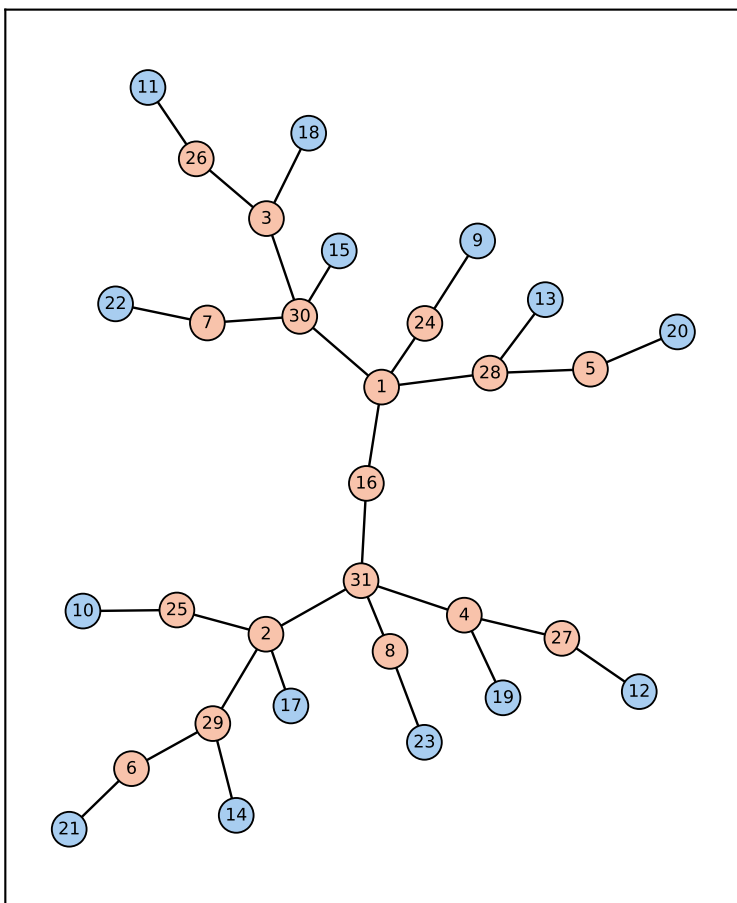
 $T_{30}(15, 20, 26)$  (L)

 $s = (15, 10, 4)$ ,  $\Delta = 3$ , caterpillar

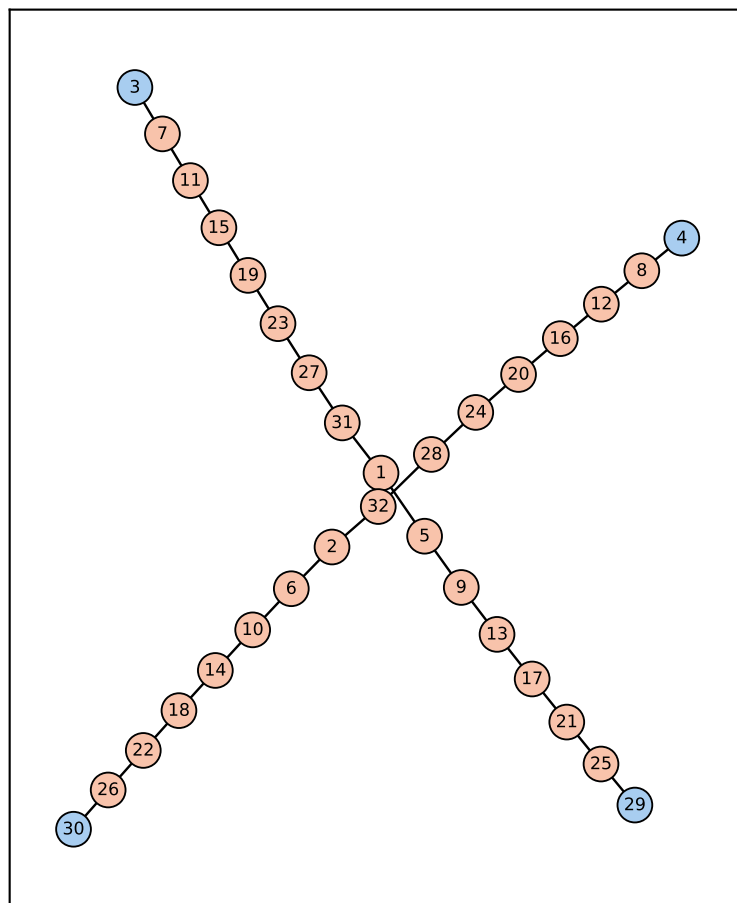
 $T_{30}(15, 22, 24)$  (M)

 $s = (15, 8, 6)$ ,  $\Delta = 3$ , caterpillar

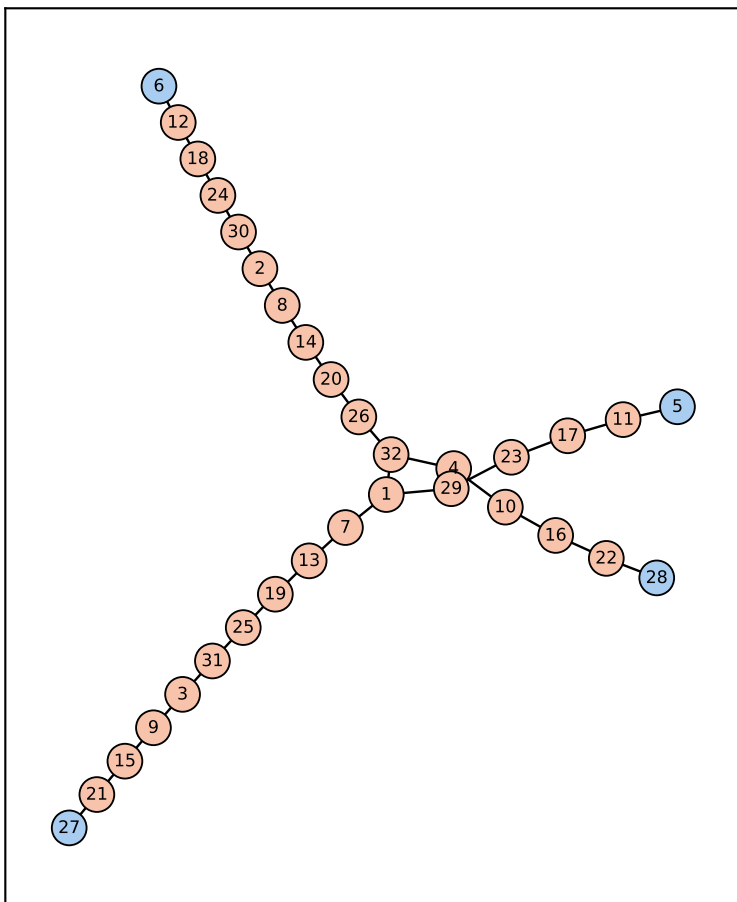
 $T_{30}(15, 22, 26, 28)$  (N)

 $s = (15, 8, 4, 2)$ ,  $\Delta = 4$ , not caterpillar

$T_{31}(15, 19, 29)$  (A)

 $s = (16, 12, 2)$ ,  $\Delta = 3$ , not caterpillar

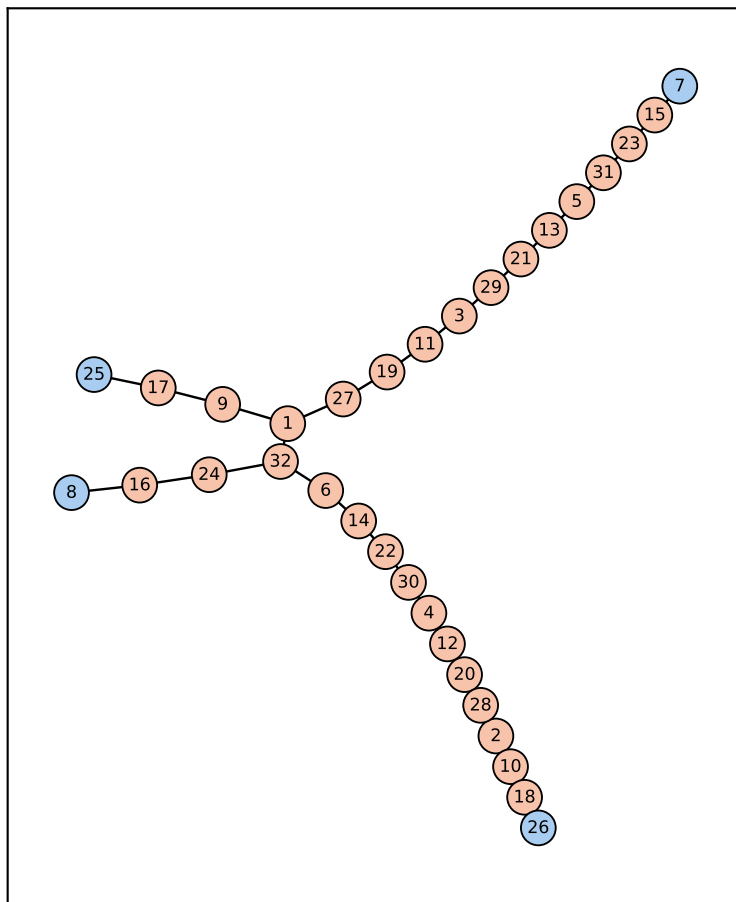
 $T_{31}(15, 23, 25)$  (B)

 $s = (16, 8, 6)$ ,  $\Delta = 3$ , not caterpillar

 $T_{31}(15, 23, 27, 29)$  (C)

 $s = (16, 8, 4, 2)$ ,  $\Delta = 4$ , not caterpillar

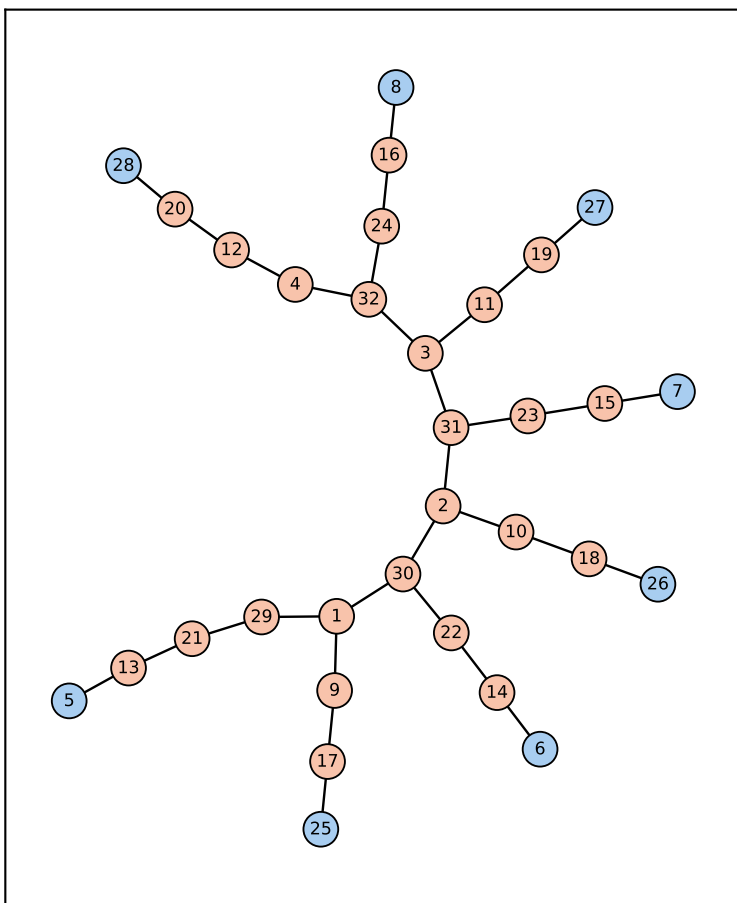
 $T_{32}(4, 30, 31)$  (A)

 $s = (28, 2, 1)$ ,  $\Delta = 3$ , not caterpillar

$T_{32}(6, 28, 31)$  (B)


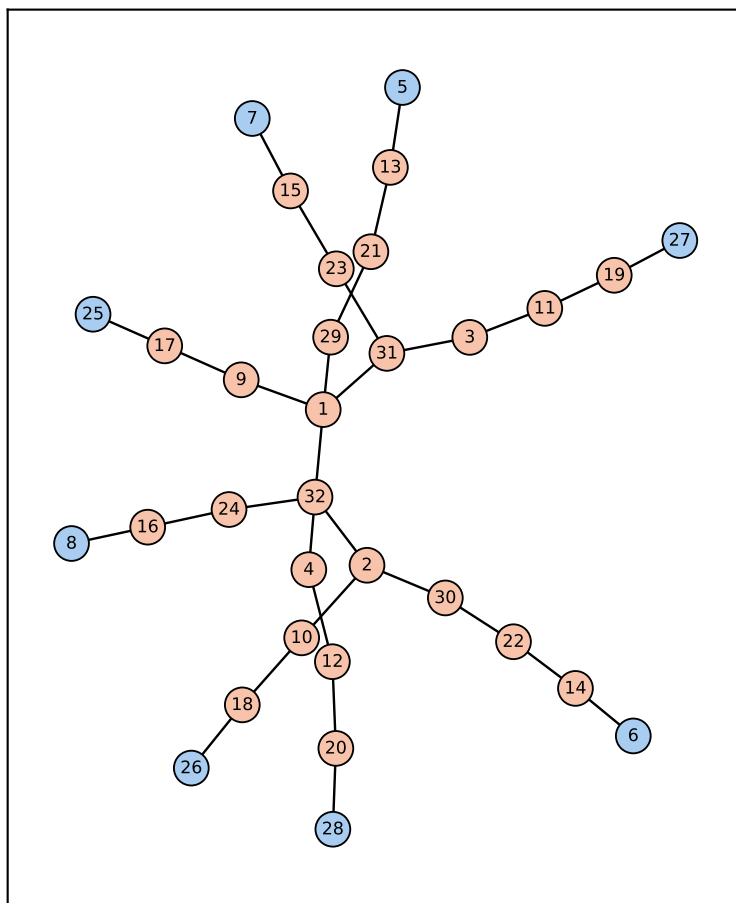
$s = (26, 4, 1)$ ,  $\Delta = 3$ , not caterpillar

 $T_{32}(8, 26, 31)$  (C)


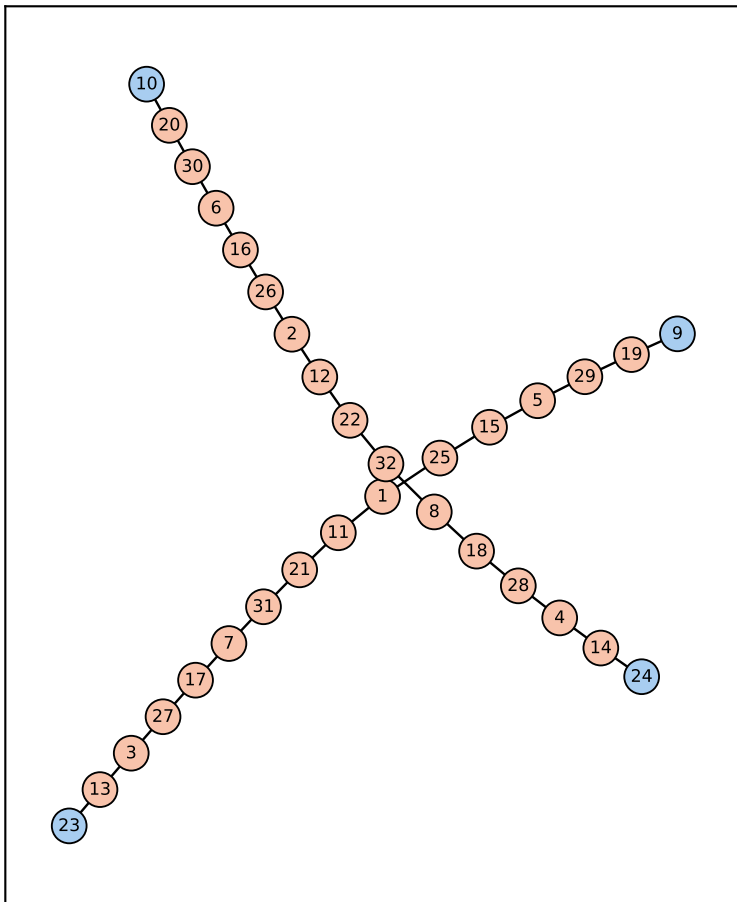
$s = (24, 6, 1)$ ,  $\Delta = 3$ , not caterpillar

 $T_{32}(8, 28, 29)$  (D)


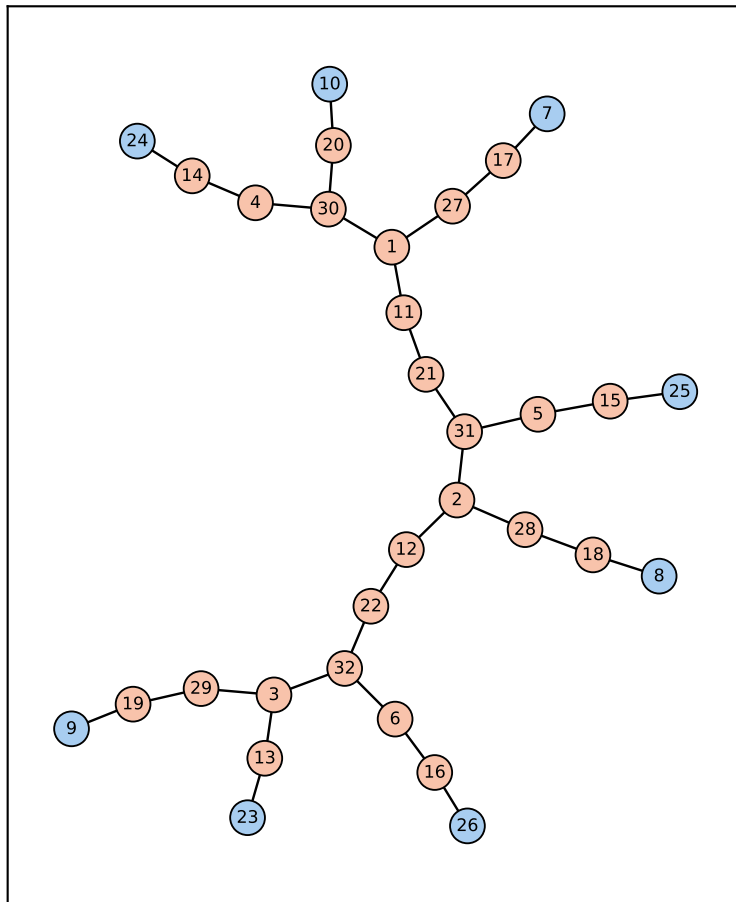
$s = (24, 4, 3)$ ,  $\Delta = 3$ , not caterpillar

 $T_{32}(8, 28, 30, 31)$  (E)


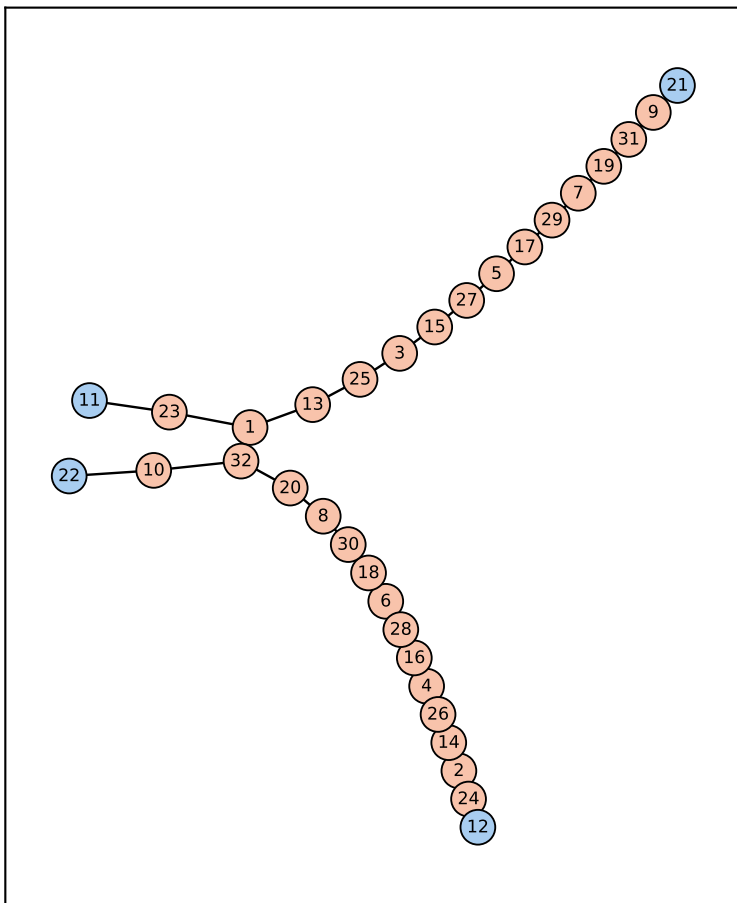
$s = (24, 4, 2, 1)$ ,  $\Delta = 4$ , not caterpillar

$T_{32}(10, 24, 31)$  (F)


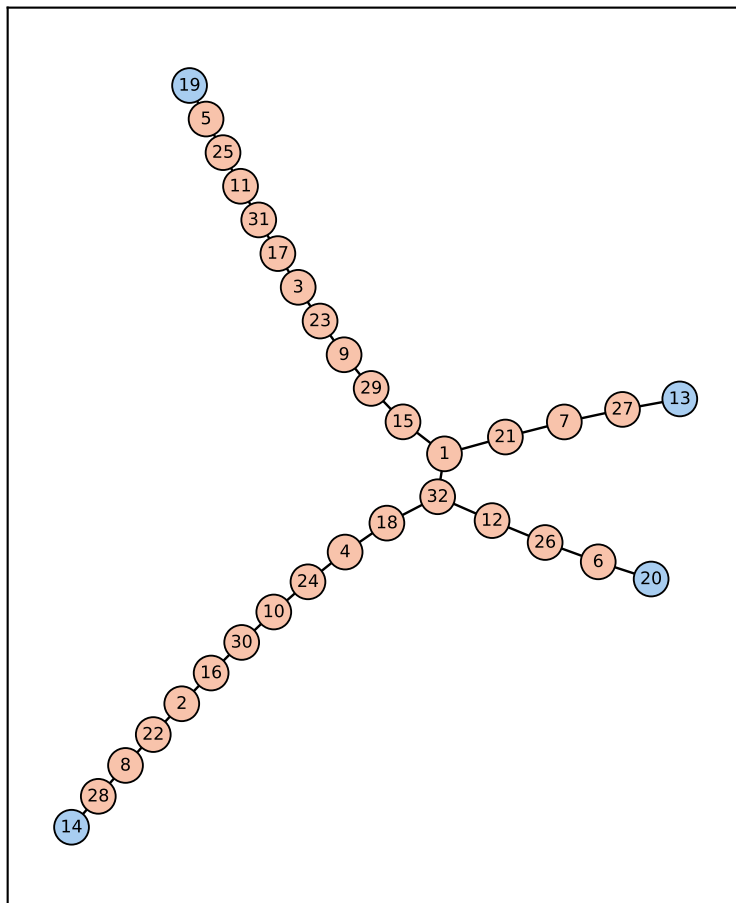
$s = (22, 8, 1)$ ,  $\Delta = 3$ , not caterpillar

 $T_{32}(10, 26, 29)$  (G)


$s = (22, 6, 3)$ ,  $\Delta = 3$ , not caterpillar

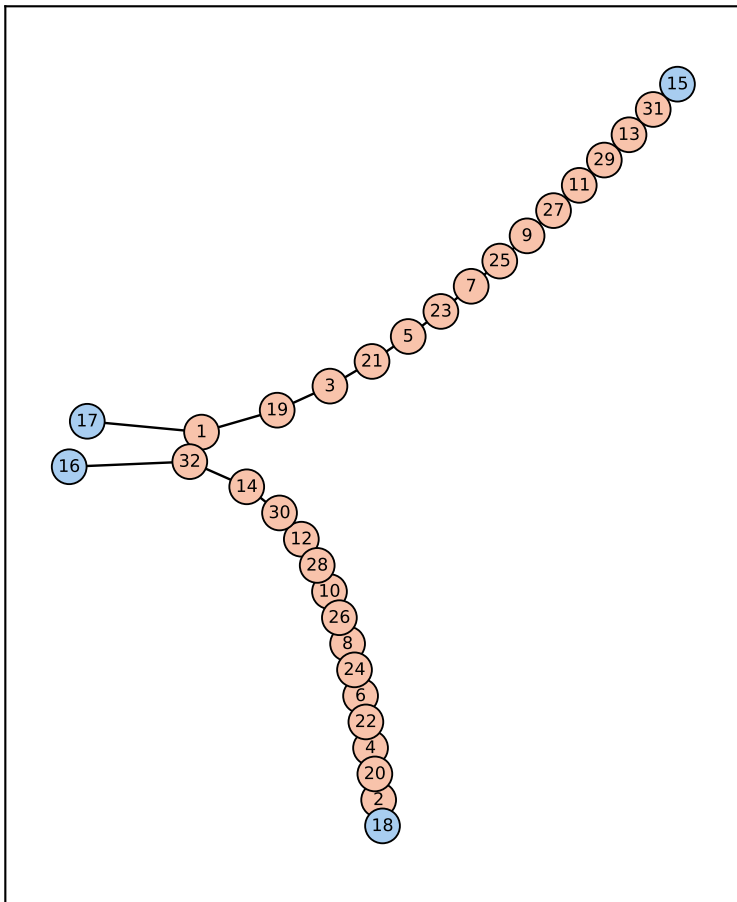
 $T_{32}(12, 22, 31)$  (H)


$s = (20, 10, 1)$ ,  $\Delta = 3$ , not caterpillar

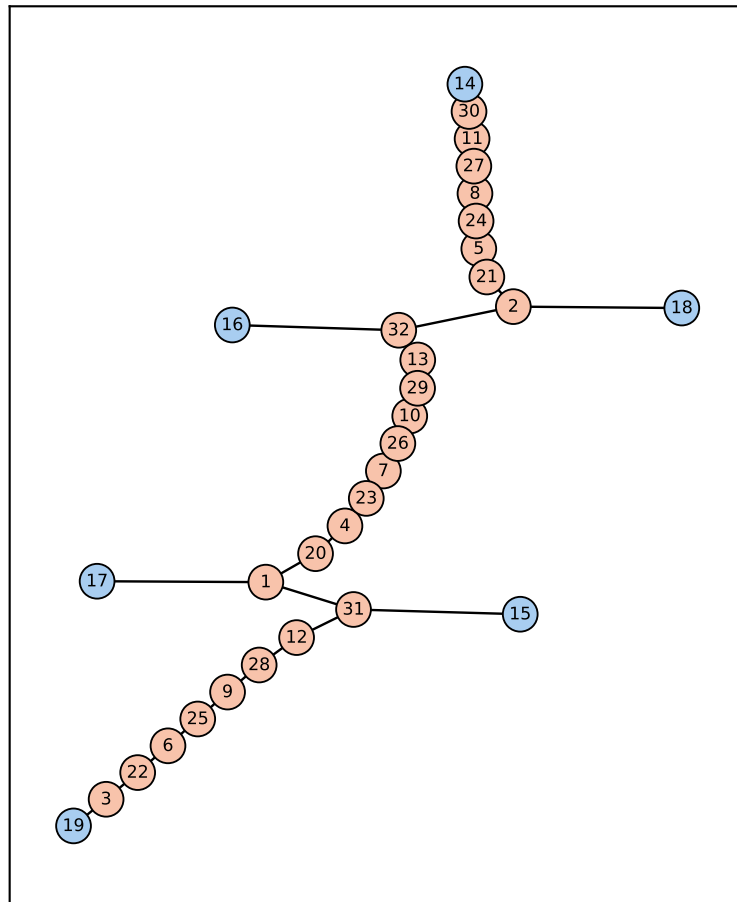
 $T_{32}(14, 20, 31)$  (I)


$s = (18, 12, 1)$ ,  $\Delta = 3$ , not caterpillar

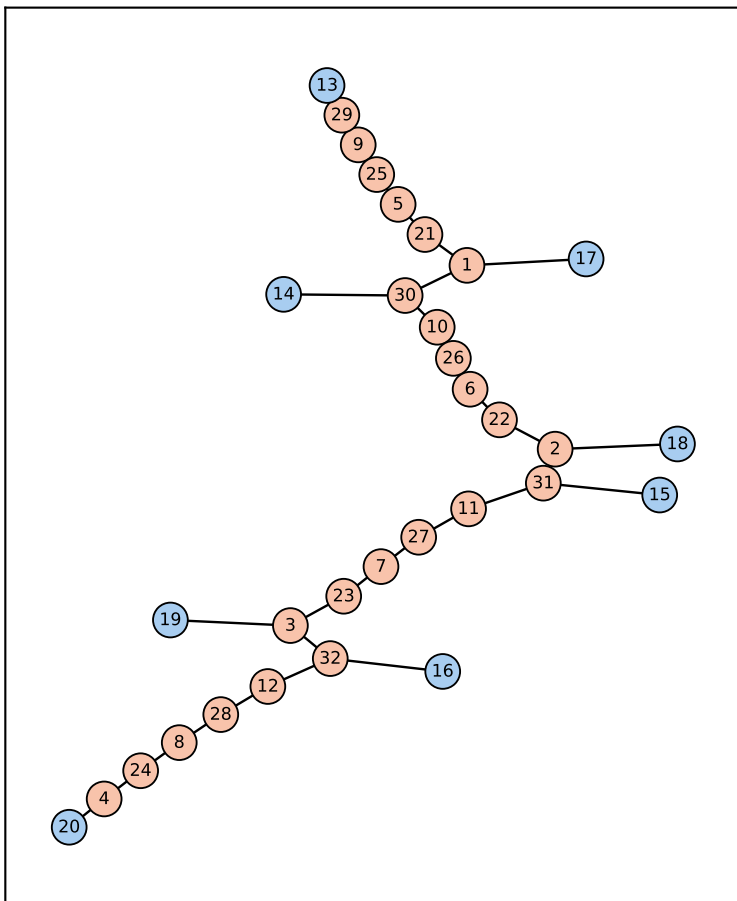


$T_{32}\langle 16, 18, 31 \rangle$  (J)


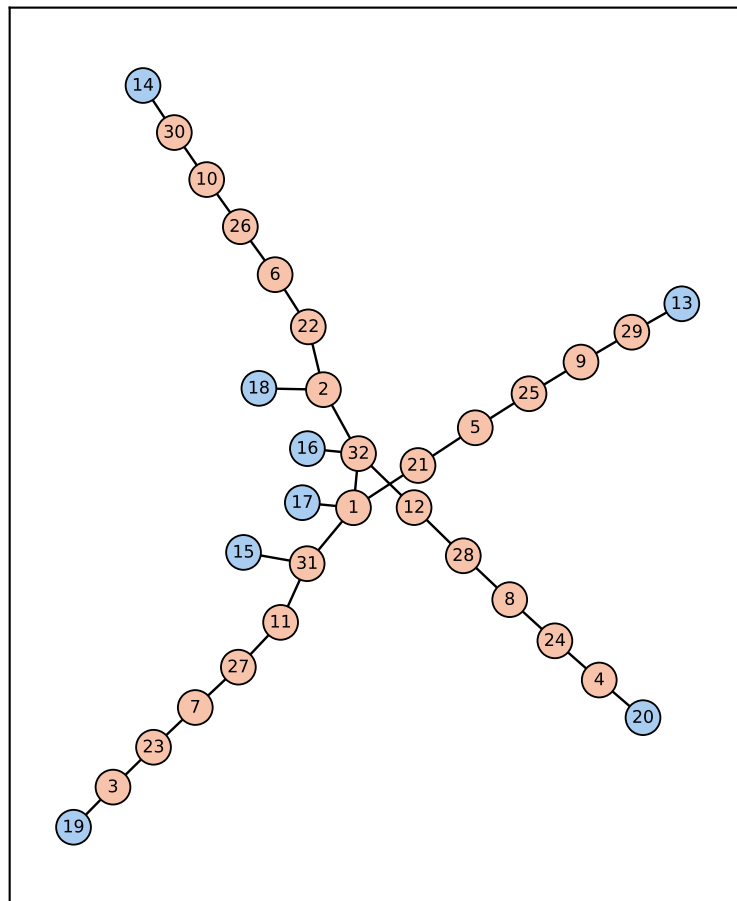
$s = (16, 14, 1)$ ,  $\Delta = 3$ , caterpillar

 $T_{32}\langle 16, 19, 30 \rangle$  (K)


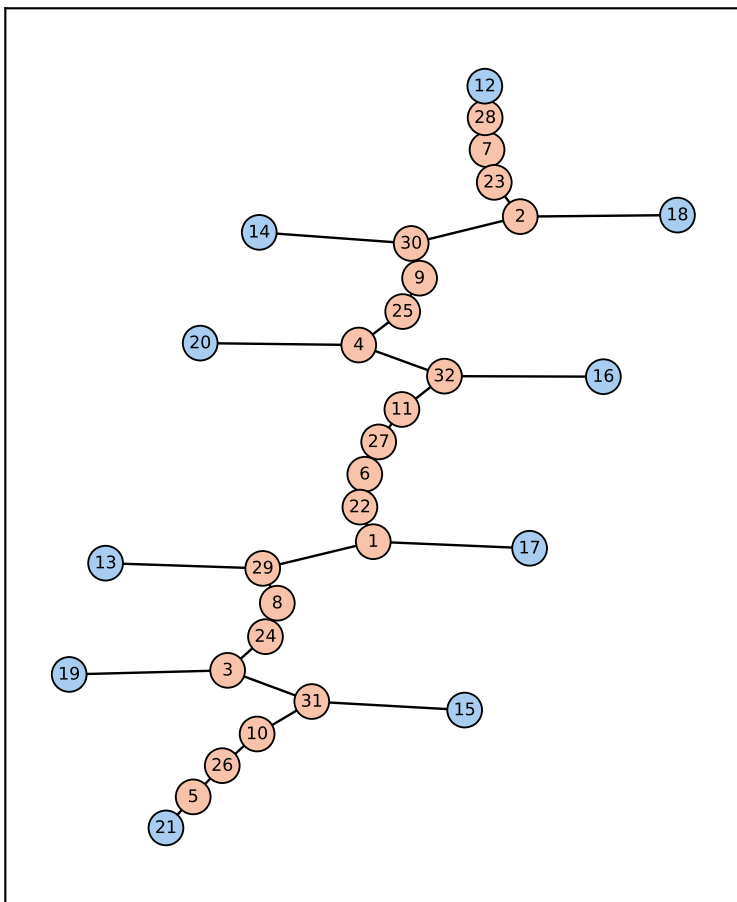
$s = (16, 13, 2)$ ,  $\Delta = 3$ , caterpillar

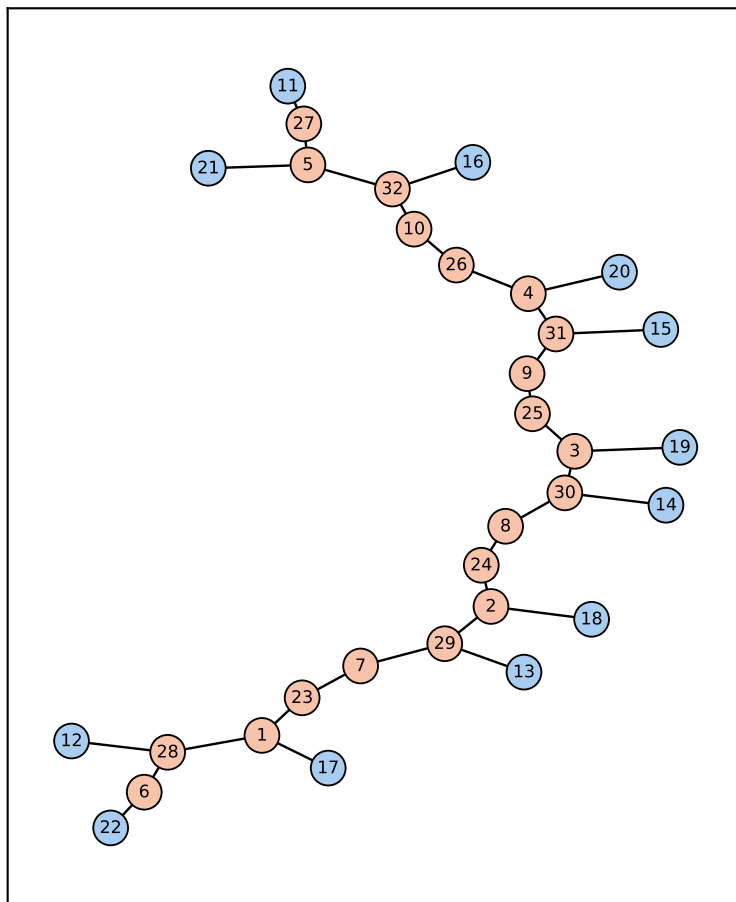
 $T_{32}\langle 16, 20, 29 \rangle$  (L)


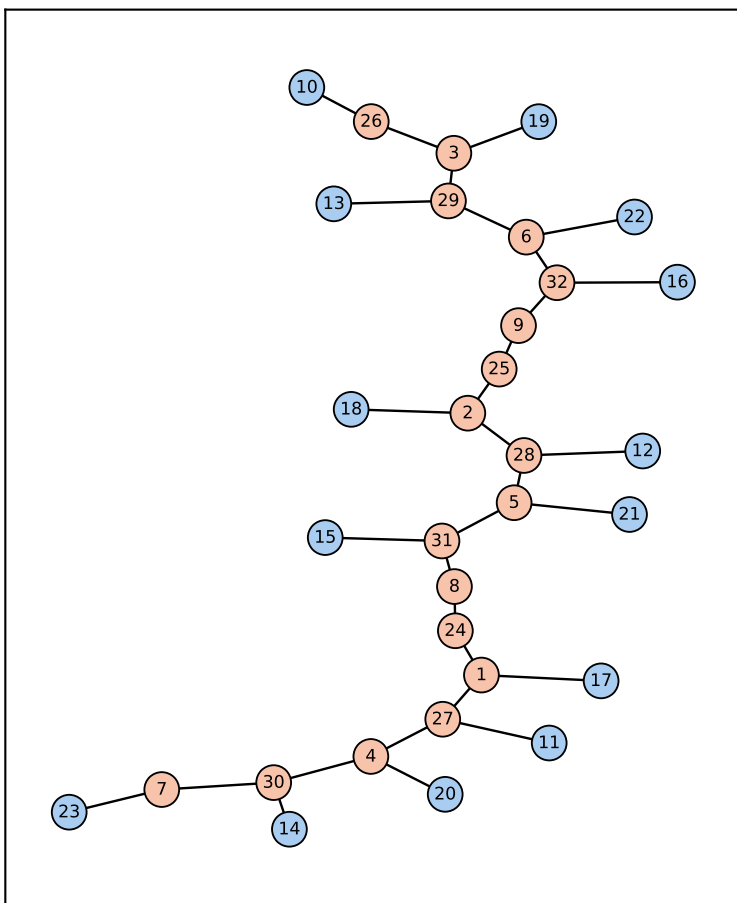
$s = (16, 12, 3)$ ,  $\Delta = 3$ , caterpillar

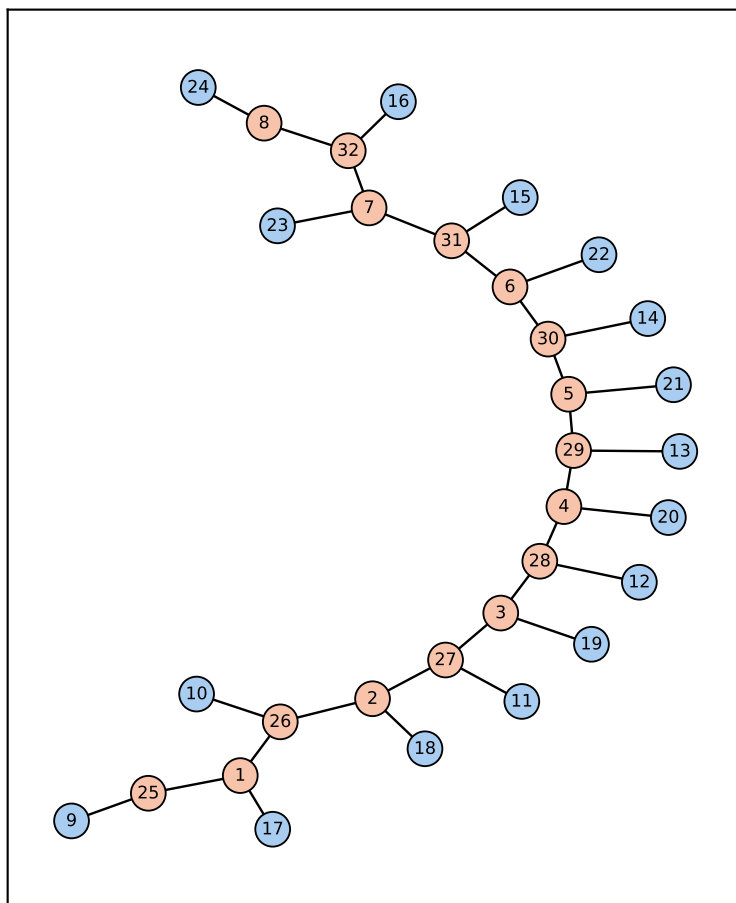
 $T_{32}\langle 16, 20, 30, 31 \rangle$  (M)


$s = (16, 12, 2, 1)$ ,  $\Delta = 4$ , not caterpillar

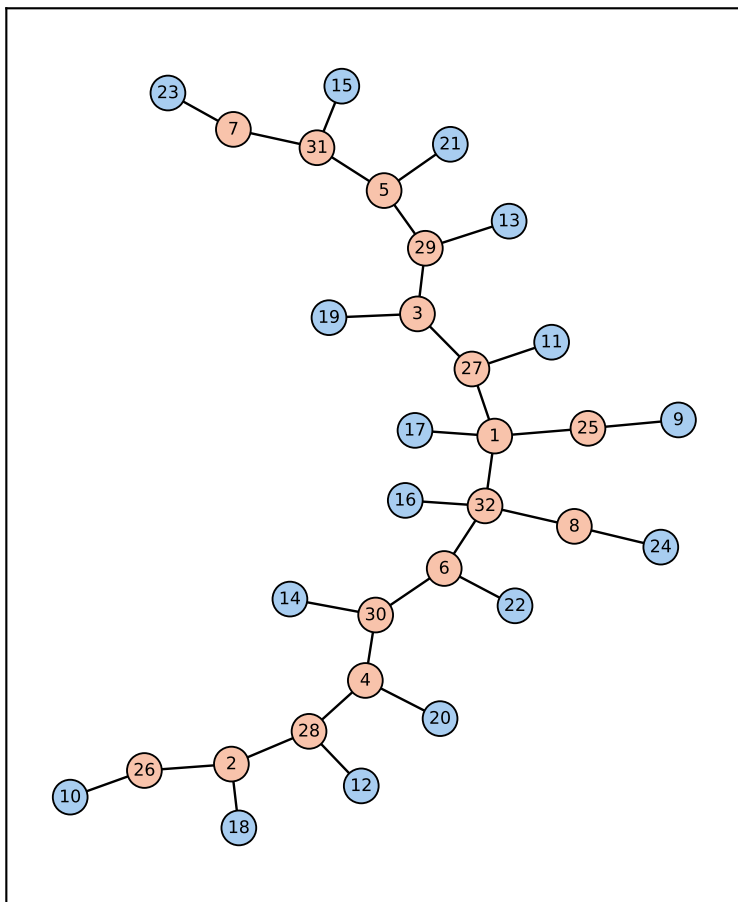
$T_{32}(16, 21, 28)$  (N)

 $s = (16, 11, 4)$ ,  $\Delta = 3$ , caterpillar

 $T_{32}(16, 22, 27)$  (O)

 $s = (16, 10, 5)$ ,  $\Delta = 3$ , caterpillar

 $T_{32}(16, 23, 26)$  (P)

 $s = (16, 9, 6)$ ,  $\Delta = 3$ , caterpillar

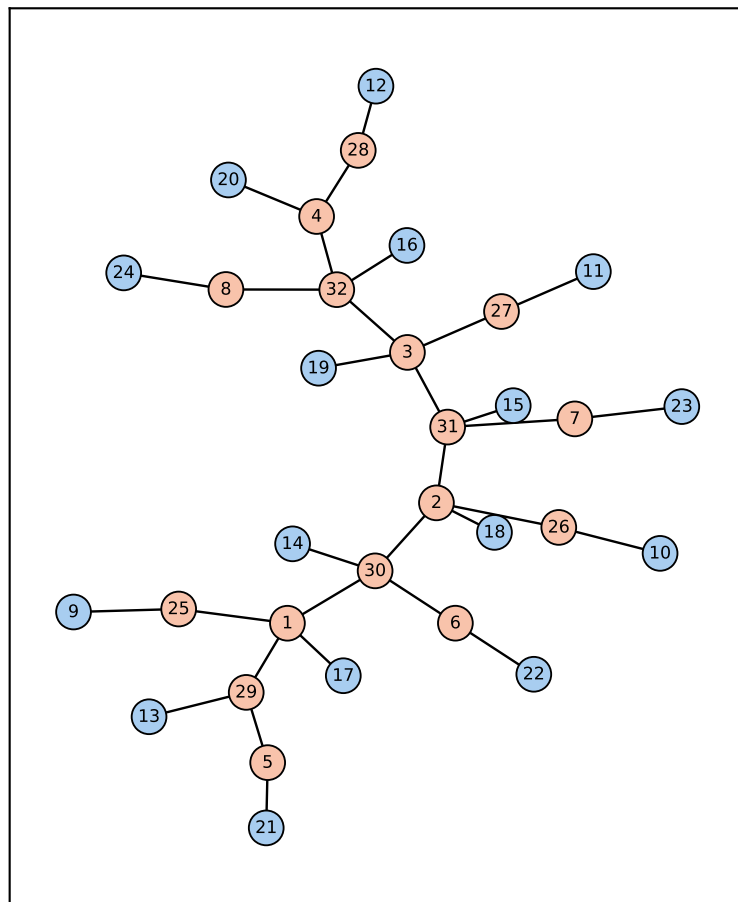
 $T_{32}(16, 24, 25)$  (Q)

 $s = (16, 8, 7)$ ,  $\Delta = 3$ , caterpillar

$T_{32}(16, 24, 26, 31)$  (R)



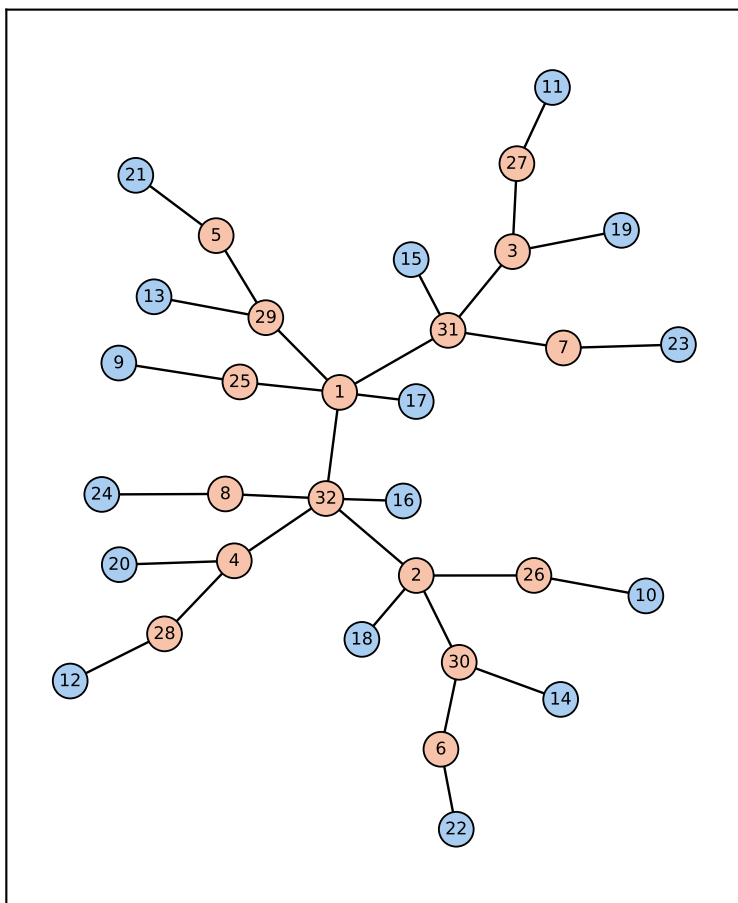
$s = (16, 8, 6, 1)$ ,  $\Delta = 4$ , not caterpillar

$T_{32}(16, 24, 28, 29)$  (S)



$s = (16, 8, 4, 3)$ ,  $\Delta = 4$ , not caterpillar

$T_{32}(16, 24, 28, 30, 31)$  (T)



$s = (16, 8, 4, 2, 1)$ ,  $\Delta = 5$ , not caterpillar